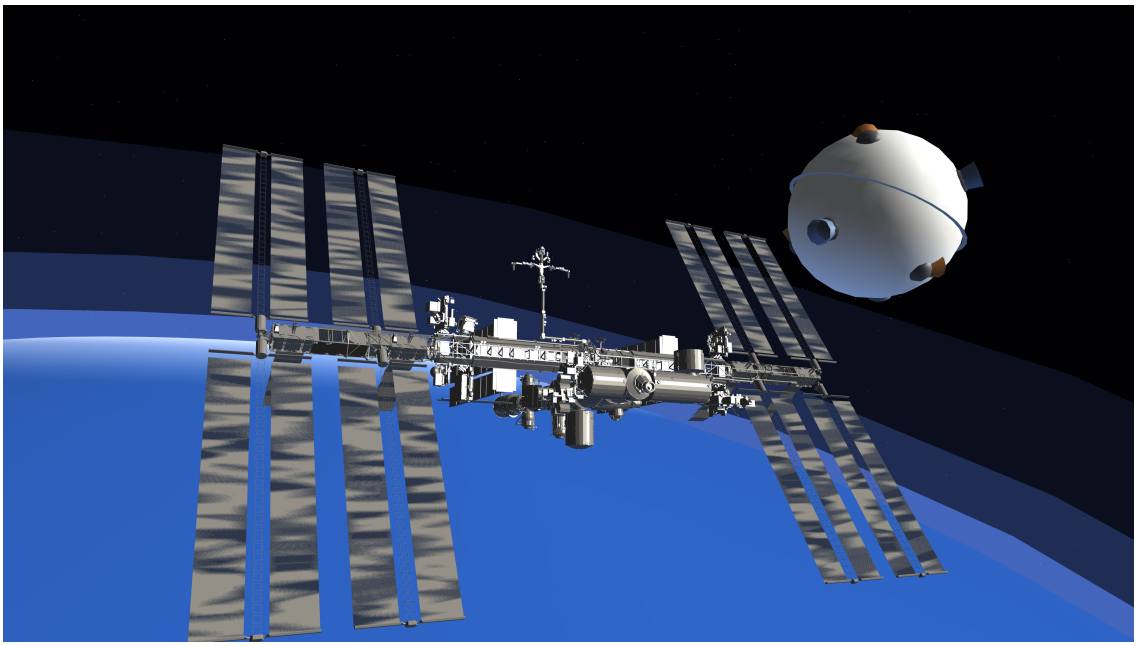
PROJECT REPORT

Sphere-1

A 500-gram cold-gas free-flying inspection drone for low Earth orbit

Propellant selection, actuator architecture, control allocation
and hardware-in-the-loop validation



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A bachelor's thesis project in mechanical engineering, defended in September 2026. The report is self-contained: figures quoted in the body are reproduced in the appendices alongside the data behind them, and the complete 24-slide defence presentation is included as Appendix A.

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CHAPTER 1

Summary

Sphere-1 is a 23 cm, 500 g free-flying inspection drone, intended to operate within a few metres of a larger orbital asset and photograph it in the visible and thermal bands. Eight cold-gas nozzles, fed from 10 g of liquid carbon dioxide, provide translation and attitude control.

This report covers the design chain end to end: the choice of propellant and the thermodynamic limit that follows from it, the reduction of the actuator count from twelve nozzles to eight under a mass budget, the closed-form resolution of the resulting redundancy at 200 Hz, and what the vehicle can do with ten grams of gas.

Principal results

The propellant fixes the nozzle geometry. Carbon dioxide deposits directly as a solid below its triple point of 216.6 K. Setting the nozzle exit temperature to that floor caps the exit Mach number at 1.589 and the expansion area ratio at 1.266. At the 0.80 mm flight throat the divergent cone that ratio permits is 0.18 mm long, a chamfer rather than a nozzle, so the nozzle is converging-only. The cost is 4.5 % of specific impulse; the return is a part that can be machined repeatably.

Valve mass sets the actuator count, not control theory. Positive spanning of the six-dimensional wrench space requires at least seven push-only thrusters. Twelve decouple cleanly and give 2.78 times the torque per pulse, but twelve valves at 25 g each put the vehicle 90 g over budget before any propellant is loaded. Seven has no antagonistic pairs, so every translation command carries a parasitic torque. Eight, as four radial and four canted nozzles, recovers the decoupling of twelve within the mass budget. Compared on lifetime angular impulse rather than torque per pulse, eight wins for any valve heavier than 14.7 g, and the lightest purchasable flight solenoid weighs 20 g.

The minimum pulse limits the mission, not the environment. One nozzle held open empties the tank in 48 s. The same ten grams sustains three-axis pointing in a $\pm 1.5°$ limit cycle for 3.3 days, a factor of about 5 900. Burn rate scales with the square of the valve opening time, so halving that time from 1.0 ms to 0.5 ms quadruples mission life at no cost in mass. Gravity-gradient torque on a near-spherical body is of order 10^{-13} N·m and drag costs roughly 27 pulses per orbit, both far below the control authority.

Trajectory and hardware results

A separate result changed the survey profile. Holding a curved path in vacuum requires continuous centripetal thrust, so a circling survey is charged $\Delta v = v\theta$ for every radian it turns, while a straight leg costs $2v$ regardless of length. Re-planning the survey as straight ballistic legs cut the propellant cost of one complete pass from 73 % of the tank to 24 %, with coverage unchanged at 98.7 % against 99.1 %.

The control loop was validated on hardware. An ESP32 running the flight controller at 200 Hz, reading a real inertial measurement unit and driving a simulated plant, reduced mean attitude tracking lag from 11° to 1.2° once gyro-rate feedforward was added, on a rig costing under $50.

> The design is complete and internally consistent, and its weakest link is identified. Every thrust and torque figure in this report scales linearly with the injector's effective flow area, which remains a datasheet value. Measuring it on a flow bench is the first item on the roadmap.

CHAPTER 2

Background: the capability gap

2.1 STS-107

On 16 January 2003, 81.7 s after the launch of STS-107, a piece of insulating foam separated from the Space Shuttle *Columbia*'s external tank at the left bipod ramp. It struck the leading edge of the left wing at approximately 500 mph, near reinforced carbon-carbon panel 8. The strike was identified the following day during routine review of launch film.

The strike was known about within roughly 24 hours. Its severity was not. STS-107 was a Spacehab science flight carrying no robotic arm. The crew could not see the wing leading edge from the flight deck windows, and no extravehicular activity was planned. The only remaining means of assessing the damage was imaging of the orbiter on orbit from the ground or from national assets.

Three separate requests for that imagery were raised: by the chair of the Intercenter Photo Working Group, through the Department of Defense Manned Space Flight Support Office, and by the Debris Assessment Team's NASA co-chair. All three were cancelled. The reasoning recorded from the Mission Management Team includes the position that the strike was not a factor during the flight because there was not much that could be done about it.

> That argument is circular. The damage cannot be inspected, therefore nothing could be done, therefore there is no value in inspecting it. The middle premise was never tested, because the information that would have tested it was the information declined.

The Columbia Accident Investigation Board examined the counterfactual in Volume II, Appendix D.13. A rescue flight by *Atlantis* was assessed as challenging but feasible, had the damage been confirmed by about flight day seven. *Columbia*'s consumables, limited by the lithium hydroxide carbon-dioxide scrubbers, would have supported the crew to roughly flight day 30. An improvised on-orbit repair was rated high risk with no means of verification. Neither option reached a decision, because the question of whether the wing was damaged was never answered.

2.2 The remedies adopted afterwards

The hardware built after the accident states the missing capability clearly. The Orbiter Boom Sensor System is a 50-foot extension carrying a Laser Dynamic Range Imager with an intensified television camera, and a Laser Camera System with a digital camera. Mounted on the Shuttle's robotic arm it extended the reach to 100 feet, and it was flown on every subsequent mission for the sole purpose of letting the orbiter inspect its own thermal protection system. It was complemented by the Rendezvous Pitch Manoeuvre, a full rotation of the orbiter beneath the International Space Station so that the station crew could photograph the heat shield with long lenses.

Both are workarounds for one absent function: a camera that can leave the vehicle and look back at it. Both require the host vehicle to carry the solution, either a 50-foot boom or a station to fly beneath.

2.3 Requirements

Sphere-1 provides that function without the host vehicle paying for it. It is not a repair system, and it would not have saved *Columbia*. That flight had no arm and no station to deploy from, and a 500 g free-flyer has no business manoeuvring near a vehicle in an unknown state. The concept addresses the narrower problem of converting an unknown into a known, early enough for a decision still to exist.

Three requirements follow, and they drive Chapters 3 to 5.

1. **No contamination of the inspected surface.** At a standoff of a few metres a chemical exhaust deposits soot and corrosive species on the optics, radiators and solar arrays the vehicle was sent to photograph. This eliminates chemical propulsion before any performance comparison.
2. **Small enough to be a payload rather than a mission.** A free-flyer is only worth carrying if it costs less than a boom.
3. **A useful pointing duration on a very small gas supply.** Inspection is a pointing task, not a manoeuvring task, so the propellant argument concerns limit cycles rather than Δv.

CHAPTER 3

Vehicle concept and budgets

3.1 Concept

The vehicle is a 23 cm sphere with a wet mass of 500 g. It carries a visible camera and a thermal camera, and is propelled by eight cold-gas nozzles mounted on the hull at a radius of 115 mm.

The spherical form was chosen for three properties. It has no preferred orientation, so an uncommanded tumble is a nuisance rather than a failure. It has no deployables to jam. Its principal moments of inertia are nearly equal, which makes the gravity-gradient torque acting on it negligible, of order 10^{-13} N·m. That last property is what makes a ten-gram tank a sensible design point, and Chapter 9 quantifies it.

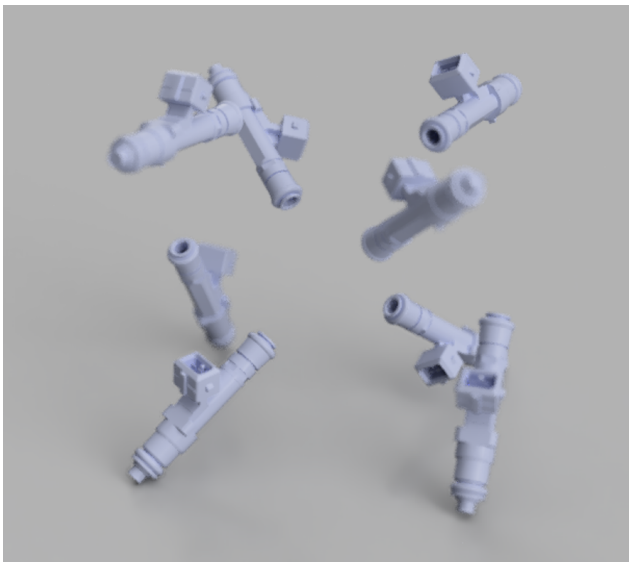


Figure 3.1. The selected eight-nozzle arrangement. Four radial nozzles on alternating corners of a cube produce force with no torque; four canted nozzles produce torque.

3.2 Mass budget

The mass budget is the most consequential table in the project, because it is what eliminates the conventional twelve-nozzle layout in Chapter 5.

Two consequences follow. First, 200 of the 490 dry grams are valves. This is a valve-limited vehicle rather than a propellant-limited one, which is why the middle of this report concerns valve count rather than tank size. Second, propellant mass is not an independent design variable. It is the remainder after the valves and the 290 g of supporting hardware are subtracted from the 500 g target.

The propellant mass of 10 g is not a physical limit. Ten grams of liquid carbon dioxide occupies about 13 cm^3, roughly 0.2 % of the sphere's internal volume, and tank mass scales at approximately 0.28 g per gram of propellant carried. The vehicle also sits deep in the linear region of the rocket equation: carrying 100 g instead of 10 g would give about 80 m/s of Δv rather than 9.18, on a 615 g vehicle. Ten grams is simply what the 500 g target leaves once eight valves have been paid for.

Table 3.1. Vehicle mass and capability budget.

Quantity	Value	Note
Sphere diameter	**23.0 cm**	6 371 cm^3 envelope; mounting radius R = 115 mm
Dry mass m_d	**490 g**	valves, electronics, tank, structure allowance
of which, eight valves	**200 g**	25 g each, flight budget
remainder	**290 g**	structure, tank, battery, electronics
Propellant m_{prop}	**10 g CO_2**	3.3 days of three-axis pointing, or 9.18 m/s of Δv
Wet mass m_w	**500 g**	design target
Control authority	**6 DOF**	3 translation and 3 rotation

3.3 Scope and exclusions

The budget covers the functional vehicle: valves, electronics, tank and propellant. It excludes structure sized for launch loads, micrometeoroid and debris shielding, and pressure-vessel certification for proximity operations near a crewed asset. None of these are estimated here. 500 g is therefore a self-imposed design target intended to keep the functional core honest, not the mass of a flight article.

The exclusions bound the claims rather than dismiss the work they represent. The items that would drive a certified design, principally containment of the pressure vessel, debris shielding and a demonstrated safe-abort behaviour, do not change the propulsion or control conclusions reached here. They would change the mass.

CHAPTER 4

Propulsion

4.1 Propellant selection

Nitrogen is the usual cold-gas propellant and on performance it is the better choice. Its ideal specific-impulse ceiling is 79.6 s against 72.5 s for carbon dioxide, and it cannot condense anywhere within the expansion range this vehicle uses.

Table 4.1. Propellant comparison at the design storage condition.

	Nitrogen N_2	Carbon dioxide CO_2
Ideal I_{sp} ceiling	79.6 s	**72.5 s**
State at 20°C	gas	**liquid**
Storage pressure	200+ bar	**≈ 57 bar**
Tank wall	thick, heavy	**thin, light**
Stored mass per litre	baseline	**several ×**
Condensation risk	none	**below 216.6 K**

Nitrogen loses on storage. It remains gaseous at room temperature, so a useful propellant mass requires roughly 200-bar tankage, and the wall thickness that implies is charged to the same 490 g dry budget that already carries 200 g of valves. Carbon dioxide liquefies at about 57 bar and stores several times the mass in the same volume behind a much thinner wall.

The trade is specific impulse for storage density. Its price is a hard thermodynamic limit, and that limit determines the nozzle geometry.

Butane and R-134a are legitimate alternatives on the same reasoning. Both store as low-pressure liquids and are flown on small satellites. Carbon dioxide was selected for availability, well-characterised properties, and inertness with respect to optical surfaces. A resistojet or warm-gas variant was rejected on electrical power: a 500 g vehicle cannot support a meaningful heater, and the thermal control it would require is additional mass.

4.2 The condensation limit

A supersonic nozzle is an efficient refrigerator. It converts thermal energy into directed kinetic energy, so the static temperature of the flow falls as it accelerates. For isentropic flow of a perfect gas,

$$T_e \;=\; T_0 \left[1 + \frac{\gamma - 1}{2} M_e^2 \right]^{-1} \tag{4.1}$$

with a stagnation temperature T_0 = 293.15 K and $\gamma = 1.280$ for carbon dioxide at room temperature, against 1.400 for nitrogen and air.

Carbon dioxide has a floor that nitrogen does not. Below its triple point at 216.6 K, carbon dioxide does not condense to a liquid; it deposits directly as a solid. Inside a throat less than a millimetre across, dry ice blocks the nozzle and changes the thrust unpredictably.

The design chain is therefore solved backwards from that floor rather than forwards from a performance target:

$T_e \geq 216.6\ \mathrm{K}$	the CO_2 triple point; below it the propellant deposits as a solid
$M_e \leq 1.589$	the resulting ceiling on exit Mach number, from (4.1) at $T_0 = 293.15\ \mathrm{K}$
$A_e/A_t \leq 1.266$	the entire expansion this vehicle is thermodynamically allowed, from the area–Mach relation

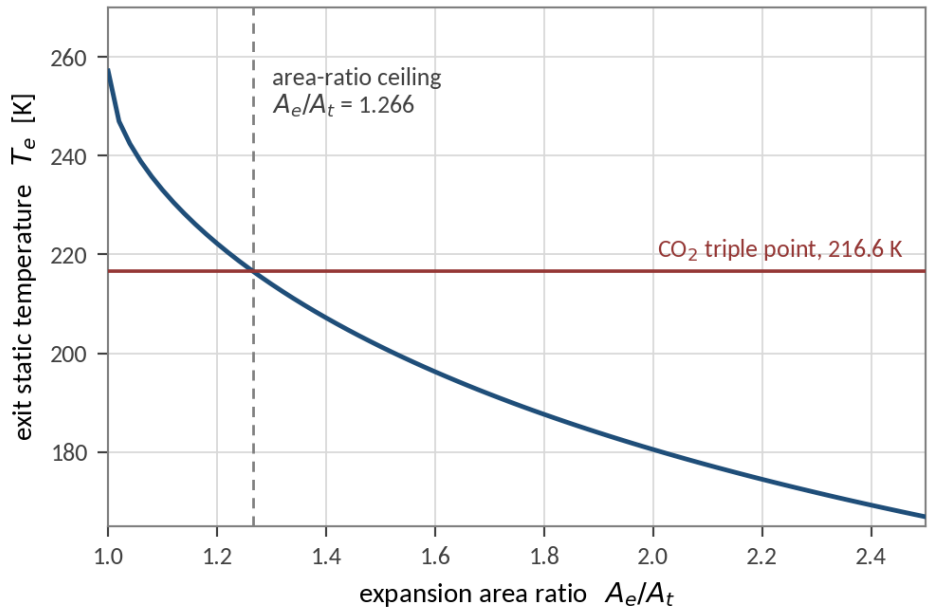


Figure 4.1. Nozzle exit static temperature against expansion area ratio for carbon dioxide at $T_0 = 293.15\ \mathrm{K}$. The curve crosses the 216.6 K triple point at $A_e/A_t = 1.266$.

The model is ideal isentropic flow with equilibrium condensation, the standard first-order treatment. Two real effects are not modelled and act in opposite directions: boundary-layer and non-equilibrium losses raise the true exit temperature above the ideal value, while real expansions supercool below the equilibrium point before nucleation begins. Sizing to the equilibrium limit is the conservative choice, and it is used here as a design rule rather than as a hard physical wall.

4.3 Consequence: the nozzle has no divergent section

An expansion ratio ceiling of 1.266 describes a very small nozzle.

Ideal specific impulse, converging-only	**46.33 s**
Ideal specific impulse, converging–diverging at $A_e/A_t = 1.25$	**48.42 s**
Gain available from the divergent cone	**+ 4.5 %**

That gain is real but unobtainable. At a 2.0 mm throat, the divergent section permitted by the area ratio is 0.44 mm long. At the 0.80 mm flight throat established in Section 4.4 it is 0.18 mm long. At that scale it is a chamfer rather than a nozzle: it cannot be machined repeatably to tolerance, and a malformed cone risks the icing the expansion was constrained to avoid.

The nozzle is converging-only. The decision is not specific impulse against specific impulse. It is a 4.5 % gain set against a feature that cannot be held to tolerance and that fails dangerously when out of tolerance.

4.4 Throat sizing: two restrictions in series

Chamber pressure is not a free parameter here. The injector orifice and the nozzle throat are two restrictions in series, and in steady state they pass identical mass flow. Writing the choked mass-flow relation for both and equating them, every gas property cancels, leaving an area ratio:

$$P_0 = P_{\text{feed}} \frac{(C_d A)_{\text{inj}}}{(C_d A)_{\text{throat}}} \tag{4.2}$$

The consequence is counter-intuitive: enlarging the throat lowers the chamber pressure, so machining a bigger hole makes the thruster weaker.

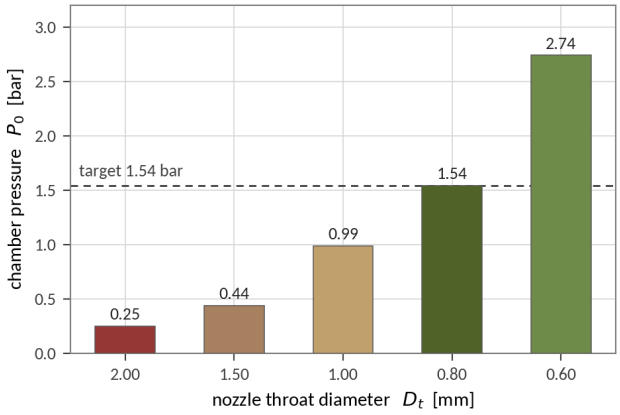


Figure 4.2. Chamber pressure against throat diameter, from (4.2). A 2.0 mm throat collapses the chamber to about 0.25 bar, a 95 % loss of thrust. The relation holds only where both restrictions choke, so the 2.00 mm point is a vacuum-only result; at the 0.80 mm flight throat the chamber sits at 1.54 bar and is comfortably choked against vacuum.

4.5 Flight operating point

Throat diameter	**0.80 mm**
Chamber pressure P_0	**1.5392 bar**
Mass flow per nozzle $\dot{m}$	**0.2073 g/s**
Thrust per nozzle F_0	**94.188 mN**
Injector effective flow area $(C_d A)_{\text{inj}}$	**0.147 mm²** (datasheet)

Every thrust, torque and mission-life figure in this report traces back through (4.2) to the injector's effective flow area, which is a datasheet number rather than a measurement. The dependence is linear, so a 10 % error there is a 10 % error in every thrust and torque quoted. Flow-benching the injector is roadmap item one.

CHAPTER 5

Actuator architecture

5.1 The positive-spanning requirement

The vehicle must control six quantities: three components of force and three of torque. Thrusters produce only positive thrust, so the set of achievable wrenches is the set of non-negative combinations of the individual thruster wrenches, not their linear span.

Positively spanning an n-dimensional space with one-sided generators requires at least $n+1$ of them. Under non-negative combination, n vectors can at best reach a cone; at least one more, in general position, is needed to reach every direction. For a six-dimensional wrench space the minimum is seven. With actuators that could both push and pull, six would suffice. The difficulty of cold-gas reaction control layout comes entirely from the one-sidedness.

Three layouts were built and modelled, in the order given below. Each failed for a different reason, and that sequence is the substance of the design work.

Table 5.1. The four configurations evaluated. Valve mass at 25 g each; propellant is what remains of the 500 g target after the valves and 290 g of structure, tank, battery and electronics.

Configuration	Valve mass	Propellant left	6-axis decoupling	Verdict
6 minimalist	150 g	+ 60 g	no, under-actuated	cannot span 6 DOF
7 minimum	175 g	+ 35 g	no, heavy coupling	loop-fighting
8 two groups	**200 g**	**+ 10 g**	**yes, decoupled**	**selected**
12 conventional RCS	300 g	− 90 g	yes, decoupled	does not close

The 80 g automotive injector used on the bench is a test article for flow characterisation. The flight budget assumes a 25 g solenoid, which is a real class of part. Both numbers appear in the presentation deliberately; the mass budget is built on the 25 g figure.

5.2 Iteration 1: twelve nozzles

The conventional reaction-control layout is six antagonistic pairs, one per axis per direction. Every pair fires directly along the axis it commands, so allocation is a lookup table with no mathematics in it. All of the fired thrust reaches the commanded axis, and the torque per pair is 21.66 mN·m, which is 2.78 times what the eight-nozzle layout achieves.

It fails on mass, and not marginally. Twelve valves at 25 g is 300 g. Adding the 290 g of structure, tank, battery and electronics gives 590 g against a 500 g budget, leaving −90 g for propellant. Lighter valves do not rescue it. At 17.5 g per valve the layout stops closing the budget entirely, and the lightest flight solenoids realistically available weigh 20 g to 30 g.



Figure 5.1. Iteration 1: twelve nozzles in six antagonistic pairs.

5.3 Iteration 2: seven nozzles

Seven is the mathematical minimum from Section 5.1, and on mass it is the better design. 175 g of valves leaves 35 g of propellant instead of 10 g, which is three and a half times the mission life.

Seven fails on control rather than on mass. With seven thrusters in general position there are no antagonistic pairs, meaning no pair whose thrust components cancel except along a single axis. Every translation command therefore carries a parasitic torque, which the allocator can null only by firing additional thrusters against the ones producing the commanded translation. The attitude loop and the translation loop then fight each other continuously, and on a 10 g tank that exchange is paid for in propellant until the tank is empty.

> Seven spans the wrench space on paper and cannot be decoupled in hardware. It buys propellant and spends it fighting itself.

A further result came out of the search for a symmetric seven-nozzle layout: the most symmetric arrangement does not work. A C_5-symmetric pentagonal bipyramid fails to positively span $\mathbb{R}^6$ at every cant angle between 30° and 80° and at every 15° azimuth step tested. A working seven-nozzle layout requires the symmetry to be deliberately staggered. In the configuration generated for comparison in Appendix B, the ring azimuth advances 228° per nozzle. Its derived authority is given in Appendix C.

5.4 Iteration 3: eight nozzles, selected

One redundant thruster above the minimum recovers the decoupling of twelve. The eight nozzles form two independent groups of four.

The radial group, R1–R4

Four nozzles mounted on four alternating corners of a cube, which places them at the vertices of a regular tetrahedron, each at $115/\sqrt{3} = 66.395$ mm on every body axis and therefore at a radius of 115 mm. Each fires radially outward, directly through the centre of mass:

$$\tau = \mathbf{r} \times \mathbf{F} = \mathbf{0} \qquad \text{for} \qquad r_{\perp} = 0.000\,mm \tag{5.1}$$

Because the thrust line passes through the centre of mass, the perpendicular moment arm is exactly zero and these four produce force with no parasitic torque. This was verified numerically to 0.000 N·m rather than to a small residual, and it is what makes splitting the vehicle into two independent control groups legitimate.

The canted group, T1–T4

Four nozzles mounted on the 115 mm hull surface and swung 38.50° away from radial. That swing creates the moment arm:

$$\text{arm} \;=\; R\sin\theta \;=\; 115\,mm \times \sin 38.50 \;=\; 71.59\,mm \tag{5.2}$$

The cant is frequently misread as a plume-avoidance compromise. It is not. A nozzle pointing straight out has no lever at all, which is what the radial group demonstrates by producing exactly zero torque. The cant is the mechanism by which torque exists.

Thrust per nozzle F_0	**94.188 mN**
Moment arm $R\sin\theta$	**71.59 mm**
Torque per nozzle τ_0	**6.743 mN·m**
Ideal geometric pair torque	**7.79 mN·m**
Achievable turn authority, Y axis	**7.45 mN·m**
Achievable turn authority, X and Z	**5.87 mN·m**
Net force per body axis, radial group	**108.76 mN**

> Eight nozzles give the decoupling of twelve at a valve count the mass budget can carry.

5.5 Why twelve does not pay

Torque per pulse is the wrong metric, and comparing configurations on it is what makes twelve look unanswerable. What a mission spends is total angular impulse: torque multiplied by how long the tank can keep supplying it.

$$H \;=\; \frac{\tau\, m_{\text{prop}}}{2\,\dot{m}} \tag{5.3}$$

The four extra valves buy 2.78 times the torque and cost 100 g, taken directly from propellant. Evaluating (5.3) for both layouts as a function of valve mass gives a crossover.

Valve unit mass	Outcome
below 14.7 g	twelve delivers more lifetime angular impulse
above 14.7 g	**eight delivers more lifetime angular impulse**
above 17.5 g	twelve does not close the mass budget at all

Every flight solenoid realistically purchasable weighs between 20 g and 30 g. The entire purchasable range sits above both thresholds, so the eight-nozzle layout wins everywhere the vehicle can actually be

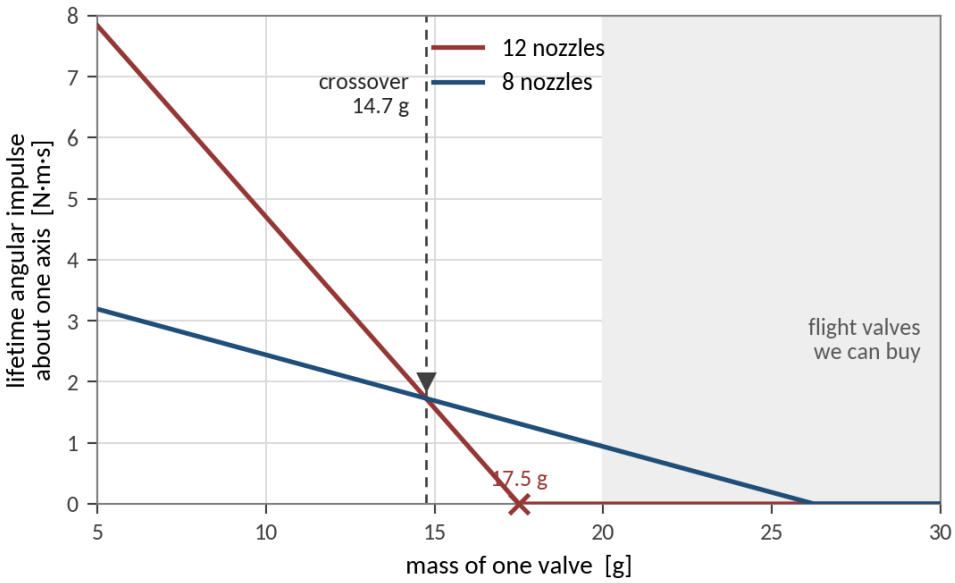


Figure 5.2. Lifetime angular impulse for the eight- and twelve-nozzle layouts against valve unit mass. The curves cross at 14.7 g. Above 17.5 g the twelve-nozzle layout does not close the mass budget at all.

built. The same trade structure rejects the fully tangential nozzle in Section 5.7: more torque per pulse, bought with propellant, on a vehicle whose binding constraint is how long the gas lasts.

H as defined in (5.3) *is an upper bound rather than a realisable budget, since it assumes all propellant is spent about one axis. It is used to compare two configurations under the same convention, so the crossover point is unaffected by that idealisation.*

5.6 The cost of the symmetry

The eight-nozzle layout is not free. Two geometric losses multiply.

Tetrahedral projection. Each radial nozzle contributes $1/\sqrt{3} = 0.577$ of its thrust to every body axis simultaneously. Firing the correct two gives $2/\sqrt{3} = 1.155$ times a single nozzle on the commanded axis, which is 57.7 % of the thrust expended.

Outward cant. $\sin 38.50^\circ = 0.623$, which applies to rotation only.

$$0.577 \times 0.623 \;=\; 0.359 \tag{5.4}$$

For rotation, the eight-nozzle layout therefore delivers about 36 % of the torque that twelve axis-aligned nozzles would produce from the same propellant. This is the strongest single argument against the selected design. The answer is Section 5.5: the 36 % figure is torque per pulse, and the mission is decided on lifetime angular impulse, where eight wins across the entire purchasable range of valve masses.

5.7 Alternative examined: recessing the hull

The obvious challenge to the canted group is to recess the nozzle into the hull and keep it fully tangential, recovering the whole 115 mm as moment arm instead of 71.59 mm. This was examined and rejected on three grounds.

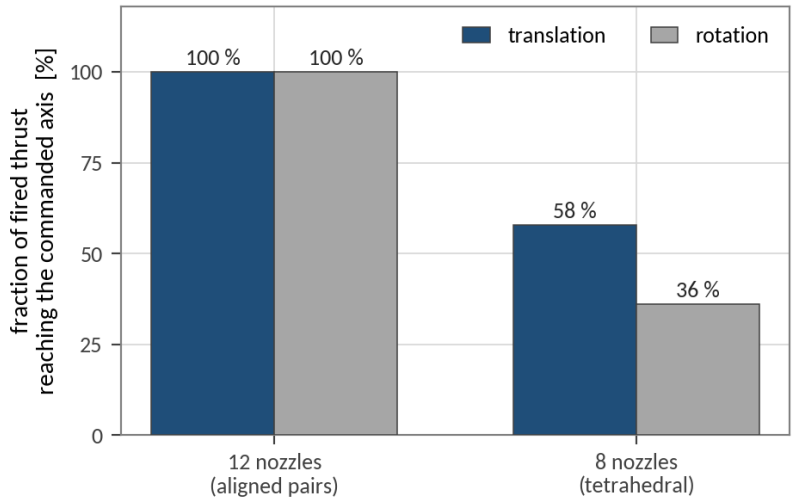


Figure 5.3. Axis efficiency of the eight-nozzle layout relative to twelve axis-aligned nozzles, for translation and for rotation.

1. **It does not buy what it appears to.** Going fully tangential recovers 1.606 times the torque. That saves 0.31 s on a 90° slew and costs 1.3 days of pointing life, because burn rate is directly proportional to torque. It is the same trade that rejected twelve nozzles, at a smaller scale.
2. **A vacuum plume does not stay in a channel.** The Prandtl–Meyer maximum turning angle for carbon dioxide at $\gamma = 1.280$ is 166.8°. The outer streamlines of an underexpanded jet can therefore turn back past perpendicular, toward the hull, and a groove shields only the first few millimetres of the expansion.
3. **The groove wall becomes the impingement surface.** A tangential jet needs a wall aligned with its own thrust direction in order to be shielded by it, so the force on that wall opposes the thrust being recovered. The change also replaces an exactly computable moment arm, $R\sin\theta$, with a quantity that requires CFD or a vacuum chamber to determine.

The cant angle of 38.50° was set by the CAD geometry, being the angle at which the plume clears the curved hull for the nozzle as built. It was not derived from an optimisation. A plume-impingement study is what would justify moving it, and that study has not been done.

CHAPTER 6

Flight dynamics model

6.1 State vector

The vehicle is represented by seventeen numbers in six blocks.

Table 6.1. The seventeen-element state vector.

Block	Size	Purpose
Position **r**	3	integrated from velocity, world frame
Velocity **v**	3	integrated from total thrust divided by current mass
Attitude quaternion q	**4**	**body-to-world rotation; the busiest quantity in the model**
Body rates ω	3	what the rate gyro measures
Wheel momentum **h**	3	held at zero; the slot allows reaction wheels to be added later
Propellant mass m	1	decreases during firing; changes both mass and inertia

Two of those choices need justifying. The wheel-momentum slot carries no hardware and is held at zero throughout. It costs three doubles and allows a momentum-exchange actuator to be added later without restructuring the state or the integrator. Propellant mass is a state rather than a constant because as the ten grams leave, both the total mass and the inertia tensor change. The same minimum pulse turns the vehicle slightly further at the end of a mission than at the start, and treating mass as constant would leave the controller progressively mistuned across a long mission.

> The thrusters act in body axes. The mission is described in world axes. The attitude quaternion converts between them, in both directions, several times per 5 ms step.

6.2 Attitude representation

Euler angles are the intuitive representation: three successive rotations, roll φ about x, pitch θ about y, yaw ψ about z. The difficulty is not describing an orientation but advancing one. Propagating Euler angles from the measured body rates p, q, r requires

$$\dot{\varphi} = p + (q\sin\varphi + r\cos\varphi)\tan\theta \tag{6.1}$$

$$\dot{\theta} = q\cos\varphi - r\sin\varphi \tag{6.2}$$

$$\dot{\psi} = \frac{q\sin\varphi + r\cos\varphi}{\cos\theta} \tag{6.3}$$

Two of those three contain $\tan\theta$ or $1/\cos\theta$. At $\theta = \pm 90^\circ$, nose straight up or straight down, $\cos\theta = 0$ and both terms are undefined. Two of the three rotation axes become parallel and a degree of freedom is lost. This is gimbal lock, and in the integrator it is a division by zero.

A unit quaternion carries the same information without the singularity. Four components $\mathbf{q} = (w, x, y, z)$ with $\|\mathbf{q}\| = 1$; for a rotation of angle α about a unit axis $\mathbf{n}$,

$$\mathbf{q} \;=\; \left(\cos\tfrac{\alpha}{2},\;\; \mathbf{n}\sin\tfrac{\alpha}{2}\right) \tag{6.4}$$

and the kinematic update is a single quaternion product,

$$\dot{\mathbf{q}} \;=\; \tfrac{1}{2}\,\mathbf{q} \otimes (0,\, \boldsymbol{\omega}) \tag{6.5}$$

with no trigonometric function, no division, and no denominator that can reach zero. It is smooth over the whole of orientation space, including straight up.

The cost is one redundant number: four components carrying three degrees of freedom, and therefore one constraint, $\|\mathbf{q}\| = 1$. Numerical integration error slowly pushes $\mathbf{q}$ off the unit sphere, so the flight code renormalises once per step,

$$\mathbf{q} \;\leftarrow\; \frac{\mathbf{q}}{\|\mathbf{q}\|} \qquad \text{every step, 200 times per second} \tag{6.6}$$

which is a single division. A rotation matrix is also singularity-free but costs nine numbers and six orthonormality constraints, making renormalisation considerably more work. The quaternion is the cheaper singularity-free option, which is why flight software generally uses it.

One property must be handled explicitly: $\mathbf{q}$ and $-\mathbf{q}$ represent the same physical rotation. The code therefore never compares two orientations by subtracting components. It forms the error quaternion and takes the shorter of the two equivalent rotations. Without this, the controller can command a 350° turn in place of a 10° one. Conversion to Euler angles happens only for display, never inside the loop.

The singular region is not a corner case for this vehicle. With the handheld controller, 90° of pitch is reached in about two seconds.

6.3 Rotational dynamics

The rotational equation of motion is Euler's equation in body axes,

$$\mathbf{I}\,\dot{\boldsymbol{\omega}} \;=\; \boldsymbol{\tau} \;-\; \boldsymbol{\omega} \times (\mathbf{I}\,\boldsymbol{\omega}) \tag{6.7}$$

where the second term is the gyroscopic coupling. Because the principal moments of inertia are not identical, the angular velocity vector wanders within the body frame even with zero external torque, which is why a thrown object wobbles rather than spinning cleanly. It requires no thruster and consumes no energy, and reproducing it correctly is one of the two verification tests in Section 10.2.

Body axes are used rather than an inertial frame for two reasons: it is the frame the rate gyro measures in, and it is the frame in which the inertia tensor is constant. Expressing the equation inertially would make $\mathbf{I}$ time-varying, which is worse.

6.4 Integration scheme

The simulation uses fixed-step fourth-order Runge–Kutta at exactly 5.00 ms, or 200 Hz. The step never adapts.

An adaptive solver shrinks its step when the dynamics stiffen, which is more accurate but has no bounded execution time. Inside a 5 ms real-time slot an unbounded solver means a missed deadline, and a late control loop is worse than a slightly inaccurate one. A fixed step also gives bit-identical replay of any recorded run, which is what makes post-flight analysis possible.

200 Hz is comfortably fast enough for this plant. The closed-loop natural frequency is 3.68 rad/s, about 0.59 Hz, so the loop runs roughly 340 times faster than the dynamics it controls. The binding constraint on performance is the 1 ms valve opening time, not the loop rate. Measured convergence order is reported in Section 10.1.

CHAPTER 7

Control

7.1 Architecture

The loop runs at 200 Hz in the following order: read the inertial measurement unit over I^2C, fuse rate and accelerometer data in a complementary filter, form the error quaternion against the commanded attitude, evaluate a quaternion PD law with gyro-rate feedforward, pass the resulting continuous wrench demand to the allocator, and convert the fractional thruster duties into whole minimum impulse bits through a sigma-delta modulator.

A complementary filter was chosen over a Kalman filter on compute budget and transparency. At 200 Hz on an ESP32 it is deterministic and easy to reason about. A Kalman filter would perform better given a characterised noise model, and is a reasonable next step once the sensor noise has been measured.

7.2 Control allocation

Each group is four thrusters solving three equations, so the allocation problem is under-determined. Infinitely many firing combinations produce exactly the same wrench, and they form a line rather than a point. The allocator's task is not to find a solution but to choose the cheapest of infinitely many, 200 times a second, for the life of the vehicle. Gas is the mission, so every gram spent on a clumsy firing choice is flight time removed.

Write the solution set as a particular solution plus the null space:

$$\mathbf{d} \;=\; \mathbf{d}_p + k\,\mathbf{n}, \qquad \mathbf{d}_p = \mathbf{A}^{\mathsf{T}}\left(\mathbf{A}\mathbf{A}^{\mathsf{T}}\right)^{-1}\mathbf{w}, \qquad \mathbf{n} = [1,1,1,1]^{\mathsf{T}} \tag{7.1}$$

Three steps close the problem.

1. **The null-space direction is physically meaningless.** $\mathbf{A}\mathbf{n} = \mathbf{0}$ exactly: firing all four thrusters of a group equally produces no net force and no net torque. Moving along $\mathbf{n}$ burns propellant and achieves nothing.
2. **Total propellant is affine in the single free parameter.** Mass flow is proportional to duty and all eight nozzles are identical, so the cost is $J = \sum_i d_{p,i} + 4k$, a linear function of k alone.
3. **A linear objective on an interval is minimised at an endpoint.** The only constraint is that every duty be non-negative, so $k_{\min} = \max_i(-d_{p,i})$ and the answer is closed-form.

> Redundant control allocation normally requires an iterative optimiser, and an optimiser has unbounded worst-case runtime, which disqualifies it from a hard real-time slot. Here the geometry reduces it to a formula that evaluates in under a microsecond.

The closed-form result was checked against a 200 001-point brute-force sweep over the null space for 400 randomly generated wrench targets. The excess propellant over the true optimum was exactly zero in every case, as reported in Section 10.3.

Saturation behaviour: if the optimal duty for any thruster exceeds 1, the command is unachievable. The whole command is then scaled down so that the direction of the commanded wrench is preserved and only its magnitude is reduced, since preserving direction matters more than magnitude for closed-loop stability.

7.3 Quantisation: the minimum impulse bit

Solenoid valves cannot be opened partially. They are shut or open, and require a finite time, 1.0 ms for the class of part assumed here, to open cleanly and repeatably. The smallest impulse the vehicle can deliver is therefore fixed:

$$\text{MIB} \;=\; F_0\, t_{\min} \;=\; 94.188\ \text{mN} \times 1.0\ \text{ms} \;=\; 94.19\ \mu\text{N·s} \tag{7.2}$$

Every command is assembled from whole copies of this quantity. One pulse on the rotation group produces:

Coast rate after one pulse, $\omega = \tau t_{\min}/(2I)$	**0.0846 °/s**
Time to cross the full $\pm 3°$ deadband, $T_c = 2\theta_{\text{db}}/\omega$	**35.5 s**
Propellant cost of one pulse, $\Delta m = 2\dot{m} t_{\min}$	**0.41 mg**

The sigma-delta modulator bridges the continuous PD demand and this quantised actuator. It accumulates the fractional pulse demands and spends them as whole minimum impulse bits, so the average delivered impulse matches the commanded one even though each individual pulse is quantised.

7.4 The attitude limit cycle

Continuous firing empties the 10 g tank in 48 s, so continuous control is not available to this vehicle at any tuning. Instead, inside a deadband of $\pm 1.5°$ in attitude and $\pm 0.20°$/s in rate, every thruster is commanded off and the vehicle drifts. Only at the boundary does the controller spend a single minimum impulse bit to reverse the drift.

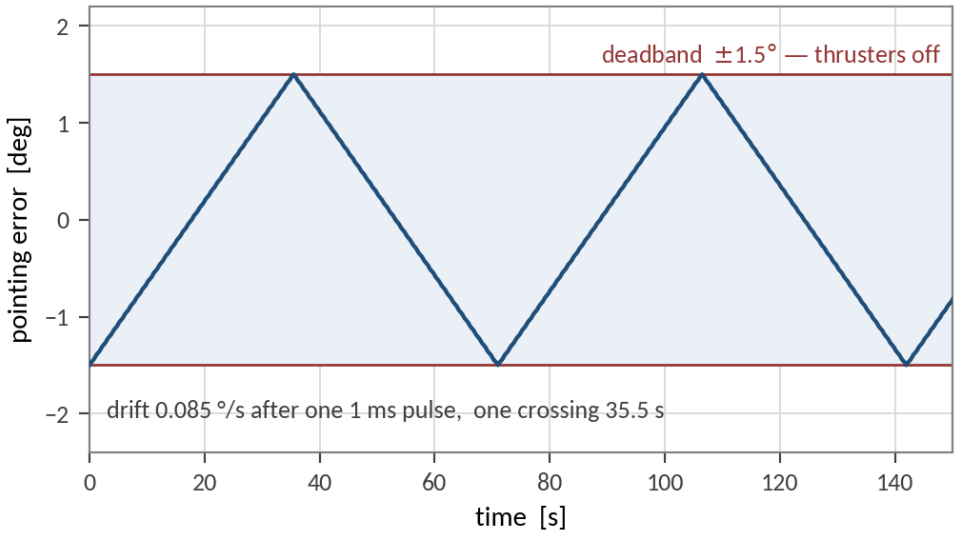


Figure 7.1. Attitude error in the limit cycle. The vehicle drifts across the band at 0.0846°/s, crossing in 35.5 s, and one minimum impulse bit reverses it at the boundary.

The resulting sawtooth is deliberate, and it is the normal operating mode of every propellant-based spacecraft attitude control system. It is stated here because an oscillating error signal is easily misread as a poorly tuned controller.

Deadband width is a mission parameter rather than a control parameter. Propellant consumption is inversely proportional to it, so tightening the band from $\pm 1.5°$ to $\pm 0.75°$ halves the mission life. It is set by what the camera's field of view can tolerate, not by how tightly the controller could hold if asked.

7.5 Gains and slew performance

Table 7.1. Controller gains and the resulting closed-loop performance, derived against the CAD inertia.

Proportional gain k_p	7.1432×10^{-2}
Derivative gain k_d	1.7469×10^{-2}
Closed-loop natural frequency ω_n	**3.680 rad/s**
Damping ratio ζ	**0.900**
Peak angular acceleration $\alpha_{\max}$	**169.2 °/s²**
Slew time, 30°	**0.84 s**
Slew time, 90°	**1.46 s**
Inertia, CAD, uniform density	2.637×10^{-3} **kg·m²**
Inertia, thin shell of the same mass	4.408×10^{-3} **kg·m²**

The gains are derived against the CAD inertia, and that inertia is the largest single modelling risk in the project. The CAD model treats the vehicle as a solid of uniform density. The real vehicle is much closer to a thin shell, not least because the 200 g of valves sit at the wall. The two values differ by a factor of exactly $5/3$, an error of +67.2 %.

The consequence is quantified rather than left open. Against the true inertia, the same gains give $\omega_n = 2.84$ rad/s and $\zeta = 0.70$. The real vehicle would slew about 30 % more slowly than the gains assume and would be underdamped, so it would overshoot. The actuator choice and the inertia error are the same problem seen twice, and measuring the inertia on a bifilar pendulum is roadmap item two.

CHAPTER 8

Achievable authority, axis by axis

8.1 Translation

The four radial nozzles sit at alternating corners of a cube, so each contributes $1/\sqrt{3} = 0.577$ of its thrust to every body axis simultaneously:

$$\mathbf{F} = \frac{F_0}{\sqrt{3}}\mathbf{A}\,\mathbf{d}, \qquad \mathbf{A} = \begin{bmatrix} -1 & -1 & +1 & +1 \\ -1 & +1 & -1 & +1 \\ -1 & +1 & +1 & -1 \end{bmatrix} \tag{8.1}$$

Firing the correct two nozzles at full duty cancels the two unwanted axes exactly and leaves $2/\sqrt{3} = 1.155$ times a single nozzle's thrust on the commanded axis.

Table 8.1. Translation firing solutions. Both nozzles run fully open.

Axis	Nozzles fired	Net force	Net torque [mN·m]
$+X$	R3 + R4	+ 108.76 mN	**0.000**
$-X$	R1 + R2	− 108.76 mN	**0.000**
$+Y$	R2 + R4	+ 108.76 mN	**0.000**
$-Y$	R1 + R3	− 108.76 mN	**0.000**
$+Z$	R2 + R3	+ 108.76 mN	**0.000**
$-Z$	R1 + R4	− 108.76 mN	**0.000**

Two points carry the argument. The unwanted axes cancel because of where the nozzles are, not because a controller noticed an error and corrected it, so no control effort is spent staying on axis. And the torque column is exactly zero rather than approximately zero, because every radial thrust line passes through the centre of mass. That exactness is what earns the split into two independent groups.

Translation authority is isotropic: 108.76 mN on all six body axes, identically. Firing all four radial nozzles equally produces zero net force, which is the null-space direction of Section 7.2 and is pure waste.

8.2 Rotation

Rotation is not as tidy. The canted four are the only source of torque, since the radial four produce none, but they are mounted on measured hardware rather than on an ideal tetrahedron. The four cant angles come out at 38.4988°, 38.4366°, 38.5013° and 38.4366°, and the T1 and T3 positions disagree by up to 0.26 mm where perfect symmetry would make them identical.

Only two of the six rotation axes come from a clean antagonistic pair. T1 and T3 together give $+Y$ at 7.472 mN·m, and T2 with T4 gives $-Y$ at 7.461 mN·m. The other pairings produce torque along diagonal directions rather than body axes. T3 with T4, for instance, gives 7.947 mN·m along $(0.47, 0, 0.88)$, which is mostly Z rather than X. Pure body-axis rotation about X or Z therefore requires three nozzles at partial duty and yields 5.87 mN·m.

Table 8.2. Rotation firing solutions, from the measured mount geometry. Four of the six axes require three nozzles at partial duty.

Axis	Nozzles fired, with duty	Torque [mN·m]	Side force [mN]
$+X$	T1 0.65 · T3 0.35 · T4 1.00	**5.87**	132
$-X$	T1 0.35 · T2 1.00 · T3 0.65	**5.84**	132
$+Y$	T1 1.00 · T3 1.00	**7.45**	156
$-Y$	T2 1.00 · T4 1.00	**7.46**	8
$+Z$	T2 0.35 · T3 1.00 · T4 0.65	**5.87**	78
$-Z$	T1 1.00 · T2 0.65 · T4 0.35	**5.87**	78

Two properties follow. Turn authority is anisotropic, 7.45 mN·m about Y against 5.87 mN·m about X and Z, a ratio of 0.79. And turning always pushes: a canted nozzle thrusts while it twists, so an attitude command leaks up to 156 mN of parasitic side force. The decoupling between the two groups is exact for torque but not for force, and the translation loop absorbs the difference.

> Four of the six rotation axes need three nozzles at partial throttle rather than a neat pair. That is why this vehicle carries an allocator and not a lookup table. Translation is a lookup table; rotation is not.

8.3 Why authority is anisotropic

The anisotropy is not a manufacturing defect, and two effects need separating.

Four push-only thrusters generate an achievable set that is a polytope rather than a sphere, so some directions are always better served than others. Sweeping the achievable authority over 1 200 directions distributed on a Fibonacci sphere gives a worst-to-best ratio of 0.719 for the radial group in force and 0.704 for the canted group in torque. The two groups are almost equally anisotropic as geometric objects.

What differs is their orientation relative to the body axes. The radial group's tetrahedron is oriented so that the six body axes fall on the peaks of its authority envelope, which is why translation comes out uniform at 108.759 mN. The canted group's tetrahedron is rotated relative to the body frame, so $\pm Y$ sits near a peak, better than 99.3 % of all directions, while $\pm X$ and $\pm Z$ sit mid-slope at about the 38th percentile. The sum of the four canted thrust directions is $(0, 1.574, 0)$, a pronounced $+Y$ bias, which is the same fact stated differently.

This suggests a design improvement at no cost in mass. Re-choosing the cant azimuths so that the torque envelope's peaks align with the body axes, as the radial group's already do, should raise the worst-axis torque by roughly 27 %. It has not been implemented or verified, and it is listed as an open item in Chapter 11.

The controller is sized on the worst axis, so 5.87 mN·m is the figure that governs. The three-nozzle solutions for X and Z are a consequence of the tetrahedral geometry rather than of manufacturing error, and they would not become a neat pair even with perfect mounts.

CHAPTER 9

Mission performance

9.1 Propellant budget and mission life

Ten grams of carbon dioxide is a small number. The useful way to present it is as the distance between two extremes.

Mode	Duration on 10 g	Ratio
One nozzle held continuously open	**48.2 s**	1 ×
Three-axis pointing, ±1.5° limit cycle	**3.30 days (79.2 h)**	**≈ 5 900 ×**

The limit-cycle figure follows by chaining the pulse consequences of Section 7.3:

$$\text{burn} = \frac{\dot{m}\,\tau\,t_{\min}^{2}}{2\,I\,\theta_{\text{db}}}, \qquad \text{life} = \frac{m_{\text{prop}}}{3\,\text{burn}} \tag{9.1}$$

The same ten grams expressed four ways:

Total impulse	**4.543 N·s**
Δv, if spent entirely on translation	**9.18 m/s**
Repositioning hops of 10 m	**46**
Three-axis pointing life	**3.30 days**

Disturbance torques are neglected in (9.1), and the neglect is justified rather than assumed. Gravity-gradient torque on a near-spherical body is of order 10^{-13} N·m, and aerodynamic drag at the intended altitude costs roughly 27 pulses per orbit. Both are about twenty times smaller than the control torque, so they perturb the deadband crossing time by a few percent rather than by a factor.

9.2 The scaling law and three levers

Equation (9.1) contains $t_{\min}$ squared, and that is the most useful result in the mission analysis. A shorter pulse leaves the vehicle drifting more slowly, so it both fires less often and spends less per firing. The two effects act in the same direction, so halving the valve opening time quarters the burn rate.

Lever	Change	Factor	Resulting life
Faster valve	1.0 → 0.5 ms	× **4**	**13.2 days**
Wider deadband	±1.5° → ±3.0°	× **2**	**6.6 days**
Both together	—	× **8**	**26.4 days**

Neither lever costs mass. The faster valve is a component substitution, and the wider deadband is free for as long as the target remains within the camera's field of view.

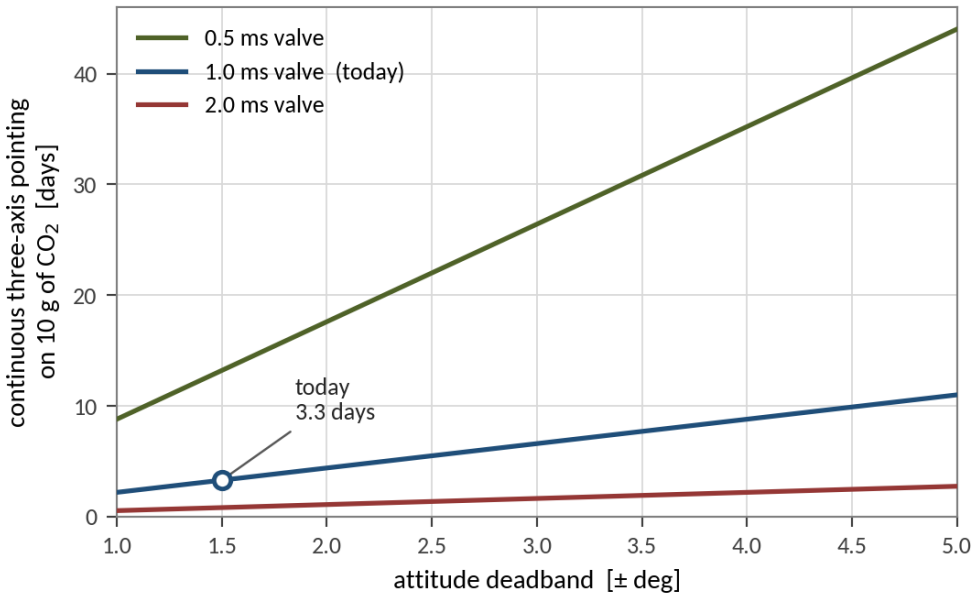


Figure 9.1. Three-axis pointing life against deadband half-width, for valve opening times of 0.5, 1.0 and 2.0 ms. The baseline design sits at ±1.5° and 1.0 ms.

> What limits this vehicle is its own minimum pulse, not the space environment. If longer flight time is wanted, the answer is a faster valve rather than orbital mechanics.

The counter-consideration is cycle life. Faster solenoids need higher drive voltages or smaller moving mass, both of which affect the number of actuations a valve survives, and a limit-cycling vehicle accumulates a very large cycle count. Characterising cycle life would have to accompany any valve substitution.

9.3 Trajectory strategy: curved against ballistic surveys

The first survey pattern flown in simulation circled the target. Re-planning the same survey as straight ballistic legs reduced the propellant cost of one complete pass from 73 % of the tank to 24 %.

The physics is elementary. Holding a curved path in vacuum requires continuous centripetal acceleration v^2/r, supplied entirely by the thrusters, since nothing else holds the vehicle in the turn. Integrating over the manoeuvre gives

$$\text{curve:}\quad \Delta v = v\,\theta \qquad\qquad \text{straight leg:}\quad \Delta v = 2v \tag{9.2}$$

where θ is the total angle turned through. Five wraps around the target is 31 rad. A coast costs nothing at all, for any duration, so a leg costs $2v$ regardless of its length: burn to speed, close every valve, glide the length of the target while scanning, burn to stop.

Table 9.1. Measured comparison of the two survey patterns, flown on the same vehicle model.

Pattern	Flight time	Mapped	Propellant	Valves shut	Passes per tank
circles	5 min	99.1 %	73 %	—	1.4
straight grid	26 min	98.7 %	**24 %**	**88 %**	**4.1**

Coverage did not move: 98.7 % against 99.1 %. The saving was not bought with a worse scan.

Attitude authority came free with it. Flying parallel to a flat face with the camera already square to that face requires no torque at all, whereas the circular pattern had to rotate continuously and spent 15 % to 25 % of its gas doing nothing but turning. The cost of the ballistic pattern is time, five times as long, and for an inspection mission time is the cheaper of the two resources.

9.4 The flown survey

The ballistic pattern was flown end to end in the simulator against a 6.6 m target satellite: 84 m of flight path, eight straight scanning legs, 25.6 minutes of flight, with every valve shut for 88 % of the run.

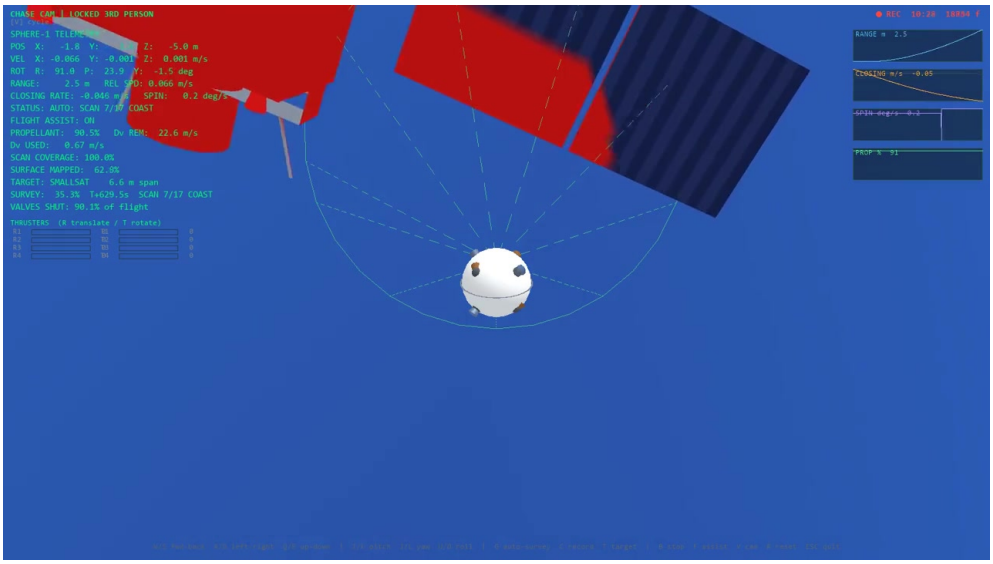


Figure 9.2. The ballistic survey in flight. Each leg is a short burn, a long unpowered glide during which the scan is taken, and a burn to stop.

Two results from that run are unfavourable and are reported here rather than in a footnote.

Standoff violation. Closest approach was 0.38 m against a planned 1.00 m standoff. The leg planner optimised coverage and propellant without a hard geometric constraint on minimum range. The fix is to add a keep-out sphere as a constraint rather than as an objective term. The violation was found by flying the pattern rather than by inspecting the plan.

Δv conversion efficiency. The rocket equation gives 24.9 m/s for this vehicle model, of which about 10 m/s proved usable, a 41 % conversion. The cause is the tetrahedral geometry of Section 5.6: the nozzles sit on cube corners, so any off-axis direction fires several of them partly against each other. The 41 % figure is that effect averaged over the directions a real survey demands rather than evaluated at a single best case.

Vehicle model caveat. The trajectory work ran on the flight simulator's vehicle model, 230 mm, 4.55 kg, eight 0.25 N nozzles and a 110 N·s tank, which is considerably heavier than the 500 g bench design of Chapter 3. The curved-against-ballistic comparison is a ratio between two flight paths on the same vehicle and therefore carries across the mass difference. The absolute Δv figures in this chapter belong to the simulator model and do not transfer to the 500 g vehicle. Reconciling the two models is outstanding work, listed in Chapter 11.

CHAPTER 10

Verification and validation

Four independent checks were run. Each was chosen so that a plausible implementation error produces a visibly wrong answer rather than a slightly degraded one.

10.1 Integrator order

A fourth-order method should reduce global error by a factor of $2^4 = 16$ each time the step is halved. Integrating a problem with a known closed-form solution and halving repeatedly gives:

Steps N	Global error	Observed order
10	2.491×10^{-6}	—
20	1.450×10^{-7}	4.103
40	8.748×10^{-9}	4.051
80	5.371×10^{-10}	4.026
160	3.328×10^{-11}	4.012
320	2.071×10^{-12}	**4.006**

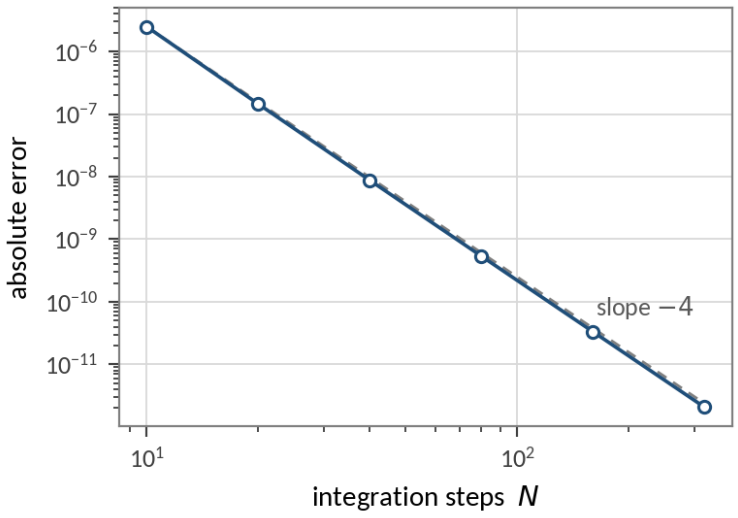


Figure 10.1. Global error against step count, logarithmic on both axes. The slope converges on 4.

10.2 Angular momentum conservation, with genuine dynamics

With every thruster off, the model was integrated for 12 000 steps. Angular momentum remained constant to one part in 10^{15}, which is machine precision, while the body rates wandered continuously.

Both halves of that sentence are necessary. A simulator that simply froze ω would also report perfectly conserved angular momentum and would be entirely wrong about the physics. The wander is the

gyroscopic term of (6.7) behaving correctly; the conserved momentum is the numerical check on it. Either result alone proves nothing.

10.3 Allocator optimality

The closed-form allocator of Section 7.2 was compared against brute force. For 400 randomly generated wrench targets the null-space parameter k was swept over 200 001 points and the total propellant cost evaluated at each. The excess of the closed-form solution over the true minimum was exactly zero in all 400 cases.

10.4 Hardware in the loop

The control loop was run on the intended flight processor against a simulated plant.

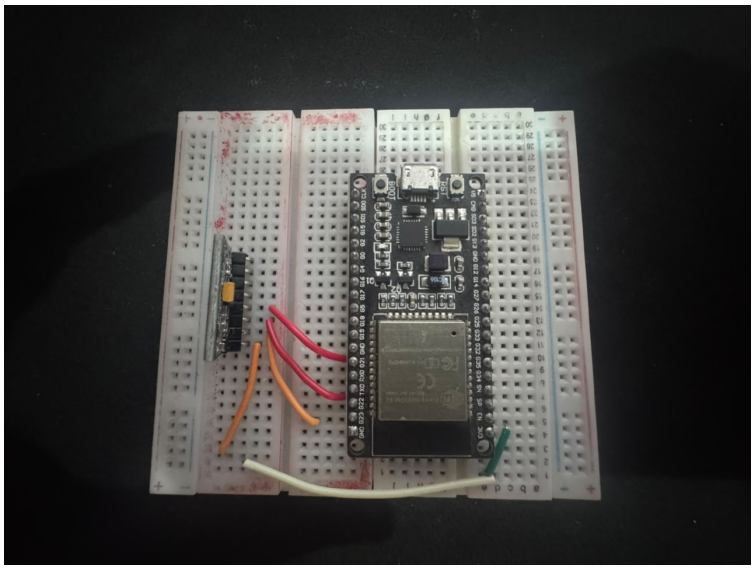


Figure 10.2. The hardware-in-the-loop rig: an ESP32 and an MPU6050 inertial measurement unit on I^{2}C at 400 kHz, driving the simulated vehicle.

Processor	**ESP32**
Sensor	**MPU6050, I^{2}C at 400 kHz**
Attitude estimation	**complementary filter at 200 Hz**
Control law	**quaternion PD with gyro-rate feedforward**
Actuator interface	**sigma-delta modulator, whole MIBs**
Allocation	**closed-form, eight thruster duties**
Mean tracking lag, without feedforward	**11°**
Mean tracking lag, with gyro-rate feedforward	**1.2°**
Hardware cost	**under $50**

Feeding the raw gyro rates forward lets the controller respond to motion as it occurs rather than waiting for a tracking error to develop, which is where the factor of nine comes from. The wider significance is cost: this rig validates a control loop that is normally exercised on an air-bearing table.

Scope of the claim. The real controller runs on the real flight processor at the real rate, driving a simulated plant, which is hardware-in-the-loop by the standard definition. What is not present is actuator hardware; the valves are modelled, not driven.

CHAPTER 11

Limitations and open items

The following are the known weaknesses of the work as it stands, with their consequences quantified where that was possible. A design is only as useful as the boundary around what it has established.

#	Item	Consequence
1	**Injector flow area is a datasheet value**	$(C_dA)_{\text{inj}} = 0.147\,\text{mm}^2$ has not been measured. Thrust and mass flow scale linearly with it, so a 10 % error is a 10 % error in every thrust, torque, minimum-impulse and mission-life figure. Roadmap item 1.
2	**Inertia from a uniform-density CAD model**	CAD gives $2.637 \times 10^{-3}\,\text{kg}\cdot\text{m}^2$; a thin shell of the same mass gives $4.408 \times 10^{-3}\,\text{kg}\cdot\text{m}^2$, an error of +67.2 %. Against the true inertia the current gains give $\omega_n = 2.84$ rad/s, $\zeta = 0.70$, underdamped. Roadmap item 2.
3	**Two vehicle models coexist**	Trajectory work ran on a 230 mm, 4.55 kg simulator model; propulsion and actuator work used the 500 g bench design. Ratios carry across; absolute Δv figures in Sections 9.3 and 9.4 do not.
4	**Thrust modelled as continuously throttleable**	The simulation uses a continuous duty from 0 to 1 carried through to mass flow; real solenoids are binary. The sigma-delta modulator bridges this, but at the current operating point it is inactive because the minimum opening time is shorter than one timestep.
5	**Leaked side force is not cancelled**	The allocator solves torque only and does not recruit the radial group to null the up to 156 mN of parasitic force from an attitude command, so a sustained slew produces a small unintended translation.
6	**No plume-impingement analysis**	The 38.50° cant was set by CAD clearance, not by optimisation. Prandtl–Meyer gives a maximum turning angle of 166.8°, which bounds the problem without solving it.
7	**Cant azimuths are not optimised**	Section 8.3 identifies a re-orientation of the canted tetrahedron that should raise worst-axis torque by roughly 27 % at no cost in mass. Not implemented or verified.
8	**Nozzle model is ideal isentropic**	No boundary-layer loss and no non-equilibrium nucleation. Both are first-order effects at this scale and act in opposite directions.
9	**Valves are modelled, not driven, in the HIL rig**	The loop, processor, sensor and rate are real; the actuator is not.
10	**Standoff constraint absent from the leg planner**	The flown survey reached 0.38 m against a 1.00 m plan. Fixed by adding a keep-out sphere as a hard constraint, not yet re-flown.
11	**Structure, shielding and certification out of scope**	500 g is a self-imposed target covering the functional vehicle only. A certified article would be heavier by an amount this work does not estimate.
12	**Measured mount asymmetry carried through as-is**	Cant angles range over 38.4366° to 38.5013° and the T1 and T3 positions disagree by up to 0.26 mm. Reported rather than corrected.

CHAPTER 12

Roadmap

Three steps separate the model in this report from a vehicle that flies. They are in this order because each feeds the next.

1. **Flow-bench the injector.** Measure $(C_dA)_{\text{inj}}$ directly instead of taking $0.147\,\text{mm}^2$ from a datasheet. That single number sets thrust, mass flow and the minimum impulse bit, so every other figure in this report moves with it.
2. **Measure the real inertia on a bifilar pendulum.** Suspend the assembled vehicle on two strings, time the torsional period about each axis, and recover the inertia from the period. It is cheap, and far more trustworthy than the uniform-density CAD assumption.
3. **Retune and fly.** Re-derive k_p and k_d against the measured inertia to restore $\zeta = 0.90$, flash the controller, and close the loop on the physical vehicle rather than on the simulated one.

$$I_{zz} = \frac{m\,g_0\,d^2\,T^2}{16\pi^2 L} \tag{12.1}$$

Beyond those three, the open items of Chapter 11 suggest a second phase: a coupled eight-column allocator to cancel the leaked side force, re-optimised cant azimuths for uniform turn authority, a plume-impingement study to justify or move the 38.50° cant, and reconciliation of the trajectory model with the 500 g bench design so that absolute Δv figures transfer.

CHAPTER 13

Methods, tools and deliverables

Area	Tools and methods
Geometry and mass	SOLIDWORKS: hull, nozzle mounts, mass properties. The thrust directions and mount positions used throughout Chapter 8 are exported from the CAD rather than regenerated from an idealised formula.
Thermodynamics	One-dimensional isentropic compressible flow; choked-flow series-restriction analysis; area–Mach relation; carbon dioxide triple-point and Prandtl–Meyer limits. Implemented in Python with NumPy.
Actuator analysis	Positive-spanning tests over Fibonacci-sphere direction sets; linear-programming authority sweeps using SciPy; null-space parametrisation and closed-form minimum-propellant allocation, verified against brute-force sweeps.
Simulation	Seventeen-state rigid-body model with fixed-step RK4 at 200 Hz; quaternion kinematics with per-step renormalisation; Euler rotational dynamics including the gyroscopic term; propellant mass as a state.
Embedded	C++ on ESP32 under the Arduino framework: I^2C at 400 kHz, complementary filter, quaternion PD with gyro-rate feedforward, sigma-delta modulator, closed-form allocator.
Figures	Matplotlib for the analysis figures; base MATLAB, no toolboxes, for the three-dimensional actuator geometry visualisations reproduced in Appendix B.

13.1 Deliverables

- This report, with the complete presentation and source listings as appendices.
- `Sphere-1_Thesis_Defense.pptx`, the 24-slide defence presentation including the embedded flight video.
- `Sphere-1_Presentation_Script_EN.docx`, the spoken script as delivered, reproduced in Appendix D.
- `Sphere-1_Slide_Explainer_EN.docx`, a slide-by-slide reference document giving the derivation behind each figure.
- `Sphere1_Figures.m`, `Sphere1_Injectors.m` and `Sphere1_Layouts_7_and_12.m`, the MATLAB sources reproduced in Appendix B.

CHAPTER 14

Nomenclature

$\mathbf{A}$	thruster influence matrix, force or torque per unit duty
(C_dA)	effective flow area of a restriction, discharge coefficient times area
$\mathbf{d}$	vector of thruster duties, $0 \le d_i \le 1$
$\mathbf{d}_p$	minimum-norm particular solution of the allocation problem
F_0	thrust of one nozzle at full opening, 94.188 mN
g_0	standard gravity, 9.80665 m/s^2
H	lifetime angular impulse, $\tau m_{\text{prop}}/(2\dot{m})$
$\mathbf{h}$	wheel momentum state, held at zero
I	moment of inertia about the axis under consideration
I_{sp}	specific impulse
J	total-duty cost function minimised by the allocator
k	free parameter along the allocation null space
k_p, k_d	proportional and derivative controller gains
M_e	nozzle exit Mach number
MIB	minimum impulse bit, $F_0 t_{\min}$
$\dot{m}$	propellant mass flow per nozzle, 0.2073 g/s
m_{prop}	propellant mass, 10 g
$\mathbf{n}$	allocation null-space direction, $[1,1,1,1]^{\mathsf{T}}$
P_0	chamber, or stagnation, pressure
p, q, r	body roll, pitch and yaw rates
$\mathbf{q}$	attitude quaternion (w,x,y,z), $\lVert\mathbf{q}\rVert = 1$
R	nozzle mounting radius, 115 mm
$\mathbf{r}$	position vector; $r_\perp$ the perpendicular moment arm
T_0	stagnation temperature, 293.15 K
T_e	nozzle exit static temperature
T_c	deadband crossing time
$t_{\min}$	minimum valve opening time, 1.0 ms
$\mathbf{w}$	commanded wrench, force and torque
α	angular acceleration; also rotation angle in the axis–angle form
γ	ratio of specific heats; 1.280 for CO_2, 1.400 for N_2
Δv	velocity increment
ζ	closed-loop damping ratio
θ	nozzle cant angle from radial, 38.50°; also pitch angle; also turned angle in (9.2)
θ_{db}	attitude deadband half-width, 1.5°
τ	torque
φ, ψ	roll and yaw Euler angles
ω	body angular rate vector; ω_n the closed-loop natural frequency

CHAPTER 15

References

The analysis follows the standard treatments listed below. Numerical values for carbon dioxide were taken from the NIST reference data.

1. Columbia Accident Investigation Board. *Report Volume I*. National Aeronautics and Space Administration and the Government Printing Office, Washington DC, August 2003.
2. Columbia Accident Investigation Board. *Report Volume II, Appendix D.13: STS-107 In-Flight Options Assessment*. October 2003.
3. G. P. Sutton and O. Biblarz. *Rocket Propulsion Elements*. John Wiley & Sons.
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7. M. D. Shuster. "A Survey of Attitude Representations." *Journal of the Astronautical Sciences*, Vol. 41, No. 4, 1993.
8. C. Davis. "Theory of Positive Linear Dependence." *American Journal of Mathematics*, Vol. 76, No. 4, 1954. The positive-spanning result underlying the $N+1$ thruster minimum.
9. P. J. Linstrom and W. G. Mallard, editors. *NIST Chemistry WebBook*, NIST Standard Reference Database Number 69. National Institute of Standards and Technology, Gaithersburg MD.

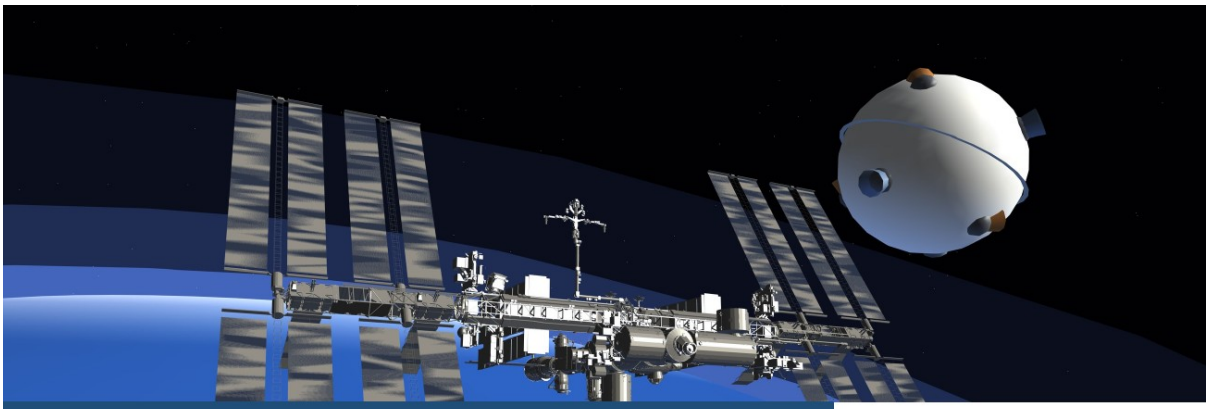
APPENDIX A

The defence presentation, slide by slide

The 24 slides below are the presentation as defended, reproduced at full size. Each is followed by a note on what the slide contains, the reasoning behind the figures on it, and, where the slide drew one, the question it raised in examination together with the answer given. A cross-reference to the chapter of this report that treats the same material in full is given for each.

Slide 20 carries an embedded video of the ballistic survey which cannot be reproduced on paper; the image shown is a frame from that run. Five further slides, covering the state vector, the integration scheme, throat sizing, the cost of the tetrahedral symmetry and the torque-free verification, were cut from the presentation for time. All five are treated in full in the body of this report, at Sections 6.1, 6.4, 4.4, 5.6 and 10.2 respectively, and are not repeated here.

A.1 Slide 1 Title



Sphere-1

TEMUR MAKHARADZE, MECHANICAL ENGINEERING
BACHELOR'S THESIS DEFENCE, TBILISI, 10 SEPTEMBER 2026

A 500-gram drone that flies free in orbit and inspects satellites from close range, pushed by cold gas instead of a chemical rocket. This talk covers the nozzle, the thrusters, the flight dynamics, and the test rig.

Supervisors: Dr. Giorgi Kakabadze · Dr. Otar Kemularia · Dr. Ioane Maisaia

The vehicle rendered against the International Space Station, the title block and the supervising staff. *Treated in full in Chapter 3.*

A.2 Slide 2 Why this needs to exist

Why this needs to exist

Columbia STS-107, or the foam strike frame,
or the 50 ft inspection boom NASA built afterwards
[picture goes here — Columbia]

January 2003. Eighty-one seconds after launch, a piece of foam came off the external tank and hit the leading edge of Columbia's left wing at around 500 mph. NASA saw it the next day in the launch film.

They knew. They could not look.

That flight carried no robotic arm. The crew could not see the leading edge from the windows. No spacewalk was planned. The only way to find out was to ask the Air Force to photograph the orbiter from the ground.

Three requests. All three cancelled.

The reasoning on record includes the line "it is not really a factor during the flight because there is not much we can do about it."

The decision chain failed at
"we cannot see it".

The assumption was wrong. The accident board found a rescue flight by Atlantis was challenging but feasible, had the damage been confirmed by flight day seven.

So NASA built the answer into every later flight: a 50-foot boom carrying cameras and laser scanners, and a full 360° backflip beneath the Station so its crew could photograph the heat shield. Both are workarounds for not having a camera that can get outside and look back.

The STS-107 foam strike, the three cancelled imagery requests, and the hardware NASA built afterwards. *Treated in full in Chapter 2.*

The sequence of events 16 January 2003, STS-107. At 81.7 s after launch a piece of insulating foam separated from the external tank's left bipod ramp and struck the leading edge of the left wing at approximately 500 mph, near reinforced carbon-carbon panel 8. The strike was identified the following day during routine review of launch film.

What was known, and what was not The strike was known about within roughly 24 hours; its severity was not. STS-107 was a Spacehab science flight carrying no robotic arm, the crew could not see the leading edge from the flight deck windows, and no extravehicular activity was planned. The only remaining assessment route was ground-based or national-asset imaging.

The three imagery requests Requests were raised by the chair of the Intercenter Photo Working Group, through the Department of Defense Manned Space Flight Support Office, and by the Debris Assessment Team's NASA co-chair. All three were cancelled; the DoD request was reversed within about 90 minutes. The Mission Management Team reasoning on record includes the position that the strike was not a factor during the flight because there was not much that could be done about it.

Why that reasoning is circular The argument runs: the damage cannot be inspected, therefore nothing could be done, therefore there is no value in inspecting it. The middle premise was never tested, because the information that would have tested it was the information declined.

Whether an alternative existed The Columbia Accident Investigation Board examined this directly in Volume II, Appendix D.13. A rescue by Atlantis was assessed as challenging but feasible if the damage

had been confirmed by about flight day seven; Columbia's consumables, limited by the lithium hydroxide CO_2 scrubbers, would have lasted to roughly flight day 30. An improvised on-orbit repair was rated high risk with no means of verification.

The remedies adopted afterwards The Orbiter Boom Sensor System — a 50-foot boom carrying a Laser Dynamic Range Imager with an intensified television camera and a Laser Camera System with a digital camera — extended the robotic arm's reach to 100 feet and was flown on every subsequent mission solely to inspect the orbiter's own thermal protection. It was complemented by the Rendezvous Pitch Manoeuvre, a 360° rotation beneath the ISS so the station crew could photograph the heat shield. Both are workarounds for one absent capability: a camera that can leave the vehicle and look back at it.

Questions raised by this material

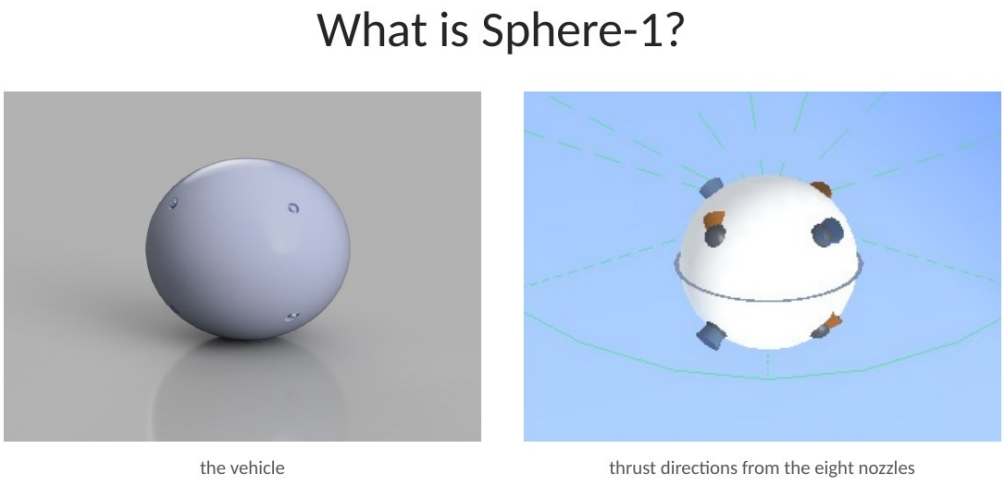
Q. Would Sphere-1 have changed the outcome of STS-107?

No, and the report does not claim otherwise. Columbia carried no arm and had no station to deploy from, and a 500 g free-flyer has no business manoeuvring near a vehicle in an unknown state. What the concept addresses is the capability that was missing — converting an unknown into a known early enough for a decision still to exist. It repairs nothing.

Q. Is this not what the Orbiter Boom Sensor System already does?

The OBSS solves the problem for a vehicle that has a 100-foot arm and can afford to carry a 50-foot boom. A free-flyer performs the same function without the host vehicle carrying either, can reach surfaces an arm cannot, and can inspect a vehicle other than the one carrying it.

A.3 Slide 3 What is Sphere-1?



Two renders — the vehicle, and its eight thrust directions — with the top-level specification. *Treated in full in Section 3.1.*

The specification A 23 cm sphere, 500 g wet, flying free within metres of the asset, carrying a visible and a thermal camera, propelled by eight cold-gas nozzles. The operative clause is the last one: nothing in the exhaust can settle on what is being photographed.

Why a sphere No preferred orientation, so an uncommanded tumble is a nuisance rather than a failure. No deployables to jam. The principal moments of inertia are nearly equal, so the gravity-gradient torque acting on the vehicle is negligible — which is what makes a ten-gram propellant budget a sensible design point at all.

A.4 Slide 4 Design constraints

Design constraints

Flying this close rules out a chemical rocket straight away. Its exhaust leaves soot and corrosive residue, and that would settle on exactly the cameras and solar panels the drone was sent to inspect. We would damage the thing we came to photograph.

> The mission requires a propellant that leaves nothing behind on the target.

Quantity	Value	Note
Sphere diameter	**23.0 cm**	6 371 cm³ envelope
Dry mass m_d	**490 g**	valves, electronics, tank
of which, eight valves	**200 g**	25 g each, flight budget
Propellant m_prop	**10 g CO_2**	3 days of pointing, or 9.2 m/s of Δv
Wet mass m_w	**500 g**	
Control	**6 DOF**	3 translation + 3 rotation

490 g empty is the number that kills several designs later in this talk.

Scope. This budget covers the functional vehicle: valves, electronics, tank and propellant. Structure, debris shielding and pressure-vessel certification for proximity operations near a crewed asset are outside this thesis and are not estimated. 500 g is a design target chosen to keep the functional core honest, not the mass of a flight article.

The contamination argument, the mass budget, and the scope boundary. *Treated in full in Sections 3.2 and 3.3.*

Contamination A hypergolic or any chemical plume deposits soot and corrosive species. At a standoff of a few metres those land on the optics and solar arrays the vehicle was sent to inspect. The propellant must therefore leave nothing behind, which eliminates chemical propulsion before any performance comparison is made.

The mass numbers 500 g wet, 490 g dry, 10 g of CO_2. Of the 490 g dry, 200 g is eight valves at 25 g each. That 200 g figure is what the entire actuator argument turns on.

Scope boundary The budget covers the functional vehicle: valves, electronics, tank and propellant. It excludes structure sized for launch loads, debris shielding, and pressure-vessel certification for proximity operations near a crewed asset. 500 g is a design target chosen to keep the functional core disciplined, not the mass of a flight article.

Questions raised by this material

Q. Where does the 500 g target come from?

It is a design target, not a derived requirement. The motivation was to keep mass as low as possible for orbital deployment and to force the functional core to be costed honestly. Propellant then falls out as the remainder after the valves.

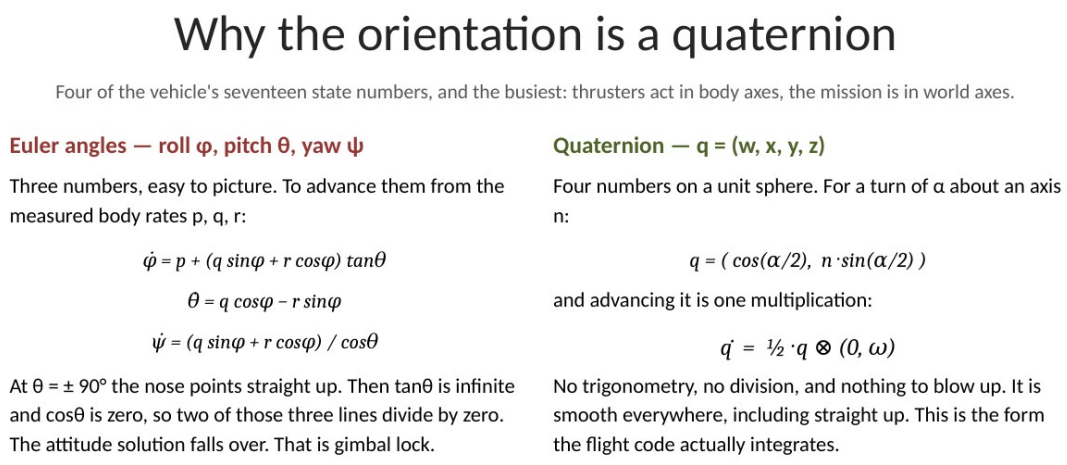
Q. Why only 10 g of propellant?

It is not a physical limit. Volume is nowhere near binding — 10 g of liquid CO_2 is about 13 cm^3, roughly 0.2 % of the sphere — and tank mass scales at about 0.28 g per gram of propellant. The vehicle is also deep in the linear region of the rocket equation, so 100 g would give about 80 m/s of Δv instead of 9.18, on a 615 g vehicle. The accurate framing is that this vehicle carries 200 g of valve and 10 g of fuel: it is valve-limited, not propellant-limited.

Q. Could this be certified to fly near the ISS?

Out of scope, and stated as such on the slide. The drivers would be containment of the pressure vessel, debris shielding, and a demonstrated safe-abort behaviour. None of them change the propulsion or control conclusions, but all of them change the mass.

A.5 Slide 5 Why the orientation is a quaternion



Euler angle kinematics and their singularity on the left, the quaternion form and its update on the right, and the renormalisation cost. *Treated in full in Section 6.2.*

Euler angles Three successive rotations: roll φ about x, pitch θ about y, yaw ψ about z. Intuitive to picture and easy to read from an instrument. The difficulty is not describing an orientation but advancing one.

Where the kinematics break $\dot{\varphi} = p + (q \sin\varphi + r \cos\varphi) \tan\theta$; $\dot{\theta} = q \cos\varphi - r \sin\varphi$; $\dot{\psi} = (q \sin\varphi + r \cos\varphi)/\cos\theta$. Two of those three contain $\tan\theta$ or $1/\cos\theta$. At $\theta = \pm 90°$, $\cos\theta = 0$ and both terms are undefined: two rotation axes become parallel and a degree of freedom is lost.

The quaternion form Four numbers on a unit sphere, $q = (w, x, y, z)$ with $\|q\| = 1$. For a rotation of angle α about a unit axis n, $q = (\cos(\alpha/2), n \sin(\alpha/2))$ — an axis and an angle, packed so that composition becomes multiplication.

The update $\dot{q} = \frac{1}{2} q \otimes (0, \omega)$, one quaternion product per step. No trigonometric function, no division, no denominator that can reach zero. Smooth over the whole of orientation space.

The cost Four numbers carry three degrees of freedom, so there is one constraint, $\|q\| = 1$. Integration error slowly pushes q off the unit sphere, and the code renormalises $q \leftarrow q/\|q\|$ once per step. That is a single division, and it is the complete price.

The operating condition that motivates it With the handheld controller, 90° of pitch is reached in about two seconds. The singular region is an operating condition for this vehicle, not a corner case.

Questions raised by this material

Q. How is the double-cover handled?

q and −q represent the same physical rotation, so the code never compares two orientations by subtracting components. It forms the error quaternion and takes the shorter of the two equivalent rotations. Without this the controller can command a 350° turn in place of a 10° one.

Q. Why not a rotation matrix?

It is also singularity-free, but it costs nine numbers instead of four and six orthonormality constraints instead of one, so renormalisation is more work. The quaternion is the cheaper singularity-free option, which is why flight software generally uses it.

Q. Are Euler angles used anywhere?

Only for display. Never inside the loop.

A.6 Slide 6 Choice of propellant

Choice of propellant

Nitrogen is the usual choice for cold gas. It gives more push per gram, and it cannot freeze anywhere inside our nozzle. On paper it wins.

	Nitrogen N_2	Carbon dioxide CO_2
Isp ceiling	79.6 s	**72.5 s**
State at 20 °C	gas	**liquid**
Storage pressure	200 + bar	**≈ 57 bar**
Tank wall	thick, heavy	**thin, light**
Stored mass per litre	baseline	**several ×**
Freezing risk	none	**below 216.6 K**

It loses on storage. Nitrogen stays a gas at room temperature, so a useful amount needs a 200-bar tank, and a tank that strong is heavy. Carbon dioxide turns liquid at about 57 bar, so the same bottle holds several times as much.

We give up push per gram to fit far more propellant in the same space. The price is a hard temperature limit.

The nitrogen-versus-CO_2 comparison, and the trade accepted. *Treated in full in Section 4.1.*

The case for nitrogen A higher specific-impulse ceiling, 79.6 s against 72.5 s, and no condensation risk anywhere in the expansion range. On performance alone it is the better propellant.

Why it was not selected Nitrogen stays gaseous at room temperature, so a useful propellant mass requires roughly 200-bar tankage, and that tank mass is charged to the same dry budget that already carries 200 g of valves. CO_2 liquefies at about 57 bar and stores several times the mass in the same volume behind a much thinner wall.

The trade Specific impulse exchanged for storage density, with a hard thermodynamic limit as the price.

Questions raised by this material

Q. Why not a warm-gas or resistojet system?

Electrical power. Heating the gas raises Isp, but the power budget on a 500 g vehicle will not support a meaningful heater, and it adds mass and thermal-control complexity.

Q. What about butane or R-134a?

Both store as low-pressure liquids and are flown on small satellites, and they are legitimate alternatives. CO_2 was chosen for availability, well-characterised properties, and complete inertness with respect to optical surfaces. Butane would be the first alternative to evaluate in a follow-on study.

A.7 Slide 7 The condensation cliff

The condensation cliff

A supersonic nozzle is a very effective refrigerator: gas cools as it expands, and it cools fast. For carbon dioxide there is a floor. Below 216.6 K it does not turn into a liquid — it freezes straight into dry ice, inside a throat less than a millimetre across.

$$Te = To \cdot [\, 1 + (\gamma - 1)/2 \cdot Me^2 \,]^{-1}$$

Setting Te equal to the triple point and solving backwards:

Te ≥ 216.6 K	the CO_2 triple point. Below it the propellant deposits as a solid
Me ≤ 1.589	the resulting ceiling on exit Mach number at T_0 = 293.15 K
Ae/At ≤ 1.266	the entire expansion this vehicle is thermodynamically allowed

γ = 1.280 for CO_2 at room temperature, against 1.400 for nitrogen and air.

The isentropic temperature relation and the three-step chain from the triple point down to the area-ratio ceiling. Treated in full in Section 4.2.

The physics A supersonic nozzle converts thermal energy into directed kinetic energy, so static temperature falls as the flow accelerates: Te = T_0 [1 + (γ−1)/2 · Me^2]$^{-1}$, with T_0 = 293.15 K and γ = 1.280 for CO_2.

Why CO_2 has a floor Below its triple point at 216.6 K, CO_2 does not condense to a liquid — it deposits directly as a solid. Inside a throat under a millimetre wide, dry ice blocks the nozzle and changes the thrust unpredictably.

The chain, solved backwards Setting Te = 216.6 K gives a maximum exit Mach number of 1.5888. Through the area–Mach relation, the maximum permissible expansion ratio is A_e/A_t = 1.2658 — the entire expansion this vehicle is thermodynamically allowed.

The role of γ γ = 1.280 for CO_2 against 1.400 for nitrogen and air. The lower ratio of specific heats means CO_2 cools less steeply for a given expansion, which helps slightly but not enough to escape the floor.

Questions raised by this material

Q. Is the flow really isentropic?

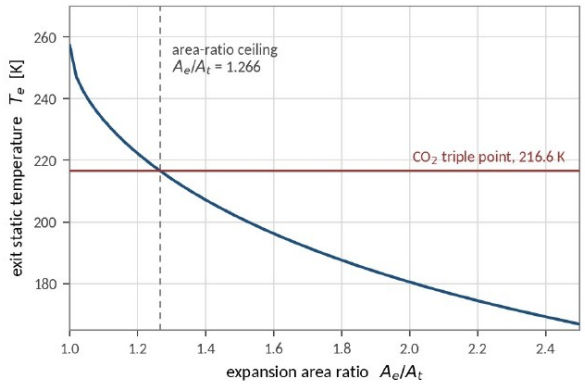
It is the standard first-order model and it is what the sizing is built on. Real losses — boundary layer, non-equilibrium condensation — make the actual exit temperature somewhat higher than ideal, so the ideal calculation errs in the conservative direction.

Q. Does condensation begin exactly at the triple point?

No. Real expansions supercool below the equilibrium point before nucleation begins, so there is margin. The design is sized to the equilibrium limit, which is the safe choice and is treated as a design rule rather than a hard physical wall.

A.8 Slide 8 The condensation cliff — consequence

The condensation cliff



At the safe area ratio:

converging-only	46.33 s
CD nozzle at 1.25	48.42 s
gain	+ 4.5 %
cone at a 2.0 mm throat	0.44 mm
cone at our 0.80 mm throat	0.18 mm

At those lengths it is a chamfer, not a nozzle: not machinable repeatably, and it risks the icing the cone was added to prevent.

The resulting operating point:

throat diameter	**0.80 mm**
chamber pressure	**1.5392 bar**
mass flow $\dot{m}$	**0.2073 g/s**
thrust per nozzle	**94.188 mN**

The cone is 0.18 mm long at our throat, so the right decision is to leave it off.

Chamber pressure is set by the ratio of two areas in series — the valve opening and the throat — so shrinking the throat is what raises it. 0.80 mm is what gives a usable 1.54 bar.

Every thrust and torque figure later in this talk traces back through that relation to the injector's flow area, which is still a datasheet value. Measuring it is roadmap item one.

Exit temperature against expansion ratio crossing the triple point, the Isp comparison, and the decision to leave the divergent cone off. *Treated in full in Sections 4.3 and 4.4.*

What the chart shows Exit static temperature against expansion area ratio, with the 216.6 K triple point drawn across it. The curve crosses at $A_e/A_t = 1.266$.

The performance numbers Converging-only gives Isp = 46.33 s. A converging–diverging nozzle at the maximum allowed ratio of 1.25 gives 48.42 s. The gain from the divergent cone is 4.5 %.

Why the cone was left off At a 2.0 mm throat the divergent section would be 0.44 mm long; at the 0.80 mm flight throat it is 0.18 mm. At that length it is a chamfer, not a nozzle: it cannot be machined repeatably to tolerance, and a badly formed cone risks precisely the icing it was added to prevent.

What fixes the throat at 0.80 mm Chamber pressure is set by the ratio of two areas in series — the injector opening and the throat — so shrinking the throat is what raises it. 0.80 mm is what gives a usable 1.54 bar and 94.188 mN per nozzle.

Questions raised by this material

Q. Is 4.5 % not worth having?

It would be if the part could be manufactured. The decision is not Isp against Isp; it is a 4.5 % gain against a feature that cannot be held to tolerance and that fails dangerously when malformed.

A.9 Slide 9 Actuator layout: three iterations

Actuator layout: three iterations

The drone has to control six things at once: push along three axes, and turn about three axes. Thrusters can only push, never pull, and that makes it harder — covering all six directions takes at least seven of them.

Configuration	Valve mass	Propellant left	6-axis decoupling	Verdict
6 minimalist	150 g	+60 g	no — under-actuated	cannot span 6 DOF
7 mathematical minimum	175 g	+35 g	no — heavy coupling	loop-fighting
8 two symmetric groups	**200 g**	**+10 g**	**yes — fully decoupled**	**selected**
12 conventional RCS	300 g	−90 g	yes — fully decoupled	does not close

Valve mass at 25 g each. Propellant is simply what is left of the 500 g after 290 g of structure, tank, battery and electronics.

All three were built and modelled, in the order shown. Each failed for a different reason: twelve was too heavy, seven could not be controlled cleanly, and eight survived both tests.

The 80 g automotive injector on the bench is a test article. The flight valve budget is 25 g.

The positive-spanning requirement, and the four-configuration comparison. *Treated in full in Section 5.1.*

The governing result The vehicle must control six quantities: three forces and three torques. Thrusters produce only positive thrust. Positively spanning an n-dimensional space with one-sided actuators requires at least n + 1 of them, so a six-dimensional wrench space requires seven.

Why it is not obvious With actuators that can both push and pull, six would suffice. The whole difficulty of cold-gas reaction-control layout comes from the one-sidedness.

The comparison Six is under-actuated and cannot span 6 DOF at all. Seven spans it but couples badly. Eight decouples cleanly and closes on mass. Twelve decouples cleanly and does not close on mass.

Questions raised by this material

Q. Where does the N + 1 result come from?

It is the standard theorem on positive spanning of $\mathbb{R}^n$. Intuitively, n vectors under non-negative combination can at best span a half-space cone; at least one more, in general position, is needed to reach every direction.

Q. The bench injector weighs 80 g, not 25 g. Which figure applies?

The 80 g automotive injector is a test article used to characterise flow. The flight budget assumes a 25 g solenoid, which is a real class of part. Both numbers appear on the slide deliberately; the 25 g is what the mass budget is built on.

A.10 Slide 10 Iteration 1 — twelve nozzles

Iteration 1 — twelve nozzles

The conventional layout: six opposing pairs, one for each direction of each axis. Every pair pushes straight along the axis it controls, so choosing which to fire is a lookup table with no real mathematics in it.

What it gets right:

100 %	of fired thrust reaches the commanded axis
21.66 mN·m	torque per pair, 2.78 × the eight-nozzle layout
trivial	allocator — no redundancy to resolve

Why it was abandoned:

Twelve valves at 25 g each is 300 g. Add the 290 g of structure, tank, battery and electronics and there is nothing left for propellant — the design is 90 g over budget before a single gram of gas goes in. It cannot be built, and it is not close.

More torque per pulse, and nothing to pulse with.



Six antagonistic pairs, the conventional RCS

The twelve-nozzle arrangement, its genuine advantages, and the mass failure. *Treated in full in Section 5.2.*

Why twelve is attractive Six antagonistic pairs, one per axis per direction. Every pair fires directly along the axis it commands, so allocation is a lookup table with no mathematics in it. 100 % of fired thrust reaches the commanded axis, and torque per pair is 21.66 mN·m, which is 2.78× the eight-nozzle layout.

Why it fails Twelve valves at 25 g is 300 g. Structure, tank, battery and electronics are 290 g. The total is 590 g against a 500 g budget, leaving −90 g for propellant. The design is 90 g over before any gas is loaded.

Questions raised by this material

Q. Could lighter valves rescue it?

At 17.5 g per valve twelve stops closing the budget entirely, and the lightest flight solenoids realistically available weigh 20 to 30 g. That bound is on the lifetime-angular-impulse slide.

A.11 Slide 11 Iteration 2 — seven nozzles

Iteration 2 — seven nozzles

Seven is the smallest number that can work at all. It also saves 25 g against eight, which is 35 g of propellant instead of 10 g — three and a half times the mission life. On mass alone, seven looks like the better design.

Why it still fails:

1. **No antagonistic pairs exist.**
   With seven thrusters in general position there is no pair whose thrusts cancel except along one axis. Every command recruits all seven.
2. **Every translation carries a parasitic torque.**
   The allocator can null it, but only by firing additional thrusters against the ones producing the translation you asked for.
3. **The attitude loop then fights the translation loop.**
   Each correction perturbs the other axis, and on a 10 g tank that exchange is paid for in propellant until the tank is empty.

Seven spans the space on paper and cannot be decoupled in hardware.
It buys propellant and spends it fighting itself.

The mass advantage of seven, and the three control reasons it was abandoned. *Treated in full in Section 5.3.*

Why seven looks better It is the mathematical minimum and it is lighter. 175 g of valves leaves 35 g of propellant instead of 10 g — three and a half times the mission life. On mass alone, seven is the superior design.

Why it fails, and it is not mass With seven there are no antipodal pairs. Every translation command therefore carries a parasitic torque, so the attitude loop permanently fights the translation loop. The propellant seven buys is spent cancelling its own cross-coupling.

The general lesson Spanning the wrench space on paper and being controllable in hardware are different requirements. Seven satisfies the first and fails the second.

Questions raised by this material

Q. Eight costs 25 g of propellant against seven. Was that worth it?

Yes, and the price is real. It was paid to keep translation and rotation decoupled, which is what makes the allocator analytic and the control loop predictable.

Q. Is there a symmetric seven-nozzle layout that works?

No. A C_5-symmetric pentagonal bipyramid fails to positively span $\mathbb{R}^6$ at every cant angle between 30° and 80° and every 15° azimuth step tested. The symmetry has to be deliberately staggered before seven

can control six axes — in the comparison layout generated for this work the ring azimuth advances 228° per nozzle.

A.12 Slide 12 Iteration 3 — eight nozzles

Iteration 3 — eight nozzles

One redundant thruster buys back the decoupling:

R1–R4 radial translation group

Mounted on four alternating corners of a cube, each pointing straight out through the centre of the sphere. Because the line of thrust passes through the centre, these push the drone without turning it at all.

$$\tau = r \times F = 0 \quad for \quad r\perp = 0.000\ mm$$

T1–T4 canted rotation group

Mounted on the 115 mm surface and swung 38.50° away from radial. That swing is what creates the lever: a nozzle pointing straight out has none. R·sin θ = 71.59 mm, against 115 mm for a nozzle firing flat along the hull.

> The decoupling of twelve, at a valve count the mass budget can carry.

The 37.7 % "lost" is measured against a nozzle firing flat along the hull — the worst possible case for the plume, not a neutral baseline. And giving it back would cost more than it buys: see the next-but-one slide.



Two symmetric groups of four

thrust per nozzle F_0	**94.188 mN**
moment arm R sin θ	**71.59 mm**
τ_0 per nozzle	**6.743 mN·m**
turn authority, Y axis	**7.45 mN·m**
turn authority, X and Z	**5.87 mN·m**

7.79 mN·m is the ideal geometric value. On the measured mounts the achievable figures are 7.45 about Y and 5.87 about X and Z.

The flight layout, the two groups described separately, and the operating point. *Treated in full in Section 5.4.*

The radial group, R1–R4 Four nozzles on alternating cube corners at $115/\sqrt{3} = 66.395$ mm on each axis, each firing radially outward through the centre of mass. Because the thrust line passes through the centre, the perpendicular moment arm is exactly zero and these produce pure force with no parasitic torque — verified to 0.000 N·m, not merely to a small residual.

The canted group, T1–T4 Mounted on the 115 mm surface and swung 38.50° away from radial. That swing is what creates the moment arm: R sin θ = 71.59 mm. A nozzle pointing straight out has no lever at all, which is precisely what the radial group demonstrates.

A point frequently misread The cant is not a concession made around the plume. It is the moment arm. The 37.7 % apparently lost is measured against a fully tangential nozzle at 115 mm — one firing flat along the hull, which is the worst possible case for impingement, not a neutral baseline.

The operating point Thrust per nozzle 94.188 mN, moment arm 71.59 mm, torque per nozzle 6.743 mN·m, turn authority 7.45 mN·m about Y and 5.87 mN·m about X and Z.

Questions raised by this material

Q. Why not recess the hull and keep the nozzle tangential, recovering the full 115 mm?

This is the strongest challenge to the layout and it has three parts to its answer. First, it does not buy what it appears to: going fully tangential recovers 1.606× the torque, which saves 0.31 s on a 90° slew

and costs 1.3 days of pointing life, because burn rate is directly proportional to torque — the same trade that rejected twelve nozzles. Second, a vacuum plume does not stay in a channel: the Prandtl–Meyer maximum turning angle for CO_2 at $\gamma = 1.280$ is 166.8°, so the outer streamlines can turn back past perpendicular toward the hull, and a groove shields only the first few millimetres. Third, the groove wall becomes the impingement surface at closer range, and because a tangential jet needs a wall aligned with its own thrust direction to be shielded, the force on that wall opposes the thrust being recovered. It also replaces an exactly computable moment arm, R sin θ, with a quantity that requires CFD or a vacuum chamber to determine.

Q. Why 38.50° specifically, and not 30° or 45°?

It is the angle at which the plume clears the curved hull for the nozzle geometry as built. It was set by the CAD, not derived from an optimisation, and a plume-impingement study is what would justify moving it.

Q. Why is the pair torque 7.45 mN·m here but 7.79 elsewhere?

7.79 mN·m is the ideal geometric value, $F_0 \times \text{arm} \times 2/\sqrt{3}$. The flight code applies discharge and alignment losses, and the measured mount geometry gives 7.45 about Y and 5.87 about X and Z. Those are the achievable values and they are the ones quoted throughout this report.

A.13 Slide 13 Moving: one axis at a time

Moving: one axis at a time

The four radial nozzles sit on alternating corners of a cube, each pointing straight out through the centre. Every one of them therefore pushes on all three body axes at once, with $1/\sqrt{3} = 0.577$ of its thrust on each.

$$F = (F_0/\sqrt{3}) \cdot A \cdot d, \quad A = [\,-1\,-1\,+1\,+1;\ -1\,+1\,-1\,+1;\ -1\,+1\,+1\,-1\,]$$

Fire the right two and the other two axes cancel each other exactly:

Axis	Nozzles fired	Net force	Net torque [mN·m]
+X	R3 + R4	+ 108.76 mN	0.000
−X	R1 + R2	− 108.76 mN	0.000
+Y	R2 + R4	+ 108.76 mN	0.000
−Y	R1 + R3	− 108.76 mN	0.000
+Z	R2 + R3	+ 108.76 mN	0.000
−Z	R1 + R4	− 108.76 mN	0.000

Both nozzles run fully open. The commanded axis gets $2/\sqrt{3} = 1.155$ times a single nozzle, and the two unwanted axes cancel because of where the nozzles are, not because a controller noticed and corrected. The torque is exactly zero, because each thrust line passes through the centre of mass.

Six clean moves, two nozzles each, no torque at all.
Moving needs no allocator — this half really is a lookup table.

The projection matrix and the six translation firing solutions, with the zero-torque column. *Treated in full in Section 8.1.*

The geometry The four radial nozzles sit on alternating corners of a cube, so each contributes $1/\sqrt{3} = 0.577$ of its thrust to every body axis simultaneously. $F = (F_0/\sqrt{3}) \cdot A \cdot d$, where A carries entries of ± 1 in the pattern of the cube corners.

How a clean axis is produced Fire two of the four. For +X that is R3 and R4: their Y components are equal and opposite and cancel, their Z components cancel, and what remains is pure X at $2/\sqrt{3} = 1.155$ times a single nozzle, or 108.759 mN.

The mechanism that matters The unwanted axes cancel by symmetry, not by feedback. No control effort is spent staying on axis, which is a stronger claim than saying the controller corrects the error.

The zero-torque column Every row reads 0.000, because every radial thrust line passes through the centre of mass. It is exactly zero, not approximately zero, and that exactness is what earns the split into two independent groups and makes the decoupled allocator legitimate.

Questions raised by this material

Q. Is translation authority the same in every direction?

Yes. All six axes give 108.759 mN, exactly. The radial group is perfectly isotropic.

Q. Why only two nozzles and not all four?

Firing all four equally produces zero net force — it is the null-space direction and is pure waste. That fact becomes step one of the allocator derivation.

A.14 Slide 14 Turning: one axis at a time

Turning: one axis at a time

The canted four are the only source of torque here, because the radial four make exactly none. But they sit on measured hardware, not a perfect tetrahedron, so turning is less tidy than moving.

Axis	Nozzles fired, with duty	Torque [mN·m]	Side push [mN]
+X	T1 0.65 · T3 0.35 · T4 1.00	**5.87**	132
−X	T1 0.35 · T2 1.00 · T3 0.65	**5.84**	132
+Y	T1 1.00 · T3 1.00	7.45	156
−Y	T2 1.00 · T4 1.00	7.46	8
+Z	T2 0.35 · T3 1.00 · T4 0.65	**5.87**	78
−Z	T1 1.00 · T2 0.65 · T4 0.35	**5.87**	78

Two things fall out of that table:

1. **Turning is not equally easy on every axis.**
   7.45 mN·m about Y, but only 5.87 about X and Z — a ratio of 0.79. Y is the axis the mounts happen to favour.

2. **Turning always pushes as well.**
   A canted nozzle thrusts while it twists, so an attitude command leaks up to 156 mN of side force. The translation loop absorbs it.

Four of the six axes need three nozzles at partial throttle, not a neat pair.
That is exactly why this vehicle has an allocator and not a lookup table.

The six rotation solutions with duty fractions, torques and parasitic side force. *Treated in full in Section 8.2.*

Why turning is different The canted four are the only source of torque, because the radial four produce none. But they are mounted on measured hardware rather than an ideal tetrahedron: the four cant angles come out at 38.4988°, 38.4366°, 38.5013° and 38.4366°, and the T1/T3 positions disagree by up to 0.26 mm where symmetry would make them identical.

The consequence Only two of the six rotation axes come from a clean pair. T1+T3 gives +Y at 7.472 mN·m and T2+T4 gives −Y at 7.461. The other four pairings produce torque along diagonal directions rather than body axes — T3+T4, for instance, gives 7.947 mN·m along (0.47, 0, 0.88), which is mostly Z rather than X.

The real solutions Pure body-axis rotation about X or Z requires three nozzles at partial duty and yields 5.87 mN·m. About Y it takes two and yields 7.45.

Observation one: authority is anisotropic 7.45 mN·m about Y against 5.87 about X and Z, a ratio of 0.79. The canted group's thrust directions sum to (0, 1.574, 0), a pronounced +Y bias, which is why Y is favoured.

Observation two: turning also pushes A canted nozzle thrusts while it twists, so an attitude command leaks up to 156 mN of parasitic side force. The decoupling is exact for torque but not for force, and the translation loop absorbs the difference.

Why this table justifies the allocator Translation genuinely is a lookup table. Rotation genuinely is not.

Questions raised by this material

Q. Does the parasitic force destabilise anything?

No, but it is not free. The allocator solves torque only and does not recruit the radial group to cancel the leaked force, so a sustained attitude manoeuvre produces a small unintended translation. On a station-keeping mission that appears as extra translation duty. A fully coupled eight-column allocation would cancel it, at the cost of the closed-form solution.

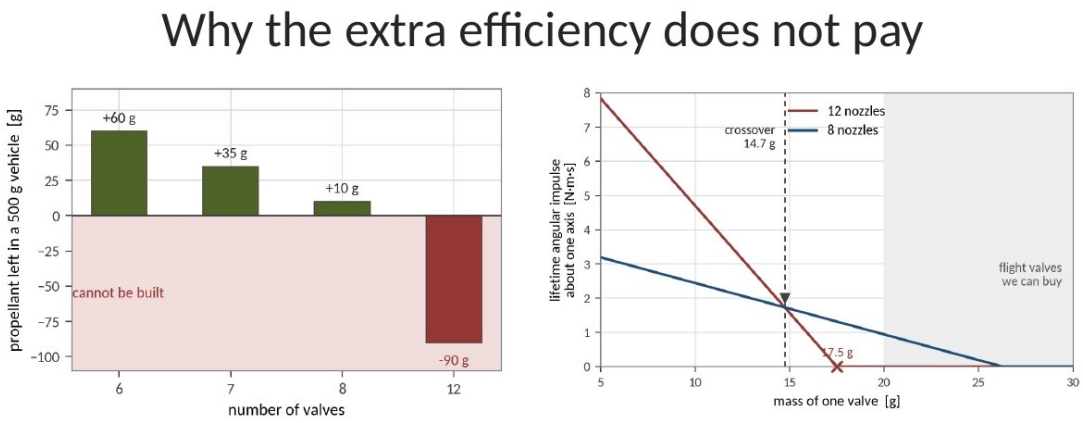
Q. Why not build the mounts more accurately?

The anisotropy would shrink, and the mounts should be improved. But the three-nozzle solutions for X and Z are a consequence of the tetrahedral geometry, not of the manufacturing error, so they would not become a neat pair even with perfect mounts.

Q. Could the anisotropy be removed by design?

Probably, at zero mass cost. A sweep over 1,200 directions shows both groups are almost equally anisotropic as geometric objects — worst-to-best 0.719 for force, 0.704 for torque. What differs is orientation: the radial tetrahedron has its envelope peaks on the body axes, the canted one does not. Re-choosing the cant azimuths to align the peaks should raise worst-axis torque by roughly 27 %. This has not been implemented or verified.

A.15 Slide 15 Why the extra efficiency does not pay



Tilting costs real torque: the tetrahedral projection is 57.7 % and the cant another 62.3 %, so eight nozzles deliver only 36 % of the twist twelve would give from the same gas. On its own that argues for going back to twelve. But twist per pulse is half the story — what a mission spends is twist multiplied by how long the tank can supply it.

below 14.7 g per valve	twelve nozzles deliver more lifetime angular impulse
above 14.7 g per valve	eight nozzles win: the propellant is worth more than the torque
above 17.5 g per valve	twelve nozzles no longer close the mass budget at all

A flight solenoid we can actually buy weighs 20 to 30 g. At every one of those masses, eight wins.

The propellant budget, the lifetime angular impulse curves, and the valve-mass thresholds. *Treated in full in Section 5.5.*

The correct metric Torque per pulse is only half of it. What a mission spends is total angular impulse: torque multiplied by how long the tank can keep supplying it. $H = \tau \cdot m_{prop} / (2\,\dot{m})$.

The trade The four extra valves buy 2.78× the torque and cost 100 g, which comes directly out of propellant.

The thresholds Below 14.7 g per valve, twelve delivers more lifetime angular impulse. Above 14.7 g, eight wins. Above 17.5 g, twelve does not close the mass budget at all.

Why that settles it Every flight solenoid realistically purchasable weighs 20 to 30 g. The entire purchasable range sits above both thresholds, so eight wins everywhere the vehicle can actually be built.

Questions raised by this material

Q. Is H an upper bound rather than a real budget?

Yes — it assumes all propellant goes to one axis. It is used to compare two configurations under the same convention, so the crossover point is unaffected by the idealisation.

A.16 Slide 16 Solving the redundancy

Solving the redundancy

Four nozzles, three equations. That means infinitely many firing combinations produce exactly the same torque — the answers form a line, not a point. So the allocator is not there to find an answer. It is there to pick the cheapest one out of infinitely many, and to do it 200 times a second without ever stopping to iterate.

Gas is the mission. Every gram wasted on a clumsy firing choice is flight time gone.

$$d = d_p + k \cdot n, \quad d_p = A^T(AA^T)^{-1}w, \quad n = [1, 1, 1, 1]^T$$

1. **Firing all four equally does nothing.**
   A · n = 0 exactly, so moving along n burns gas and produces no force and no torque at all.

2. **The cost is a straight line in k.**
   Total propellant is Σ d_p + 4k — an ordinary straight line in the single free parameter.

3. **So the cheapest answer is at an end.**
   A straight line over an interval is smallest at one of its two endpoints. No search needed.

$$k_min = max_i (- d_p,i)$$

A formula, not a search. Under a microsecond, every cycle, at 200 Hz.

The under-determined allocation problem, the null-space parametrisation, and the three-step closed-form solution. Treated in full in Section 7.2.

Why there is a problem Each group is four thrusters solving three equations. The system is under-determined: infinitely many firing combinations produce exactly the same wrench, and they form a line rather than a point. The allocator's task is not to find an answer but to choose the cheapest of infinitely many, 200 times a second.

Why cheapest matters Gas is the mission. Every gram spent on a clumsy firing choice is flight time removed, and this is the one decision the vehicle makes continuously for its entire life.

The parametrisation $\mathrm{d} = d_p + \mathrm{k}\cdot\mathrm{n}$, where $d_p = \mathbf{A}^\mathsf{T}(\mathbf{A}\mathbf{A}^\mathsf{T})^{-1}\mathbf{w}$ is the minimum-norm particular solution and $\mathrm{n} = [1,1,1,1]^\mathsf{T}$ spans the null space.

Step one $\mathrm{A}\cdot\mathrm{n} = 0$ exactly. Firing all four equally produces no net force and no net torque, so moving along n burns propellant and achieves nothing.

Step two Total propellant is $\mathrm{J} = \sum_i d_{p,i} + 4\mathrm{k}$, an affine function of the single free parameter k.

Step three A linear objective on an interval attains its minimum at an endpoint, so $k_{\min} = \max_i(-d_{p,i})$ and the answer is closed-form.

Why that matters Redundant allocation normally requires an iterative optimiser, and an optimiser has unbounded worst-case runtime, which disqualifies it from a hard real-time slot. Here the geometry reduces it to a formula that evaluates in under a microsecond.

Verification Checked against a 200,001-point brute-force sweep over the null space for 400 random targets. Excess over the true optimum: exactly 0.000×10^{0}.

Questions raised by this material

Q. What happens if the optimal duty exceeds 1?

The command is unachievable and is saturated. The whole command is scaled down so the direction is preserved and only the magnitude is reduced, since preserving direction matters more than magnitude for stability.

Q. Is minimising total duty the same as minimising propellant?

Yes here, because mass flow is proportional to duty and all eight nozzles are identical with a single maximum-thrust value. If the nozzles differed, the objective would need weighting by each one's mass flow.

Q. Is the thrust really continuously throttleable?

In the simulation, yes — duty is continuous from 0 to 1 and the fractional value carries through to mass flow. The hardware is not: solenoid injectors are binary. The sigma-delta modulator exists to bridge the two, and at the current operating point it is inactive because the minimum opening time is shorter than one timestep. Shrinking the throat or lengthening the valve time restores the real pulsing behaviour. This is a known model–hardware mismatch and it is listed in Section 11.

A.17 Slide 17 The smallest pulse we can make

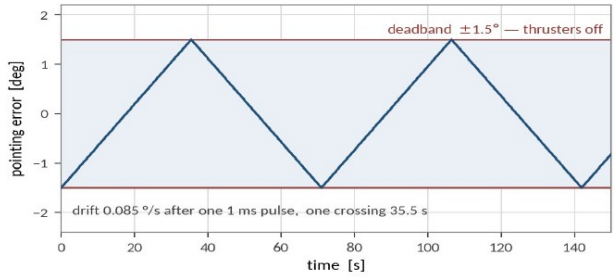
The smallest pulse we can make

The valves cannot be opened halfway. They are shut or open, and need 1.0 ms to open cleanly. So there is a smallest push the vehicle can ever deliver, and every command is built out of whole copies of it.

$$MIB = F_0 \cdot t_min = 94.188\ mN \times 1.0\ ms = 94.19\ \mu Ns$$

One pulse from a canted pair, and what follows from it:

$\omega = \tau \cdot t_min / (2 \cdot I)$	**0.085 °/s**	the drift rate one pulse leaves behind
$T_c = 2\,\theta_db / \omega$	**35.5 s**	how long it then takes to cross the deadband
$\Delta m = 2\,\dot{m} \cdot t_min$	**0.41 mg**	the propellant that one pulse costs



Firing without pause empties the tank in 48 seconds.

So while the drone points within ± 1.5° of target, every thruster is off and it simply drifts. Only at the edge does it spend one pulse to turn itself around.

That sawtooth is not a badly tuned controller. It is the propellant budget being protected.

The minimum impulse bit, the three consequences that follow from it, and the deadband limit cycle. *Treated in full in Sections 7.3 and 7.4.*

Why there is a minimum at all Solenoid valves cannot be opened partially. They are shut or open, and require 1.0 ms to open cleanly and repeatably, so the smallest impulse the vehicle can deliver is fixed.

The minimum impulse bit MIB = $F_0 \cdot t_{\min}$ = 94.188 mN × 1.0 ms = 94.19 μN·s. Every command is assembled from whole copies of this.

What one pulse produces $\omega = \tau\, t_{\min}/(2I)$ = 0.0846 °/s, the rate the vehicle coasts at afterwards. $T_c = 2\theta_{\mathrm{db}}/\omega$ = 35.5 s, the time to cross the full ±3° band. $\Delta m = 2\dot{m}\, t_{\min}$ = 0.41 mg, the propellant one pulse costs.

Why the deadband exists Continuous firing empties the 10 g tank in 48 seconds. Inside a ±1.5° attitude and ±0.20°/s rate deadband every thruster is commanded off and the vehicle drifts. Only at the boundary does the controller spend a single MIB to reverse it.

Why the sawtooth is deliberate An oscillating error signal is easily misread as a poorly tuned controller. Every operational spacecraft attitude control system limit-cycles for exactly this reason.

Questions raised by this material

Q. Why $\pm 1.5^\circ$ and not tighter?

Propellant consumption is inversely proportional to the deadband. Tightening it to $\pm 0.75^\circ$ halves the mission life. The band is set by what the camera's field of view can tolerate, not by how tightly the controller could hold.

Q. What is the sigma-delta modulator for?

The PD controller asks for fractional pulses. The modulator accumulates those fractions and spends them as whole minimum impulse bits, so the average delivered impulse matches the commanded one even though each individual pulse is quantised.

A.18 Slide 18 How long does Sphere-1 actually fly?

How long does Sphere-1 actually fly?

Everything in this talk meets on this slide. The nozzle sets how fast gas leaves, the valve sets the smallest nudge we can deliver, the mass distribution sets how far that nudge turns the drone, and the tolerance band sets how often it has to fire. Mission life is where all four land.

The number people expect, and the number that matters:

one nozzle firing continuously	**48 seconds**	the tank drained flat out
three-axis pointing, limit cycle	**3.3 days**	79 hours, thrusters off inside the deadband

The deadband makes the same ten grams last roughly 5 900 times longer.

What ten grams of CO_2 actually buys:

4.54 N·s	total impulse
9.18 m/s	Δv if it were all spent moving
46	repositioning hops of 10 m, accelerate and stop
3.3 days	or continuous three-axis pointing instead

A realistic split: 6 g on pointing and 4 g on moving gives two days on station and eighteen repositioning hops. How far that goes depends on the flight path, and the next slide is where we won the most.

The 48-seconds-against-3.3-days contrast, and the ten-gram budget expressed four ways. *Treated in full in Section 9.1.*

The two numbers One nozzle firing continuously drains the tank in 48 seconds. Three-axis pointing in a limit cycle lasts 3.30 days, or 79 hours. The same ten grams, roughly 5,900 times longer.

Where 3.3 days comes from burn = $\dot{m}\,\tau\,t_{\min}{}^2/(2\,I\,\theta_{db})$ per axis; life = $m_{prop}/(3\cdot\text{burn})$ for three axes.

The budget 4.54 N·s total impulse. 9.18 m/s of Δv if spent entirely on translation, which is 46 repositioning hops of 10 m each. Or 3.3 days of continuous pointing.

Questions raised by this material

Q. Are disturbance torques ignored?

They are neglected, and the neglect is justified. Gravity gradient on a near-spherical body is of order 10^{-13} N·m, and drag costs about 27 pulses per orbit. Both are roughly twenty times smaller than the control torque, so they perturb the crossing time by a few percent rather than by a factor.

A.19 Slide 19 Circles cost, straight lines do not

Circles cost, straight lines do not

Flying the same scan in straight lines instead of circling the target dropped the cost of one full pass from 73 % of the tank to 24 %.

The first pattern circled the target. Going round a curve means pushing sideways the entire time — nothing holds you in the turn but your own thrusters, so a curved path charges you for every second it stays curved. Five laps is 31 radians of turning.

$$\textit{curve: } \Delta v = v \cdot \theta \qquad \textit{straight leg: } \Delta v = 2v$$

A coast costs nothing at all, for as long as you like. So a leg — burn to speed, shut every valve, glide the length of the target scanning, burn to stop — costs 2v whatever its length.

Flight pattern	Flight time	Mapped	Propellant	Valves shut	Passes per tank
circles	5 min	99.1 %	73 %	—	1.4
straight grid	26 min	98.7 %	24 %	88 %	4.1

Coverage did not move: 98.7 % against 99.1 %. The saving is not a worse scan.

Attitude came free with it. Flying parallel to a flat face with the camera already square to that face needs no torque at all, while the orbit had to rotate continuously and spent 15 to 25 % of its gas doing nothing but turning.

Flown on the simulator's vehicle model: 230 mm, 4.55 kg, eight 0.25 N nozzles, 110 N·s tank — heavier than the 500 g bench design on slide 3. The comparison is a ratio of two patterns, so it carries across the mass.

The $\Delta v = v \cdot \theta$ against $\Delta v = 2v$ argument, the measured comparison, and the vehicle-model caveat. *Treated in full in Section 9.3.*

The result Flying the same scan as straight legs instead of circling dropped the propellant cost of one full pass from 73 % of the tank to 24 %. One tank is four passes instead of one.

The physics Holding a curve in vacuum requires continuous centripetal acceleration v^2/r; nothing else holds the vehicle in the turn. Integrating over the manoeuvre gives $\Delta v = v \cdot \theta$, where θ is the total angle turned. Five wraps is 31 radians. A straight coast costs nothing at all, for any duration, so a leg costs 2v — burn to speed, burn to stop — regardless of length.

The measured comparison Circles: 5 min, 99.1 % mapped, 73 % of tank, 1.4 passes per tank. Straight grid: 26 min, 98.7 % mapped, 24 % of tank, valves shut 88 % of the time, 4.1 passes per tank.

Coverage 98.7 % against 99.1 %. The saving was not bought with a worse scan.

The attitude benefit Flying parallel to a flat face with the camera already square to it requires no torque at all. The circular pattern had to rotate continuously and spent 15 to 25 % of its gas doing only that.

The caveat This ran on the simulator's vehicle model: 230 mm, 4.55 kg, eight 0.25 N nozzles, 110 N·s tank — heavier than the 500 g bench design. The comparison is a ratio between two flight paths on the same vehicle, so it carries across the mass, but the absolute Δv figures belong to the simulator model.

Questions raised by this material

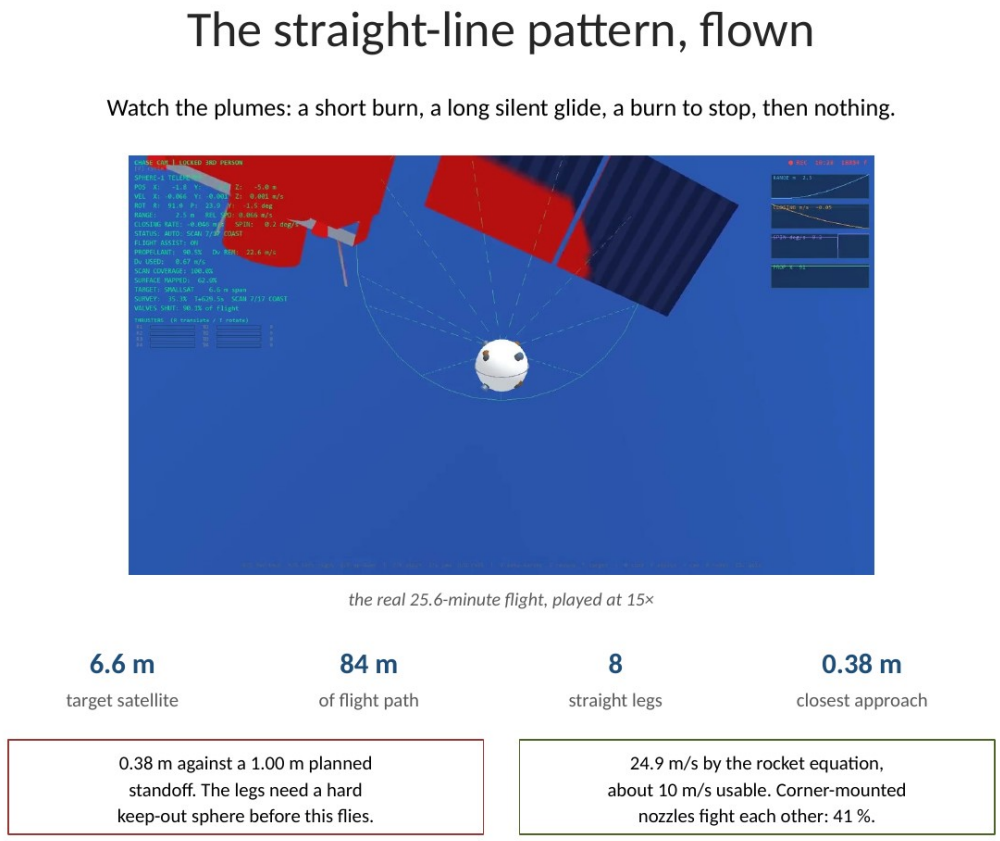
Q. Why is the simulation vehicle 4.55 kg when the design is 500 g?

Because the flight simulator runs a heavier model than the bench design, and reconciling the two is outstanding work. The ballistic result is a ratio and survives; the absolute numbers do not transfer to the 500 g vehicle.

Q. Why does the straight pattern take five times longer?

Because it coasts. The saving comes precisely from spending time instead of propellant, and for an inspection mission time is the cheaper of the two.

A.20 Slide 20 The straight-line pattern, flown



A looping video of the flown survey, the four flight facts, and two unfavourable results reported openly. *Treated in full in Section 9.4.*

What the video shows A short burn, a long silent glide, a burn to stop. Every valve is shut for 88 % of the flight. It is the real 25.6-minute run played at 15×.

The flight 6.6 m target satellite, 84 m of flight path, eight straight scanning legs.

Result one: standoff violation Closest approach was 0.38 m against a 1.00 m planned standoff. The legs need re-planning with a hard keep-out sphere enforced.

Result two: Δv conversion The rocket equation gives 24.9 m/s. Only about 10 m/s is usable — a 41 % conversion — because the nozzles sit on cube corners and any off-axis direction fires several of them partly against each other. This is the same effect as the 36 % rotational axis efficiency, averaged over the directions a real scan requires rather than a single best case.

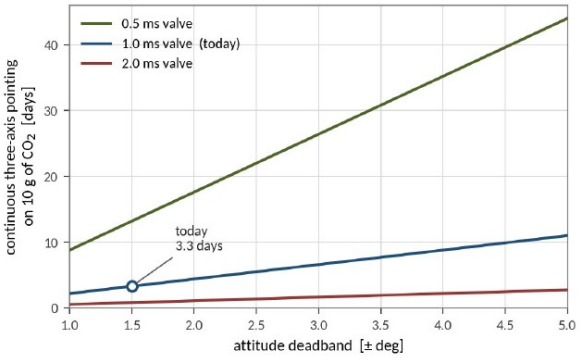
Questions raised by this material

Q. Why did the trajectory violate the standoff?

The leg planner optimised coverage and propellant without a hard geometric constraint on minimum range. Adding a keep-out sphere as a constraint rather than an objective term is the fix, and the violation was found by flying the pattern rather than by inspecting the plan.

A.21 Slide 21 Making the ten grams last

Making the ten grams last



Chain the three lines from the last slide together and the burn rate falls out:

$$burn = 2\,\dot{m}\,t_min\,/\,T_c$$

$$= \dot{m}\cdot\tau\cdot t_min^2\,/\,(2\,I\,\theta_db)$$

$$life = m_prop\,/\,(3\cdot burn)$$

t_min is squared. Halving the valve time cuts the burn by four, so the same ten grams last four times as long — and a faster valve weighs nothing extra. The deadband only helps linearly.

Three levers, in order of how much they buy:

× 4 **Faster valve, 1.0 → 0.5 ms**	3.3 → 13.2 days. The strongest lever by far, and it costs nothing in mass.
× 2 **Wider deadband, ±1.5 → ±3.0°**	3.3 → 6.6 days. Free as long as the target stays inside the camera's field of view.
× 8 **Both together**	26 days on the same ten grams.

What limits us is our own smallest pulse, not space itself. A sphere is barely tugged by gravity, and air drag costs only about 27 pulses per orbit.

Mission life against deadband, the derived scaling law, and the three levers. *Treated in full in Section 9.2.*

The scaling law Chaining the pulse consequences: burn = $2\dot{m}\,t_{\min}/T_c = \dot{m}\,\tau\,t_{\min}^2/(2\,I\,\theta_{db})$, and life = $m_{prop}/(3 \cdot \text{burn})$. The valve opening time appears squared.

Why squared A shorter pulse leaves the vehicle drifting more slowly, so it takes longer to cross the band and fires less often. Both effects act in the same direction, so halving $t_{\min}$ quarters the burn rate.

The three levers Faster valve, 1.0 → 0.5 ms: ×4, taking 3.3 days to 13.2, at no cost in mass. Wider deadband, ±1.5° → ±3.0°: ×2, to 6.6 days, free as long as the target stays in the field of view. Both together: ×8, 26 days on the same ten grams.

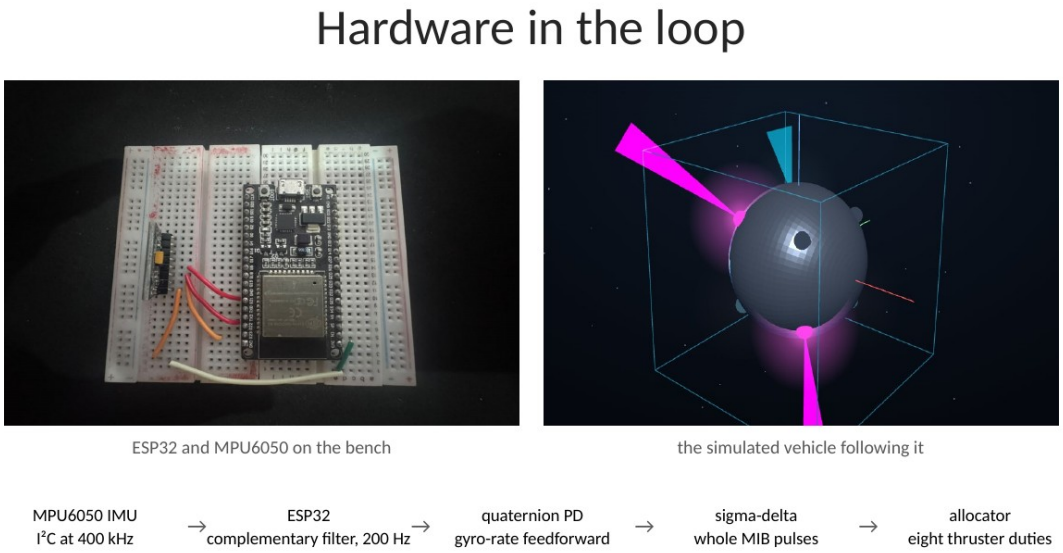
The conclusion What limits the mission is the vehicle's own minimum pulse, not the space environment. A sphere has almost no gravity-gradient torque and drag costs about 27 pulses per orbit. If longer flight time is wanted, the answer is a faster valve, not orbital mechanics.

Questions raised by this material

Q. Would a faster valve reduce reliability?

It is the right concern. Faster solenoids need higher drive voltages or smaller moving mass, both of which affect cycle life, and a limit-cycling vehicle accumulates a very large number of cycles. Characterising cycle life would have to accompany any valve change.

A.22 Slide 22 Hardware in the loop



The breadboard rig, the simulated vehicle following it, the signal chain, and the tracking-lag result. *Treated in full in Section 10.4.*

The rig An ESP32 and an MPU6050 IMU on I^2C at 400 kHz, running a complementary filter at 200 Hz, a quaternion PD controller with gyro-rate feedforward, a sigma-delta modulator emitting whole minimum impulse bits, and the allocator producing eight thruster duties.

The result Feeding the raw gyro rates forward lets the controller respond to motion as it happens rather than waiting for a tracking error to develop. Mean tracking lag falls from 11° to 1.2°.

Why it matters Under $50 of hardware validating a control loop that is normally tested on an air-bearing table.

Questions raised by this material

Q. Is this hardware-in-the-loop, or a sensor test?

The real controller runs on the real flight processor at the real rate, driving a simulated plant, which is hardware-in-the-loop by the standard definition. What is not present is actuator hardware — the valves are modelled, not driven.

Q. Why a complementary filter and not a Kalman filter?

Compute budget and transparency. At 200 Hz on an ESP32 a complementary filter is deterministic and easy to reason about. A Kalman filter would perform better in the presence of a characterised noise model, and is a reasonable next step once the IMU noise has been measured.

A.23 Slide 23 What comes next

What comes next

Three things stand between the model in this talk and a vehicle that flies, and they are in this order because each one feeds the next.

1. **Flow-bench the injector.**
   Measure how much gas it actually passes, instead of trusting the datasheet. That single number sets the thrust, the mass flow and the minimum pulse, so every other figure in this talk moves with it.

2. **Measure the real inertia.**
   Hang the assembled vehicle on two strings, time how long it takes to twist back and forth about each axis, and recover the inertia from the period. Cheap, and far more trustworthy than what the CAD assumes.

3. **Retune the gains and fly it.**
   Re-derive the controller against the measured inertia, flash it, and close the loop on the physical vehicle instead of on the simulated one.

$$I_{zz} = m \cdot g_0 \cdot d^2 \cdot T^2 / (16\,\pi^2\,L)$$

The bifilar pendulum, which is step two.

Three roadmap items in dependency order, and the bifilar pendulum relation. *Treated in full in Chapter 12.*

Item one: flow-bench the injector Measure $(C_dA)_{\text{inj}}$ directly instead of taking 0.147 mm^2 from a datasheet. It sets thrust, mass flow and minimum impulse bit, so every other figure moves with it. It is first because everything else depends on it.

Item two: measure the real inertia Suspend the assembled vehicle on two strings, time the torsional period about each axis, and recover the inertia from I_{zz} = m g_0 d^2 T^2/(16π^2 L). Cheap, and far more trustworthy than the CAD assumption.

Item three: retune and fly Re-derive k_p and k_d against the measured inertia to restore $\zeta = 0.90$, then close the loop on the physical vehicle rather than the simulated one.

Questions raised by this material

Q. The CAD models a solid uniform sphere but the real vehicle is a thin shell. Is that not a large error?

It is, and it is quantified. CAD gives 2.637 $\times$ 10^{-3} kg·m^2; a thin shell of the same mass gives 4.408 $\times$ 10^{-3}, a ratio of exactly 5/3 and an error of +67.2 %. The consequence is that the real vehicle slews about 30 % more slowly than the gains assume, with ω_n falling from 3.68 to 2.84 rad/s and ζ from 0.90 to 0.70 — underdamped, so it would overshoot. The 200 g of valves sit at the wall, so the actuator choice and the inertia error are the same problem seen twice. This is why the bifilar pendulum is roadmap item two.

A.24 Slide 24 Reference figures

Thank you.

I would be glad to take questions.

wet / dry mass	500 / 490 g	propellant	10 g CO_2
valve mass, flight / bench	25 g / 80 g	γ · R	1.280 · 188.9
triple point	216.6 K	Me / (Ae/At) limit	1.589 / 1.266
Isp conv / CD	46.33 / 48.42 s	throat Ø	0.80 mm
chamber pressure	1.5392 bar	mass flow	0.2073 g/s
thrust per nozzle	94.188 mN	force per axis	108.76 mN
minimum impulse bit	94.19 µN·s	cant · moment arm	38.50° · 71.59 mm
pointing life, 3 axes	3.3 days	Δv / 10 m hops	9.18 m/s / 46
turn authority Y / X,Z	7.45 / 5.87	axis efficiency, rot / trans	36 % / 58 %
12-nozzle torque advantage	2.78 ×	crossover valve mass	14.7 g
αmax	169.2 °/s²	slew 30° / 90°	0.84 / 1.46 s
I CAD / shell	2.637 / 4.408 e-3	kp · kd	7.1432e-2 · 1.7469e-2
ωn · ζ nominal	3.680 rad/s · 0.900	integrator	RK4, 5 ms, 200 Hz
survey, orbit / ballistic	73 % / 24 % tank	surveys per tank	1.4 / 4.1
valves shut, ballistic	88 %	Δv conversion, measured	41 %

The closing slide, carrying every quoted figure in one place. *Treated in full in Appendix C.10.*

APPENDIX B

MATLAB source listings

Three MATLAB scripts accompany this report. They require base MATLAB only, with no toolboxes, and run on R2016b or newer. Each opens its figures with a single Run and prints a geometry table to the console.

File	Produces
`Sphere1_Figures.m`	The eight analysis figures reproduced in this report: the condensation cliff, the chamber-pressure curve, the axis-efficiency bars, the propellant budget, lifetime angular impulse, the deadband limit cycle, mission life against deadband, and RK4 convergence.
`Sphere1_Injectors.m`	Three-dimensional visualisation of the eight flight injectors: where each sits on the hull, where its gas goes, and the reaction force on the body. The geometry is the measured flight geometry, not an idealised regeneration.
`Sphere1_Layouts_7_and_12.m`	The seven- and twelve-nozzle layouts generated for comparison, plus a three-way figure. Includes the search result that a symmetric seven-nozzle arrangement fails to positively span $\mathbb{R}^6$.

B.1 Sphere1_Figures.m

```
% Sphere-1 -- all presentation figures
%
%  Temur Makharadze, bachelor's thesis
%
%  ------------------------------------------------------------------------
%  HOW TO RUN
%
%  First time, and any time you have restarted MATLAB:
%      press the green Run button in the toolbar, or F5.
%  That executes the whole file and opens all eight figures at once.
%
%  After that, to redo one figure only:
%      click anywhere inside its %% section, then press Ctrl+Enter.
%
%  Ctrl+Enter runs ONE section, and every figure section needs the values
%  defined in CONSTANTS below. If the workspace is empty the section stops
%  with a message telling you to run CONSTANTS first. Run it (click in it,
%  Ctrl+Enter) and the figure sections will work for the rest of the session.
%  ------------------------------------------------------------------------
%
%  Everything is computed from the physics. Nothing is hard-coded except the
%  measured RK4 convergence data in Figure 3 and the injector flow area, which
%  is a datasheet value.
%
%  To save a figure for the slides:
%      exportgraphics(gcf,'fig_name.png','Resolution',300)
%  A screenshot works too, but exportgraphics gives a cleaner edge.
%
%  No toolboxes needed, base MATLAB only. Tested on R2023b, needs R2020a or
%  newer: exportgraphics is R2020a, xline/yline are R2018b, compose is R2016b,
%  per-bar CData is R2017b. On an older release, replace exportgraphics with
%  print(gcf,'-dpng','-r300',...) and the x/yline calls with plot() of a
%  two-point line.


%% ===== CONSTANTS ============================================================
% Run this section before any figure section.

clear; clc; close all

% Gas
gam   = 1.280;          % CO2 ratio of specific heats at room temperature
Rgas  = 188.9;          % J/(kg K)    = 8314/44.01
T0    = 293.15;         % K           chamber stagnation temperature, 20 C
Ttrip = 216.6;          % K           CO2 triple point
Cd    = 0.95;           % discharge coefficient (boundary-layer blockage)

% Feed system
Pfeed  = 5.0;           % bar         regulator setting
CdA_in = 0.147;         % mm^2        injector effective flow area (datasheet)

% Vehicle
Rsph  = 0.115;          % m           sphere radius
cant  = deg2rad(38.50); % rad         outward cant of the rotation group
Iner  = 2.6372e-3;      % kg m^2      CAD inertia
mProp = 0.010;          % kg          propellant
mWet  = 500;            % g
mStruct = 290;          % g           everything that is not a valve
Isp   = 46.33;          % s           converging-only
g0    = 9.80665;        % m/s^2

% Thruster operating point
F0    = 94.188e-3;      % N           thrust per nozzle
mdot  = 0.2073e-3;      % kg/s        choked mass flow per nozzle
tmin  = 1.0e-3;         % s           minimum clean valve opening
dbDeg = 1.5;            % deg         attitude deadband

% Derived
tau8  = (2/sqrt(3))*F0*Rsph*sin(cant);   % N m, one canted pair
tau12 = 2*F0*Rsph;                       % N m, one aligned pair

% Plot style
BLUE  = [ 31  78 121]/255;
RED   = [149  55  53]/255;
GREEN = [ 79  98  34]/255;
GREY  = [ 89  89  89]/255;
FS    = 11;             % font size
FN    = 'Calibri';      % font name
```

```
79
80  fprintf('Derived quantities\n');
81  fprintf('  moment arm  R sin(cant)   : %6.2f mm\n', Rsph*sin(cant)*1e3);
82  fprintf('  torque, 8-nozzle pair     : %6.3f mN m\n', tau8*1e3);
83  fprintf('  torque, 12-nozzle pair    : %6.3f mN m\n', tau12*1e3);
84  fprintf('  12/8 torque ratio         : %6.3f\n', tau12/tau8);
85  fprintf('  limit-cycle drift rate    : %6.4f deg/s\n', rad2deg(tau8*tmin/(2*Iner)));
86
87
88  %% ===== FIGURE 1 -- the condensation cliff ================================
89  %  Exit static temperature against expansion area ratio, with the CO2 triple
90  %  point. Below 216.6 K the propellant deposits as dry ice, so the area ratio
91  %  where the curve crosses that line is a hard ceiling on the nozzle.
92
93  if ~exist('gam','var')
94      error('Sphere1:noConstants', ...
95            ['Run the CONSTANTS section first: click inside it near the top ' ...
96             'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
97             'make every figure at once.']);
98  end
99
100 ar = linspace(1.0, 2.5, 400);
101 Te = zeros(size(ar));
102 for k = 1:numel(ar)
103     if ar(k) <= 1
104         M = 1;
105     else
106         M = machFromArea(ar(k), gam);       % supersonic root
107     end
108     Te(k) = T0/(1 + (gam-1)/2*M^2);
109 end
110
111 % the ceiling: set Te = Ttrip, solve back for Mach, then for area ratio
112 MeMax  = sqrt( 2/(gam-1) * (T0/Ttrip - 1) );
113 arMax  = areaFromMach(MeMax, gam);
114
115 figure('Color','w','Position',[100 100 760 470]);
116 plot(ar, Te, 'Color', BLUE, 'LineWidth', 2); hold on
117 yline(Ttrip, '-', 'Color', RED, 'LineWidth', 1.4);
118 xline(arMax, '--', 'Color', GREY, 'LineWidth', 1.1);
119 text(2.45, Ttrip+4, sprintf('CO_2 triple point, %.1f K', Ttrip), ...
120      'Color', RED, 'HorizontalAlignment','right', 'FontName',FN,'FontSize',FS-1);
121 text(arMax+0.04, 256, sprintf('ceiling  A_e/A_t = %.3f', arMax), ...
122      'Color', [0.25 0.25 0.25], 'FontName',FN,'FontSize',FS-1);
123 xlabel('expansion area ratio   A_e / A_t');
124 ylabel('exit static temperature   T_e   [K]');
125 xlim([1.0 2.5]); ylim([165 270]);
126 styleAxes(FN, FS);
127
128 fprintf('\nFig 1: Me,max = %.4f, area-ratio ceiling = %.4f\n', MeMax, arMax);
129
130
131 %% ===== FIGURE 2 -- what sets the chamber pressure ========================
132 %  The injector orifice and the nozzle throat are two CHOKED restrictions in
133 %  series. In steady state they pass the same mass flow, every gas property
134 %  cancels, and only the two effective areas are left:
135 %      P0 = Pfeed * (Cd A)injector / (Cd A)throat
136
137 if ~exist('gam','var')
138     error('Sphere1:noConstants', ...
139           ['Run the CONSTANTS section first: click inside it near the top ' ...
140            'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
141            'make every figure at once.']);
142 end
143
144 Dt   = [2.00 1.50 1.00 0.80 0.60];        % mm
145 CdAt = Cd * pi/4 .* Dt.^2;                % mm^2
146 P0   = Pfeed * CdA_in ./ CdAt;            % bar
147
148 figure('Color','w','Position',[100 100 760 470]);
149 xpos = 1:numel(Dt);                       % numeric x, labelled afterwards
150 b = bar(xpos, P0, 0.55, 'EdgeColor',[0.35 0.35 0.35], 'LineWidth', 0.6);
151 b.FaceColor = 'flat';
152 b.CData = [RED; 0.65 0.50 0.37; 0.75 0.63 0.42; GREEN; 0.43 0.55 0.29];
153 hold on
154 yline(P0(4), '--', 'Color', [0.25 0.25 0.25], 'LineWidth', 1.0);
155 text(0.55, P0(4)+0.11, sprintf('target %.2f bar', P0(4)), ...
156      'FontName',FN,'FontSize',FS-1,'Color',[0.25 0.25 0.25]);
157 for k = 1:numel(P0)
158     text(xpos(k), P0(k)+0.07, sprintf('%.2f', P0(k)), 'HorizontalAlignment','center', ...
159          'FontName',FN,'FontSize',FS-1);
```

```
end
xlabel('nozzle throat diameter   D_t   [mm]');
ylabel('chamber pressure   P_0   [bar]');
xlim([0.4 numel(Dt)+0.6]); ylim([0 3.2]);
set(gca, 'XTick', xpos, 'XTickLabel', compose('%.2f', Dt));
styleAxes(FN, FS);

fprintf('\nFig 2: '); fprintf('%.2fmm->%.2fbar  ', [Dt; P0]); fprintf('\n');
fprintf('  CAVEAT: this assumes BOTH orifices choke. At 2.0 mm the model gives\n');
fprintf('  0.25 bar, which would not choke against 1 bar on the bench, so that\n');
fprintf('  point is only valid in vacuum.\n');


%% ===== FIGURE 3 -- RK4 convergence ======================================
%  Integrate y'' = -1.7 y over 1 s, for which the exact answer is known, and
%  halve the step repeatedly. Classical RK4 has global error O(h^4), so the
%  error should fall by 2^4 = 16 each time. These are the measured values.

if ~exist('gam','var')
    error('Sphere1:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

Nsteps = [10 20 40 80 160 320];
errRK4 = [2.491e-6 1.450e-7 8.748e-9 5.371e-10 3.328e-11 2.071e-12];
ideal  = errRK4(1) * (Nsteps(1)./Nsteps).^4;      % perfect 4th-order slope

figure('Color','w','Position',[100 100 700 470]);
loglog(Nsteps, ideal, '--', 'Color', GREY, 'LineWidth', 1.1); hold on
loglog(Nsteps, errRK4, 'o-', 'Color', BLUE, 'LineWidth', 1.5, ...
       'MarkerSize', 7, 'MarkerFaceColor','w', 'MarkerEdgeColor', BLUE);
text(190, 4e-11, 'slope -4', 'Color', GREY, 'FontName',FN,'FontSize',FS-1);
xlabel('number of integration steps   N');
ylabel('absolute error at t = 1 s');
legend({'ideal 4th order','measured'}, 'Location','southwest', ...
       'Box','off', 'FontName',FN, 'FontSize',FS-1);
styleAxes(FN, FS);

order = log2(errRK4(1:end-1)./errRK4(2:end));
fprintf('\nFig 3: observed order of convergence = '); fprintf('%.3f  ', order); fprintf('\n');


%% ===== FIGURE 4 -- how much thrust reaches the commanded axis ============
%  A tetrahedral nozzle never points along the axis it commands, so only its
%  projection is useful. Two losses multiply for rotation:
%     tetrahedral projection  (2/sqrt3)/2 = 57.7 %
%     outward cant            sin(38.5)   = 62.3 %

if ~exist('gam','var')
    error('Sphere1:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

effTrans12 = 100;
effRot12   = 100;
effTrans8  = (2/sqrt(3))/2 * 100;
effRot8    = (2/sqrt(3))/2 * sin(cant) * 100;

vals = [effTrans12 effRot12; effTrans8 effRot8];

figure('Color','w','Position',[100 100 700 470]);
b = bar(vals, 0.7, 'EdgeColor',[0.35 0.35 0.35], 'LineWidth',0.6);
b(1).FaceColor = BLUE;
b(2).FaceColor = [0.65 0.65 0.65];
hold on
for i = 1:2
    for j = 1:2
        text(i + (j-1.5)*0.35, vals(i,j)+2.5, sprintf('%.0f %%', vals(i,j)), ...
             'HorizontalAlignment','center','FontName',FN,'FontSize',FS-1);
    end
end
set(gca,'XTickLabel',{sprintf('12 nozzles\n(aligned pairs)'), ...
                      sprintf('8 nozzles\n(tetrahedral)')});
ylabel(sprintf('fraction of fired thrust\nreaching the commanded axis   [%%]'));
ylim([0 118]); yticks(0:25:100);
legend({'translation','rotation'}, 'Location','northeast', 'Orientation','horizontal', ...
       'Box','off','FontName',FN,'FontSize',FS-1);
```

```
styleAxes(FN, FS);

fprintf('\nFig 4: translation %.1f %%, rotation %.1f %% (= %.3f x %.3f)\n', ...
        effTrans8, effRot8, (2/sqrt(3))/2, sin(cant));


%% ===== FIGURE 5 -- propellant left against valve count ==================
%  Total wet mass is capped at 500 g. Everything that is not a valve is 290 g.
%  So whatever the valves do not take is what is left for propellant.

if ~exist('gam','var')
    error('Spherel:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

mValve = 25;                              % g per flight valve
Nv     = [6 7 8 12];
mLeft  = mWet - mStruct - mValve*Nv;      % g

figure('Color','w','Position',[100 100 720 470]);
xpos = 1:numel(Nv);
yl   = [-110 90];

% shade the impossible region FIRST, so the bars are drawn on top of it
patch([0.4 numel(Nv)+0.6 numel(Nv)+0.6 0.4], [yl(1) yl(1) 0 0], ...
      [0.95 0.87 0.87], 'EdgeColor','none'); hold on

b = bar(xpos, mLeft, 0.5, 'EdgeColor',[0.35 0.35 0.35], 'LineWidth',0.6);
b.FaceColor = 'flat';
b.CData = repmat(GREEN, numel(Nv), 1);
b.CData(mLeft < 0, :) = repmat(RED, sum(mLeft<0), 1);

yline(0, '-', 'Color', [0.25 0.25 0.25], 'LineWidth', 1.0);
for k = 1:numel(Nv)
    off = 7; if mLeft(k) < 0, off = -17; end
    if mLeft(k) < 0, col = RED; else, col = [0.15 0.15 0.15]; end
    text(xpos(k), mLeft(k)+off, sprintf('%+d g', mLeft(k)), ...
         'HorizontalAlignment','center','FontName',FN,'FontSize',FS-1,'Color',col);
end
text(0.55, -58, 'cannot be built', 'Color', RED, 'FontName',FN,'FontSize',FS-1);
xlabel('number of valves');
ylabel('propellant left in a 500 g vehicle   [g]');
xlim([0.4 numel(Nv)+0.6]); ylim(yl);
set(gca, 'XTick', xpos, 'XTickLabel', compose('%d', Nv));
styleAxes(FN, FS);

fprintf('\nFig 5: '); fprintf('N=%d:%+dg  ', [Nv; mLeft]); fprintf('\n');


%% ===== FIGURE 6 -- why twelve nozzles do not pay ========================
%  Twelve nozzles give 2.78x the torque per pulse. They also cost four extra
%  valves, and that mass comes straight out of propellant. What the mission
%  actually spends is TOTAL ANGULAR IMPULSE = torque x how long it can fire.
%
%  H assumes all the propellant goes to one axis. That is an upper bound, but
%  it uses the same convention for both configurations, so the comparison holds.

if ~exist('gam','var')
    error('Spherel:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

mv   = linspace(5, 30, 400);                    % g per valve
prop8  = max(mWet - mStruct - 8*mv,  0)/1000;   % kg
prop12 = max(mWet - mStruct - 12*mv, 0)/1000;
H8   = tau8  .* prop8  / (2*mdot);              % N m s
H12  = tau12 .* prop12 / (2*mdot);

% analytic crossover and the point where twelve stops fitting at all
mCross = (tau12 - tau8)*(mWet-mStruct) / (12*tau12 - 8*tau8);
mClose = (mWet - mStruct)/12;

figure('Color','w','Position',[100 100 780 480]);
patch([20 30 30 20], [0 0 8 8], [0.93 0.93 0.93], 'EdgeColor','none', ...
      'HandleVisibility','off'); hold on
h12 = plot(mv, H12, 'Color', RED,  'LineWidth', 2);
h8  = plot(mv, H8,  'Color', BLUE, 'LineWidth', 2);
```

```
xline(mCross, '--', 'Color', [0.25 0.25 0.25], 'LineWidth', 1.0);
plot(mClose, 0, 'x', 'Color', RED, 'MarkerSize', 10, 'LineWidth', 1.8, ...
     'HandleVisibility','off');
text(mCross-0.4, 6.4, sprintf('crossover\n%.1f g', mCross), ...
     'HorizontalAlignment','right','FontName',FN,'FontSize',FS-1);
text(mClose, 0.32, sprintf('%.1f g', mClose), 'HorizontalAlignment','center', ...
     'Color', RED, 'FontName',FN,'FontSize',FS-1);
text(29.4, 3.1, sprintf('flight valves\nwe can buy'), 'HorizontalAlignment','right', ...
     'VerticalAlignment','top','Color',GREY,'FontName',FN,'FontSize',FS-1);
xlabel('mass of one valve   [g]');
ylabel(sprintf('lifetime angular impulse\nabout one axis   [N m s]'));
xlim([5 30]); ylim([0 8]);
legend([h12 h8], {'12 nozzles','8 nozzles'}, 'Location','north', 'Box','off', ...
       'FontName',FN,'FontSize',FS-1);
styleAxes(FN, FS);

fprintf('\nFig 6: crossover %.2f g, twelve stops closing above %.2f g\n', mCross, mClose);


%% ===== FIGURE 7 -- the deadband limit cycle ==============================
%  CORRECTED. The drift rate is not arbitrary: one minimum pulse from a pair
%  reverses the drift, so the post-pulse rate is
%       w = tau * tmin / (2 I)
%  and the time to cross the full 2*deadband is 2*db/w. Everything below comes
%  from that, so this figure now agrees with the mission-life figure.

if ~exist('gam','var')
    error('Sphere1:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

wRad  = tau8*tmin/(2*Iner);          % rad/s
wDeg  = rad2deg(wRad);               % deg/s
Tc    = 2*dbDeg/wDeg;                % s, one crossing

t   = linspace(0, 150, 4000);
ph  = mod(t, 2*Tc);
err = -dbDeg + wDeg*ph;
sel = ph > Tc;
err(sel) = dbDeg - wDeg*(ph(sel) - Tc);

figure('Color','w','Position',[100 100 780 460]);
patch([0 150 150 0], [-dbDeg -dbDeg dbDeg dbDeg], [0.91 0.94 0.97], ...
      'EdgeColor','none'); hold on
plot(t, err, 'Color', BLUE, 'LineWidth', 1.8);
yline( dbDeg, '-', 'Color', RED, 'LineWidth', 1.1);
yline(-dbDeg, '-', 'Color', RED, 'LineWidth', 1.1);
text(148, dbDeg+0.16, sprintf('deadband \\pm%.1f\\circ - all thrusters off inside', dbDeg), ...
     'HorizontalAlignment','right','Color',RED,'FontName',FN,'FontSize',FS-1);
text(3, -dbDeg-0.30, sprintf('drift %.3f\\circ/s, one crossing %.1f s', wDeg, Tc), ...
     'Color',[0.25 0.25 0.25],'FontName',FN,'FontSize',FS-1);
xlabel('time   [s]');
ylabel('pointing error   [deg]');
xlim([0 150]); ylim([-2.3 2.3]); yticks(-2:1:2);
styleAxes(FN, FS);

fprintf('\nFig 7: drift %.4f deg/s, crossing %.1f s, %.1f full cycles in 150 s\n', ...
        wDeg, Tc, 150/(2*Tc));


%% ===== FIGURE 8 -- how long the ten grams last ===========================
%  In a minimum-impulse limit cycle:
%     propellant per pulse = 2 mdot tmin
%     post-pulse rate      = tau tmin /(2 I)
%     crossing time        = 2 db / rate
%     burn rate            = mdot tau tmin^2 / (2 I db)
%  So the burn scales as the SQUARE of the valve opening time and inversely
%  with the deadband. Divided by three because all three axes limit-cycle.

if ~exist('gam','var')
    error('Sphere1:noConstants', ...
          ['Run the CONSTANTS section first: click inside it near the top ' ...
           'of this file and press Ctrl+Enter. Or just press Run (F5) to ' ...
           'make every figure at once.']);
end

dbSweep = linspace(1.0, 5.0, 400);              % deg
tvals   = [0.5 1.0 2.0]*1e-3;                   % s
cols    = [GREEN; BLUE; RED];
```

```
labs    = {'0.5 ms valve','1.0 ms valve  (today)','2.0 ms valve'};

figure('Color','w','Position',[100 100 780 480]); hold on
lifeDays = zeros(numel(tvals), numel(dbSweep));
for i = 1:numel(tvals)
    rate = mdot*tau8*tvals(i)^2 ./ (2*Iner*deg2rad(dbSweep));   % kg/s, one axis
    lifeDays(i,:) = mProp ./ (rate*3) / 86400;                  % three axes
    plot(dbSweep, lifeDays(i,:), 'Color', cols(i,:), 'LineWidth', 2);
end
% mark today's design point
rate0 = mdot*tau8*tmin^2/(2*Iner*deg2rad(dbDeg));
life0 = mProp/(rate0*3)/86400;
plot(dbDeg, life0, 'o', 'MarkerSize', 8, 'MarkerFaceColor','w', ...
     'MarkerEdgeColor', BLUE, 'LineWidth', 1.6);
text(dbDeg+0.28, life0+5.0, sprintf('today\n%.1f days', life0), ...
     'FontName',FN,'FontSize',FS-1);
xlabel('attitude deadband   [\pm deg]');
ylabel(sprintf('continuous three-axis pointing\non 10 g of CO_2   [days]'));
xlim([1 5]); ylim([0 46]);
legend(labs, 'Location','northwest','Box','off','FontName',FN,'FontSize',FS-1);
styleAxes(FN, FS);

fprintf('\nFig 8: today %.2f days; 0.5 ms valve %.2f d; 3 deg deadband %.2f d\n', ...
        life0, ...
        mProp/(mdot*tau8*(0.5e-3)^2/(2*Iner*deg2rad(dbDeg))*3)/86400, ...
        mProp/(mdot*tau8*tmin^2/(2*Iner*deg2rad(3.0))*3)/86400);
fprintf('  continuous single-nozzle firing would last %.1f s\n', mProp/mdot);
fprintf('  so the deadband buys a factor of %.0f\n', (life0*86400)/(mProp/mdot));


%% ===== LOCAL FUNCTIONS ================================================
% (MATLAB requires these at the end of the script.)

function styleAxes(FN, FS)
% Plain academic look: white ground, thin box, faint grid.
    ax = gca;
    ax.FontName   = FN;
    ax.FontSize   = FS;
    ax.Box        = 'on';
    ax.XGrid      = 'on';
    ax.YGrid      = 'on';
    ax.GridColor  = [0.82 0.82 0.82];
    ax.GridAlpha  = 1;
    ax.XColor     = [0.25 0.25 0.25];
    ax.YColor     = [0.25 0.25 0.25];
    ax.LineWidth  = 0.7;
    ax.Layer      = 'top';
end

function ar = areaFromMach(M, g)
% Isentropic area-Mach relation.
    ar = (1./M) .* ((2/(g+1))*(1 + (g-1)/2*M.^2)).^((g+1)/(2*(g-1)));
end

function M = machFromArea(ar, g)
% Supersonic root of the area-Mach relation, by bracketing on M > 1.
    f = @(m) areaFromMach(m, g) - ar;
    M = fzero(f, [1 + 1e-9, 12]);
end
```

B.2 Sphere1_Injectors.m

```
% Sphere-1 -- the eight injectors: where they sit and where they point
%
%  Temur Makharadze, bachelor's thesis
%
%  ------------------------------------------------------------------------
%  HOW TO RUN
%      Press Run (F5). It opens two figures and prints a geometry table.
%      Figure 1: all eight on the sphere, jets and reaction forces.
%      Figure 2: the two groups drawn separately.
%
%      Drag inside a figure to rotate it. To save one for a slide:
%          exportgraphics(gcf,'injectors.png','Resolution',300)
%  ------------------------------------------------------------------------
%
%  The numbers below are the flight geometry as measured, not idealised.
%  R1-R4 are analytic (four alternating cube corners). T1-T4 come from
%  SOLIDWORKS and are deliberately NOT regenerated from a formula: the
%  four cant angles differ slightly (38.4366 to 38.5013 deg) and the
%  T1/T3 positions disagree by up to 0.26 mm where perfect symmetry
%  would make them identical. That asymmetry is real and it is why
%  rotation authority is not equal on every axis.
%
%  No toolboxes needed. Base MATLAB, R2016b or newer.

clear; clc; close all


%% ===== GEOMETRY ============================================================

R_MM = 115.0;              % mm, mounting radius (sphere surface)
F0   = 94.188e-3;          % N,  thrust per nozzle, full open

NAME = {'R1','R2','R3','R4','T1','T2','T3','T4'};

% Thrust direction unit vectors, body frame.
% NOTE: these are the direction of the FORCE ON THE BODY.
% The gas leaves in the opposite direction (see JET below).
DIR = [ ...
 -0.577350 -0.577350  0.577350  0.577350 -0.076090  0.317520  0.076100 -0.317529
 -0.577350  0.577350 -0.577350  0.577350  0.829232 -0.041990  0.829231 -0.041990
 -0.577350  0.577350  0.577350 -0.577350  0.553701 -0.947321 -0.553701  0.947318 ];

% Nozzle positions, body frame, mm from the centre of mass
POS = [ ...
  66.39528  66.39528 -66.39528 -66.39528 -62.36530 -65.39570  62.44840  65.39610
  66.39528 -66.39528  66.39528 -66.39528 -69.30520  63.22120 -69.48640  63.22130
  66.39528 -66.39528 -66.39528  66.39528 -67.32270  70.36670  67.05850 -70.36630 ];

JET = -DIR;                % where the gas actually goes
isR = [true true true true false false false false];

% colours
BLUE  = [ 31  78 121]/255;   % translation group
RED   = [149  55  53]/255;   % rotation group
GREY  = [140 140 140]/255;
FN    = 'Calibri';


%% ===== DERIVED GEOMETRY, PRINTED ===========================================

fprintf('Sphere-1 injector geometry\n');
fprintf('%5s %26s %26s %8s %9s %10s\n', ...
        'noz', 'position [mm]', 'thrust direction', 'cant', 'arm [mm]', 'torque');
fprintf('%s\n', repmat('-', 1, 92));

cant = zeros(1,8); arm = zeros(1,8);
for k = 1:8
    r = POS(:,k);  d = DIR(:,k);
    inward = -r/norm(r);                    % straight in through the centre
    cant(k) = acosd(max(min(dot(d, inward), 1), -1));
    m   = cross(r/1000, d*F0);              % N m
    arm(k) = norm(m)/F0*1000;               % mm
    fprintf('%5s  (%7.2f %7.2f %7.2f)  (%6.3f %6.3f %6.3f) %7.2f° %8.2f %8.3f mN·m\n', ...
        NAME{k}, r(1), r(2), r(3), d(1), d(2), d(3), cant(k), arm(k), norm(m)*1e3);
end

fprintf('\n  every nozzle sits at |r| = %.3f mm\n', mean(vecnorm(POS)));
fprintf('  radial group  R1-R4 : cant %.2f°, arm %.3f mm -> zero torque, pure push\n', ...
```

```
        mean(cant(isR)), mean(arm(isR)));
fprintf('  canted group  T1-T4 : cant %.2f° to %.2f°, arm %.2f to %.2f mm\n', ...
        min(cant(~isR)), max(cant(~isR)), min(arm(~isR)), max(arm(~isR)));
fprintf('  R sin(38.50°) = %.2f mm, the nominal moment arm\n', R_MM*sind(38.50));

fprintf('\n  sum of R thrust directions : (%6.3f %6.3f %6.3f)\n', sum(DIR(:,isR),2));
fprintf('  sum of T thrust directions : (%6.3f %6.3f %6.3f)  <- the +Y bias\n', sum(DIR(:,~isR),2))
↪ ;
fprintf('  that bias is why turning about Y is stronger than about X or Z.\n');


%% ===== FIGURE 1 -- all eight on the sphere ==============================
%  Solid arrows are the escaping gas. Faint arrows are the reaction force
%  on the vehicle, which is what actually moves it.

figure('Color','w','Position',[80 80 900 780]);
drawSphere(R_MM);
hold on

LJ = 70;      % mm, length of the jet arrow
LF = 45;      % mm, length of the force arrow

for k = 1:8
    c = BLUE; if ~isR(k), c = RED; end
    p = POS(:,k);

    % the gas leaving the nozzle
    quiver3(p(1), p(2), p(3), JET(1,k)*LJ, JET(2,k)*LJ, JET(3,k)*LJ, 0, ...
        'Color', c, 'LineWidth', 2.0, 'MaxHeadSize', 0.55);

    % the reaction force on the body, drawn in a lighter tint of the same colour
    % (a solid tint rather than an alpha value, which older releases reject)
    quiver3(p(1), p(2), p(3), DIR(1,k)*LF, DIR(2,k)*LF, DIR(3,k)*LF, 0, ...
        'Color', c*0.40 + 0.60, 'LineWidth', 1.4, 'MaxHeadSize', 0.5);

    plot3(p(1), p(2), p(3), 'o', 'MarkerSize', 7, ...
        'MarkerFaceColor', c, 'MarkerEdgeColor', 'w', 'LineWidth', 1.0);

    lab = p + JET(:,k)*(LJ + 22);
    text(lab(1), lab(2), lab(3), NAME{k}, 'FontName', FN, 'FontSize', 12, ...
        'FontWeight', 'bold', 'Color', c, ...
        'HorizontalAlignment', 'center', 'VerticalAlignment', 'middle');
end

bodyAxes(R_MM, FN);
title({'Sphere-1 -- eight injectors', ...
       'solid arrow = escaping gas,  faint arrow = force on the vehicle'}, ...
      'FontName', FN, 'FontSize', 13, 'FontWeight', 'normal');
finishAxes(R_MM, FN);


%% ===== FIGURE 2 -- the two groups, side by side ==========================

figure('Color','w','Position',[80 80 1200 620]);

subplot(1,2,1);
drawSphere(R_MM); hold on
for k = find(isR)
    p = POS(:,k);
    quiver3(p(1),p(2),p(3), JET(1,k)*LJ, JET(2,k)*LJ, JET(3,k)*LJ, 0, ...
        'Color', BLUE, 'LineWidth', 2.0, 'MaxHeadSize', 0.55);
    plot3([0 p(1)],[0 p(2)],[0 p(3)], ':', 'Color', GREY, 'LineWidth', 1.0);
    plot3(p(1),p(2),p(3),'o','MarkerSize',7, ...
        'MarkerFaceColor',BLUE,'MarkerEdgeColor','w','LineWidth',1.0);
    lab = p + JET(:,k)*(LJ+22);
    text(lab(1),lab(2),lab(3), NAME{k}, 'FontName',FN,'FontSize',12, ...
        'FontWeight','bold','Color',BLUE,'HorizontalAlignment','center');
end
bodyAxes(R_MM, FN);
title({'R1-R4  translation group', ...
       'four alternating cube corners, firing straight out', ...
       'thrust line passes through the centre, so torque = 0'}, ...
      'FontName',FN,'FontSize',12,'FontWeight','normal');
finishAxes(R_MM, FN);

subplot(1,2,2);
drawSphere(R_MM); hold on
for k = find(~isR)
    p = POS(:,k);
    quiver3(p(1),p(2),p(3), JET(1,k)*LJ, JET(2,k)*LJ, JET(3,k)*LJ, 0, ...
        'Color', RED, 'LineWidth', 2.0, 'MaxHeadSize', 0.55);
```

```
    plot3([0 p(1)],[0 p(2)],[0 p(3)], ':', 'Color', GREY, 'LineWidth', 1.0);
    plot3(p(1),p(2),p(3),'o','MarkerSize',7, ...
        'MarkerFaceColor',RED,'MarkerEdgeColor','w','LineWidth',1.0);
    lab = p + JET(:,k)*(LJ+22);
    text(lab(1),lab(2),lab(3), NAME{k}, 'FontName',FN,'FontSize',12, ...
        'FontWeight','bold','Color',RED,'HorizontalAlignment','center');
end
bodyAxes(R_MM, FN);
title({'T1-T4  rotation group', ...
       sprintf('swung %.2f° off radial, moment arm %.2f mm', 38.50, R_MM*sind(38.50)), ...
       'the dotted line is the radial the nozzle is swung away from'}, ...
      'FontName',FN,'FontSize',12,'FontWeight','normal');
finishAxes(R_MM, FN);


%% ===== LOCAL FUNCTIONS ================================================

function drawSphere(R)
% faint hull so the mounting points read as being on a surface
    [sx, sy, sz] = sphere(60);
    surf(sx*R, sy*R, sz*R, 'FaceColor', [0.82 0.85 0.90], 'FaceAlpha', 0.25, ...
         'EdgeColor', 'none');
    hold on
end

function bodyAxes(R, FN)
% the body X, Y, Z reference triad
    L = R*1.55;
    ax = {'X','Y','Z'};
    for i = 1:3
        v = zeros(3,1); v(i) = L;
        plot3([-v(1) v(1)], [-v(2) v(2)], [-v(3) v(3)], '-', ...
              'Color', [0.72 0.72 0.72], 'LineWidth', 0.8);
        text(v(1)*1.06, v(2)*1.06, v(3)*1.06, ax{i}, 'FontName', FN, ...
             'FontSize', 11, 'Color', [0.45 0.45 0.45]);
    end
end

function finishAxes(R, FN)
    axis equal; grid on; box on
    L = R*2.0;
    xlim([-L L]); ylim([-L L]); zlim([-L L]);
    xlabel('x  [mm]'); ylabel('y  [mm]'); zlabel('z  [mm]');
    set(gca, 'FontName', FN, 'FontSize', 10, ...
             'GridColor', [0.85 0.85 0.85], 'GridAlpha', 1);
    view(128, 22);
    camlight('headlight'); lighting gouraud
    rotate3d on
end
```

B.3 Sphere1_Layouts_7_and_12.m

```
% Sphere-1 -- the seven- and twelve-nozzle layouts, for comparison with the
% eight that was built
%
%  Temur Makharadze, bachelor's thesis
%
%  ------------------------------------------------------------------
%  HOW TO RUN
%      Press Run (F5). Three figures and a comparison table.
%        Figure 1  twelve nozzles, six opposed pairs
%        Figure 2  seven nozzles, the minimum that can work at all
%        Figure 3  all three layouts side by side
%      Drag inside a figure to rotate. To save one:
%         exportgraphics(gcf,'layout12.png','Resolution',300)
%  ------------------------------------------------------------------
%
%  ON THE TWELVE
%  Two nozzles at each of the six face centres, every one firing tangentially
%  so its moment arm is the full radius. The two on the +X face push +Y and
%  -Y; the two on the +Y face push +Z and -Z; the two on the +Z face push
%  +X and -X, and likewise on the negative faces. Fire an antipodal pair the
%  same way and you get pure force with the torques cancelling; fire them
%  opposite and you get a pure couple of 2*F0*R = 21.66 mN·m. That is where
%  the 2.78x torque advantage over the eight-nozzle layout comes from.
%
%  ON THE SEVEN
%  Seven is the theoretical minimum: push-only thrusters need N+1 = 7 to
%  positively span a six-dimensional wrench. Finding a seven that actually
%  works is harder than it sounds, and the search is in the header of the
%  section below. The short version: the MOST symmetric arrangement does
%  not work at all. A pentagonal bipyramid with every nozzle canted the same
%  way in its own local frame fails to span -- the symmetry that makes it
%  elegant is exactly what stops it reaching every direction. The cant
%  pattern has to be deliberately staggered before seven can control six
%  axes, and even then it is far weaker than eight.
%
%  No toolboxes needed. Base MATLAB, R2016b or newer.

clear; clc; close all


%% ===== CONSTANTS ============================================================

R_MM = 115.0;              % mm, mounting radius
F0   = 94.188e-3;          % N,  thrust per nozzle at full open

BLUE  = [ 31  78 121]/255;
RED   = [149  55  53]/255;
GREEN = [ 79  98  34]/255;
GREY  = [140 140 140]/255;
FN    = 'Calibri';

LJ = 66;   % mm, jet arrow length
LF = 42;   % mm, force arrow length


%% ===== FIGURE 1 -- twelve nozzles ===========================================
%  Face axis carries thrusters along the NEXT axis, cyclically:
%     +/-X faces push along +/-Y,  +/-Y faces push +/-Z,  +/-Z faces push +/-X.
%  Both nozzles at a face centre are co-located in the model; they are drawn
%  a few millimetres apart so you can see both.

if ~exist('R_MM','var'), error('Run the CONSTANTS section first.'); end

I3 = eye(3);
P12 = zeros(3,12); D12 = zeros(3,12); N12 = cell(1,12);
AX  = 'XYZ';  c = 0;
for fi = 1:3
    ti = mod(fi,3) + 1;                       % thrust axis follows the face axis
    for fs = [1 -1]
        for ts = [1 -1]
            c = c + 1;
            P12(:,c) = R_MM * fs * I3(:,fi);
            D12(:,c) = ts * I3(:,ti);
            N12{c}   = sprintf('%c%c:%c%c', sgn(fs), AX(fi), sgn(ts), AX(ti));
        end
    end
end

```

```
reportLayout('TWELVE', P12, D12, N12, F0);

figure('Color','w','Position',[60 60 900 780]);
drawSphere(R_MM); hold on
for k = 1:12
    p = P12(:,k);
    off = 9 * cross(P12(:,k)/R_MM, D12(:,k));      % nudge co-located pairs apart
    drawNozzle(p + off, D12(:,k), LJ, LF, BLUE, N12{k}, FN, 9);
end
bodyAxes(R_MM, FN);
title({'Twelve nozzles -- six opposed pairs', ...
       'every nozzle tangential, moment arm = R = 115 mm', ...
       'pair together = pure force,  pair opposed = pure couple of 21.66 mN·m'}, ...
      'FontName', FN, 'FontSize', 12, 'FontWeight', 'normal');
finishAxes(R_MM, FN);


%% ===== FIGURE 2 -- seven nozzles ========================================
%  Positions: pentagonal bipyramid, five on the equator plus two poles. That
%  is the most symmetric way to place seven points on a sphere.
%
%  Cant: 65 deg off radial for all seven, so every moment arm is 104.23 mm.
%  The azimuth of the cant advances 228 deg per nozzle around the ring, and
%  the poles are canted 120 deg apart. Those numbers come from a search over
%  cant angle and azimuth pattern, keeping only configurations whose wrench
%  matrix positively spans R^6. A uniform azimuth -- the symmetric choice --
%  does not span, which is the whole point.

if ~exist('R_MM','var'), error('Run the CONSTANTS section first.'); end

BETA  = 65;      % deg, cant off radial
AZ0   = 0;       % deg, first ring nozzle
DAZ   = 228;     % deg, azimuth advance per ring nozzle
APOL  = 120;     % deg, north pole cant azimuth

P7 = zeros(3,7); D7 = zeros(3,7); N7 = cell(1,7);
b  = deg2rad(BETA);
for k = 1:5
    th   = 2*pi*(k-1)/5;
    rh   = [cos(th); sin(th); 0];
    east = [-sin(th); cos(th); 0];
    nor  = [0; 0; 1];
    a    = deg2rad(AZ0 + (k-1)*DAZ);
    tang = cos(a)*east + sin(a)*nor;
    P7(:,k) = R_MM * rh;
    D7(:,k) = -rh*cos(b) + tang*sin(b);       % force on body
    N7{k}   = sprintf('E%d', k);
end
for j = 1:2
    s  = 3 - 2*j;                              % +1 then -1
    rh = [0; 0; s];
    a  = deg2rad(APOL + (j-1)*180);
    tang = cos(a)*[1;0;0] + sin(a)*[0;1;0];
    P7(:,5+j) = R_MM * rh;
    D7(:,5+j) = -rh*cos(b) + tang*sin(b);
    N7{5+j}   = sprintf('P%d', j);
end

reportLayout('SEVEN', P7, D7, N7, F0);

fprintf('  and the reason seven was abandoned:\n');
best = inf;
for i = 1:6
    for j = i+1:7
        t = norm(cross(P7(:,i)/1000, D7(:,i)) + cross(P7(:,j)/1000, D7(:,j)));
        best = min(best, t);
    end
end
fprintf('    smallest leftover torque from ANY pair : %.4f mN·m\n', best*F0*1e3);
fprintf('    the eight-nozzle radial pairs leave    : 0.0000 mN·m\n');
fprintf('    nothing cancels, so every push also twists.\n\n');

figure('Color','w','Position',[60 60 900 780]);
drawSphere(R_MM); hold on
for k = 1:7
    c = BLUE; if k > 5, c = RED; end
    plot3([0 P7(1,k)],[0 P7(2,k)],[0 P7(3,k)], ':', 'Color', GREY, 'LineWidth', 1.0);
    drawNozzle(P7(:,k), D7(:,k), LJ, LF, c, N7{k}, FN, 11);
end
bodyAxes(R_MM, FN);
title({'Seven nozzles -- the theoretical minimum', ...
```

```
        sprintf('pentagonal bipyramid, canted %d° off radial, arm %.2f mm', ...
                BETA, R_MM*sind(BETA)), ...
        'the cant azimuth is staggered on purpose: a symmetric one does not span'}, ...
      'FontName', FN, 'FontSize', 12, 'FontWeight', 'normal');
finishAxes(R_MM, FN);


%% ===== FIGURE 3 -- all three side by side ================================

if ~exist('P7','var'), error('Run the two layout sections first.'); end

% the eight that was actually built
D8 = [ ...
 -0.577350 -0.577350  0.577350  0.577350 -0.076090  0.317520  0.076100 -0.317529
 -0.577350  0.577350 -0.577350  0.577350  0.829232 -0.041990  0.829231 -0.041990
 -0.577350  0.577350  0.577350 -0.577350  0.553701 -0.947321 -0.553701  0.947318 ];
P8 = [ ...
  66.39528  66.39528 -66.39528 -66.39528 -62.36530 -65.39570  62.44840  65.39610
  66.39528 -66.39528  66.39528 -66.39528 -69.30520  63.22120 -69.48640  63.22130
  66.39528 -66.39528 -66.39528  66.39528 -67.32270  70.36670  67.05850 -70.36630 ];

figure('Color','w','Position',[40 60 1500 560]);
sets = {P7, D7, 'seven'; P8, D8, 'eight -- built'; P12, D12, 'twelve'};
cols = {RED, GREEN, BLUE};
for q = 1:3
    subplot(1,3,q);
    drawSphere(R_MM); hold on
    Pq = sets{q,1}; Dq = sets{q,2};
    for k = 1:size(Pq,2)
        off = [0;0;0];
        if q == 3, off = 9*cross(Pq(:,k)/R_MM, Dq(:,k)); end
        drawNozzle(Pq(:,k)+off, Dq(:,k), 58, 0, cols{q}, '', FN, 0);
    end
    bodyAxes(R_MM, FN);
    title(sprintf('%s  (%d nozzles)', sets{q,3}, size(Pq,2)), ...
          'FontName', FN, 'FontSize', 13, 'FontWeight', 'normal');
    finishAxes(R_MM, FN);
end


%% ===== LOCAL FUNCTIONS ===================================================

function s = sgn(v)
    if v > 0, s = '+'; else, s = '-'; end
end

function reportLayout(label, P, D, N, F0)
% print position, direction, cant off radial and moment arm for each nozzle
    n = size(P,2);
    fprintf('\n%s -- %d nozzles\n', label, n);
    fprintf('%7s %24s %26s %8s %10s\n', 'noz', 'position [mm]', 'thrust direction', ...
            'cant', 'arm [mm]');
    fprintf('%s\n', repmat('-', 1, 80));
    for k = 1:n
        r = P(:,k); d = D(:,k);
        inward = -r/norm(r);
        cant = acosd(max(min(dot(d, inward), 1), -1));
        arm  = norm(cross(r, d));
        fprintf('%7s (%6.1f %6.1f %6.1f)  (%7.4f %7.4f %7.4f) %6.2f° %9.2f\n', ...
                N{k}, r(1), r(2), r(3), d(1), d(2), d(3), cant, arm);
    end
    fprintf('  sum of thrust directions: (%6.3f %6.3f %6.3f)\n', sum(D,2));
    fprintf('  torque from one nozzle at full open: %.3f mN·m\n', ...
            norm(cross(P(:,1)/1000, D(:,1)*F0))*1e3);
end

function drawNozzle(p, d, LJ, LF, c, lab, FN, fs)
    quiver3(p(1), p(2), p(3), -d(1)*LJ, -d(2)*LJ, -d(3)*LJ, 0, ...
        'Color', c, 'LineWidth', 1.9, 'MaxHeadSize', 0.55);
    if LF > 0
        quiver3(p(1), p(2), p(3), d(1)*LF, d(2)*LF, d(3)*LF, 0, ...
            'Color', c*0.40 + 0.60, 'LineWidth', 1.3, 'MaxHeadSize', 0.5);
    end
    plot3(p(1), p(2), p(3), 'o', 'MarkerSize', 6, ...
        'MarkerFaceColor', c, 'MarkerEdgeColor', 'w', 'LineWidth', 0.9);
    if ~isempty(lab)
        q = p - d*(LJ + 20);
        text(q(1), q(2), q(3), lab, 'FontName', FN, 'FontSize', fs, ...
            'FontWeight', 'bold', 'Color', c, 'HorizontalAlignment', 'center');
    end
end
```

```
241
242 function drawSphere(R)
243     [sx, sy, sz] = sphere(60);
244     surf(sx*R, sy*R, sz*R, 'FaceColor', [0.82 0.85 0.90], 'FaceAlpha', 0.25, ...
245          'EdgeColor', 'none');
246     hold on
247 end
248
249 function bodyAxes(R, FN)
250     L = R*1.55; ax = {'X','Y','Z'};
251     for i = 1:3
252         v = zeros(3,1); v(i) = L;
253         plot3([-v(1) v(1)], [-v(2) v(2)], [-v(3) v(3)], '-', ...
254               'Color', [0.72 0.72 0.72], 'LineWidth', 0.8);
255         text(v(1)*1.06, v(2)*1.06, v(3)*1.06, ax{i}, 'FontName', FN, ...
256              'FontSize', 11, 'Color', [0.45 0.45 0.45]);
257     end
258 end
259
260 function finishAxes(R, FN)
261     axis equal; grid on; box on
262     L = R*2.0;
263     xlim([-L L]); ylim([-L L]); zlim([-L L]);
264     xlabel('x  [mm]'); ylabel('y  [mm]'); zlabel('z  [mm]');
265     set(gca, 'FontName', FN, 'FontSize', 10, ...
266              'GridColor', [0.85 0.85 0.85], 'GridAlpha', 1);
267     view(128, 22);
268     camlight('headlight'); lighting gouraud
269     rotate3d on
270 end
```

APPENDIX C

Verification data and full geometry tables

C.1 As-built nozzle geometry

Thrust directions and mount positions exported from CAD, in the body frame, with the derived cant angle, perpendicular moment arm and torque per nozzle at full opening. Direction vectors give the direction of the force on the body; the gas leaves in the opposite direction. Positions are millimetres from the centre of mass.

Noz.	Position [mm]			Thrust direction			$\lvert r \rvert$	Cant [°]	Arm [mm]
R1	66.395	66.395	66.395	-0.57735	-0.57735	-0.57735	115.0000	0.0553	0.000
R2	66.395	-66.395	-66.395	-0.57735	0.57735	0.57735	115.0000	0.0553	0.000
R3	-66.395	66.395	-66.395	0.57735	-0.57735	0.57735	115.0000	0.0553	0.000
R4	-66.395	-66.395	66.395	0.57735	0.57735	-0.57735	115.0000	0.0553	0.000
T1	-62.365	-69.305	-67.323	-0.07609	0.82923	0.55370	114.9999	38.4988	71.587
T2	-65.396	63.221	70.367	0.31752	-0.04199	-0.94732	115.0000	38.4367	71.490
T3	62.448	-69.486	67.058	0.07610	0.82923	-0.55370	115.0000	38.5013	71.591
T4	65.396	63.221	-70.366	-0.31753	-0.04199	0.94732	115.0000	38.4366	71.489

Sum of the four radial thrust directions	**(0.000, 0.000, 0.000)**
Sum of the four canted thrust directions	**(0.000, 1.5745, 0.000)**
Mean mounting radius	**115.0000 mm**
Nominal moment arm $R \sin 38.50°$	**71.5892 mm**
Torque per canted nozzle at full opening	**6.7335 to 6.7430 mN·m**

The radial group's residual cant of $0.0553°$ *is an artefact of the six-decimal direction vectors exported from CAD rather than a real misalignment; the computed moment arm is 0.000 mm and the torque 0.0000 mN·m to the precision carried. The canted group's spread, 38.4366° to 38.5013° with the T1 and T3 positions differing by up to 0.26 mm where symmetry would make them identical, is real, and it is why turn authority is uneven as described in Section 8.3.*

C.2 Translation solutions, full precision

Axis	Nozzles and duty	Net force [N]	Net torque [N·m]
$+X$	R3 1.00 · R4 1.00	0.1087588836	0.0000000000
$-X$	R1 1.00 · R2 1.00	0.1087588836	0.0000000000
$+Y$	R2 1.00 · R4 1.00	0.1087588836	0.0000000000
$-Y$	R1 1.00 · R3 1.00	0.1087588836	0.0000000000
$+Z$	R2 1.00 · R3 1.00	0.1087588836	0.0000000000
$-Z$	R1 1.00 · R4 1.00	0.1087588836	0.0000000000

C.3 Rotation solutions, full precision

Axis	Nozzles and duty	Torque [N·m]
$+X$	T1 0.6516 · T3 0.3470 · T4 1.0000	5.873779×10^{-3}
$-X$	T1 0.3476 · T2 1.0000 · T3 0.6508	5.841348×10^{-3}
$+Y$	T1 0.9998 · T2 0.0055 · T3 1.0000	7.450913×10^{-3}
$-Y$	T2 1.0000 · T4 1.0000	7.461145×10^{-3}
$+Z$	T2 0.3505 · T3 1.0000 · T4 0.6514	5.866644×10^{-3}
$-Z$	T1 1.0000 · T2 0.6538 · T4 0.3473	5.867896×10^{-3}

Duties are the linear-programming solution for maximum torque about the named body axis subject to every duty lying in [0, 1]. Only $\pm Y$ is served by a clean two-nozzle pair; the other four axes require three nozzles at partial duty.

C.4 Three-layout authority comparison

Guaranteed authority for the seven-, eight- and twelve-nozzle layouts, computed by linear programming over 200 directions distributed on a Fibonacci sphere. For each direction the solver maximises the achievable magnitude subject to every thruster duty lying in [0, 1] and to the complementary wrench component being exactly zero. The table reports the worst direction found.

Layout	Valve mass	Guaranteed pure force	Guaranteed pure torque
seven	175 g	21.63 mN	2.537 mN·m
eight, built	**200 g**	**76.94 mN**	**2.952 mN·m**
twelve	300 g	188.50 mN	21.677 mN·m

These figures are stricter than the per-axis numbers in Chapter 8 and do not contradict them. Chapter 8 reports the best achievable along a body axis using the relevant group alone: 108.76 mN of force, and 5.87 mN·m of torque on the worst body axis. This table reports the worst direction anywhere on the sphere, with the complementary wrench component driven to exactly zero, which additionally

> requires the radial group to cancel the side force leaked by the canted group. The two answer different questions.

The comparison also makes the seven-nozzle failure concrete. Seven gives roughly a quarter of the eight-nozzle guaranteed force and about 86% of its guaranteed torque, while saving 25 g of valve. The smallest leftover torque from any pair of the seven is non-zero, which is the same statement as “no antagonistic pairs exist”, against exactly 0.0000 mN·m for the eight-nozzle radial pairs.

C.5 RK4 convergence, raw

Steps N	Global error	Ratio to previous
10	2.491×10^{-6}	—
20	1.450×10^{-7}	17.18 ×
40	8.748×10^{-9}	16.58 ×
80	5.371×10^{-10}	16.29 ×
160	3.328×10^{-11}	16.14 ×
320	2.071×10^{-12}	16.07 ×

A fourth-order method should give a ratio of $2^4 = 16$ per halving of the step. The observed orders are 4.103, 4.051, 4.026, 4.012 and 4.006.

C.6 Nozzle exit temperature against expansion ratio

Carbon dioxide at $T_0 = 293.15$ K, $\gamma = 1.280$, ideal isentropic expansion. The triple point at 216.6 K is crossed at $A_e/A_t = 1.2658$.

A_e/A_t	Exit static temperature [K]
1.00	257.1
1.20	222.2
1.25	217.9
1.30	**214.0**
1.40	**207.2**
1.60	**196.3**
1.80	**187.6**
2.00	**180.5**
2.20	**174.5**
2.40	**169.3**
2.60	**164.7**
2.80	**160.6**
3.00	**156.9**

C.7 Chamber pressure against throat diameter

Throat diameter [mm]	Chamber pressure [bar]
2.00	0.25
1.50	0.44
1.00	0.99
0.80	**1.54**
0.60	2.74

From $P_0 = P_{\text{feed}}\,(C_dA)_{\text{inj}}/(C_dA)_{\text{throat}}$ with $(C_dA)_{\text{inj}} = 0.147\ mm^2$. The relation is valid only where both restrictions choke, so the 2.00 mm point is a vacuum-only result.

C.8 Lifetime angular impulse against valve unit mass

Equation (5.3) evaluated for the eight- and twelve-nozzle layouts. The eight-nozzle curve is the one that survives across the purchasable range.

Valve mass [g]	H, eight nozzles [N·m·s]	H, twelve nozzles [N·m·s]
5.00	3.193	**7.838**
10.00	2.441	**4.703**
12.50	2.066	**3.135**
14.75	**1.728**	1.724
15.00	**1.690**	1.568
17.50	**1.315**	does not close
20.00	**0.939**	does not close
25.00	**0.188**	does not close
30.00	does not close	does not close

Crossover valve mass	**14.74 g**
Valve mass above which twelve does not close the budget	**17.5 g**
Purchasable flight solenoid range	**20 to 30 g**

C.9 Three-axis pointing life against deadband

Deadband half-width [°]	$t_{\min}$ = 0.5 ms	$t_{\min}$ = 1.0 ms	$t_{\min}$ = 2.0 ms
±1.00	8.80 d	2.20 d	0.55 d
±1.50	13.20 d	**3.30 d**	0.83 d
±2.00	17.60 d	4.40 d	1.10 d
±2.50	22.00 d	5.50 d	1.38 d
±3.00	26.40 d	6.60 d	1.65 d
±4.00	35.21 d	8.80 d	2.20 d
±5.00	44.01 d	11.00 d	2.75 d

Baseline design point	**±1.50°, $t_{\min}$ = 1.0 ms**
Baseline three-axis pointing life	**3.30 days**
Single nozzle held continuously open	**48.2 s**
Ratio	**5 912 ×**
Total impulse	**4.543 N·s**
Δv if spent entirely on translation	**9.18 m/s**
Repositioning hops of 10 m	**45**

C.10 Consolidated reference figures

Every quoted figure in this report, in one place. This is the table carried on the closing slide of the presentation.

wet / dry mass	**500 / 490 g**	propellant	**10 g CO_2**
valve mass, flight / bench	**25 g / 80 g**	γ · specific gas constant	**1.280 · 188.9**
CO_2 triple point	**216.6 K**	M_e / (A_e/A_t) limit	**1.589 / 1.266**
I_{sp}, conv. / conv.–div.	**46.33 / 48.42 s**	throat diameter	**0.80 mm**
chamber pressure	**1.5392 bar**	mass flow per nozzle	**0.2073 g/s**
thrust per nozzle	**94.188 mN**	force per body axis	**108.76 mN**
minimum impulse bit	**94.19 μN·s**	cant · moment arm	**38.50° · 71.59 mm**
three-axis pointing life	**3.3 days**	Δv / 10 m hops	**9.18 m/s / 46**
turn authority, Y / X, Z	**7.45 / 5.87 mN·m**	axis efficiency, rot. / trans.	**36 / 58 %**
twelve-nozzle torque advantage	**2.78 ×**	crossover valve mass	**14.7 g**
peak angular acceleration	**169.2 °/s^2**	slew time, 30° / 90°	**0.84 / 1.46 s**
inertia, CAD / thin shell	**2.637 / 4.408 $\times 10^{-3}$**	$k_p \cdot k_d$	**7.1432 · 1.7469 $\times 10^{-2}$**
$\omega_n \cdot \zeta$, nominal	**3.680 rad/s · 0.900**	integrator	**RK4, 5 ms, 200 Hz**
survey cost, orbit / ballistic	**73 / 24 % of tank**	surveys per tank	**1.4 / 4.1**
valves shut, ballistic survey	**88 %**	Δv conversion, measured	**41 %**

APPENDIX D

Defence script, as delivered

The spoken script for the 24 slides, with the timing marks used in rehearsal. It is included as a record of how the material was presented, and because the phrasing of several arguments, in particular the mass budget, the condensation limit and the allocator, is more compact here than in the body of this report. Stage directions have been removed; what follows is the spoken text only.

Slides	**24**
Spoken words	**2 741**
Approximate spoken duration	**18 min 40 s**

Slide 1 Title 0:00 - 0:11

Hi, I'm Temur. This is Sphere-1.

Before I show you what it is, I want to show you why it should exist.

Slide 2 Why this needs to exist 0:11 - 2:03

January 2003. Eighty-one seconds after launch, a piece of foam came off Columbia's external tank and hit the leading edge of the left wing at about 500 miles an hour. NASA saw it the next day, in the launch film.

So they knew something had hit the wing. What they couldn't tell was whether it mattered.

And they had no way to find out. That flight didn't carry the robotic arm. The crew couldn't see the leading edge from the windows. No spacewalk was planned. The only option left was to ask the Air Force to photograph the orbiter from the ground.

Three separate requests for that were made. All three were cancelled. One of the reasons on record is that there wasn't much they could do about it anyway.

[pause]

That turned out to be wrong. The accident board later found a rescue flight by Atlantis was challenging but feasible, if the damage had been confirmed by flight day seven. But nobody could look, so the question never got asked properly.

And look at what NASA built afterwards. A fifty-foot boom with cameras and laser scanners, carried on every flight just to inspect its own heat shield. And a full backflip under the Space Station so the crew there could photograph the belly.

Those are both workarounds for one missing thing: a camera that can get outside the vehicle and look back at it.

Sphere-1 wouldn't have flown on Columbia. No arm, nothing to deploy from. But the reason that crew never got an answer is that nobody could look, and that's the gap this fills.

Slide 3 What is Sphere-1? 2:03 - 2:31

So that is what it is for. This is what it is.

A 23-centimetre sphere, 500 grams, flying on its own within a few metres of the asset, with a visible camera and a thermal one. Eight cold-gas nozzles, and nothing in the exhaust that can settle on what it is photographing.

Which raises the obvious question: why not just use a small rocket?

Slide 4 Design constraints 2:31 - 3:12

Because the exhaust leaves soot, and it lands on the cameras and solar panels we came to photograph. We would wreck the thing we came to look at. So the propellant has to leave nothing behind.

Then the mass. 490 grams empty, and 200 of those are the eight valves. 10 grams of CO2.

I should say what's not in there: structure, shielding, anything you'd need to certify it near a crewed station. I don't know how to estimate that. 500 grams is a target I set myself to keep the design honest.

Slide 5 Why the orientation is a quaternion 3:12 - 4:58

The vehicle is seventeen numbers. Four of them are the orientation, and those four do the most work, because the thrusters act in body axes and the mission is described in world axes.

So how do you store an orientation. Two options.

Euler angles are the obvious one: roll, pitch, yaw. Three numbers, easy to picture. The problem is advancing them. To get from the measured body rates to the rate of change of the angles, you need those three lines on the left, and two of them have cos theta in the denominator and tan theta in them.

At ninety degrees of pitch, nose straight up, cos theta is zero. So the computer divides by zero and the attitude solution falls over. That is gimbal lock.

A quaternion is four numbers instead. It's just an axis and an angle, packed together: cosine of half the angle, and the axis times sine of half the angle. And advancing it is one multiplication, q-dot equals a half q times omega. No trigonometry and no division anywhere in it. Smooth everywhere, including straight up.

What it costs is one extra number. Four numbers for three degrees of freedom, so there's a constraint, the length has to stay one. Integration error slowly pushes it off, so the code divides q by its own length once every step. That's the entire price.

And this is not a corner case. With the controller in my hand I reach ninety degrees of pitch in about two seconds.

Slide 6 Choice of propellant 4:58 - 5:29

Nitrogen is what people normally use. More push per gram, and it can't freeze. On paper it's better.

It loses on storage. It stays a gas, so you need a 200-bar tank, and a tank that strong is heavy. CO2 goes liquid at 57 bar and the same bottle holds several times more.

So I gave up efficiency for storage. The price is a temperature limit, and that's the next slide.

Slide 7 The condensation cliff 5:29 - 5:59

A supersonic nozzle is a very good fridge. Gas cools as it expands, and it cools fast.

CO2 has a floor. Below 216 kelvin it doesn't turn into liquid, it goes straight to dry ice. Inside a hole less than a millimetre wide.

So I set the exit temperature to that floor and worked backwards. Maximum Mach 1.589. Maximum expansion ratio 1.266. That's all the expansion I'm allowed.

Slide 8 The condensation cliff - consequence 5:59 - 6:49

Follow the blue line down to the red one. That's where it freezes.

Now in metal. A converging-diverging cone would buy 4.5% more efficiency. At my throat that cone is 0.18 millimetres long.

That's a chamfer, not a nozzle. You can't machine it repeatably and it risks the icing it's meant to prevent. So I left it off. I pay 4.5% and get a part I can actually make.

And that fixes the throat at 0.8 millimetres, because chamber pressure comes from the ratio of two areas in series, the valve opening and the throat. That gives 1.54 bar and 94 millinewtons per nozzle, which is where every thrust and torque number after this comes from.

Slide 9 Actuator layout: three iterations 6:49 - 7:11

Now the thrusters.

It has to control six things: push on three axes, turn on three axes. Thrusters only push, never pull, and that's what makes it hard. You need at least seven to cover six directions.

I built and modelled three layouts, and each one failed differently.

Slide 10 Iteration 1 - twelve nozzles 7:11 - 7:39

Twelve is the conventional layout. Six opposing pairs, each pushing straight down its own axis. No maths, just a lookup table. Full thrust on the axis you want, and 2.78 times the turning force of what I ended up with.

It does not fit. Twelve valves is 300 grams. Structure and electronics are 290. That's 90 grams over before any gas goes in.

Slide 11 Iteration 2 - seven nozzles 7:39 - 8:10

Seven is the minimum that works at all, and it's lighter. 35 grams of propellant instead of 10, so three and a half times the life. On mass alone it's the better design.

It fails on control. With seven you can't make opposing pairs, so every time you push you also twist. The attitude controller ends up fighting the translation controller. It buys propellant and then spends it fighting itself.

Slide 12 Iteration 3 - eight nozzles 8:10 - 9:04

Eight works, as two groups of four doing two different jobs.

Four sit on alternating corners of a cube, pointing straight out through the centre. Because the thrust line goes through the centre, they push without turning it at all.

The other four are swung 38 and a half degrees away from radial. And that swing is not a compromise, it is what makes the torque exist at all. A nozzle pointing straight out has no lever. Swinging it gives me 71.6 millimetres.

The theoretical maximum is 115 millimetres, from a nozzle firing flat along the hull. So the 37.7 percent I am supposedly losing is measured against the worst possible case for the plume, not against a neutral design.

Now let me show you what each group actually does.

Slide 13 Moving: one axis at a time 9:04 - 10:02

The four radial nozzles sit on alternating corners of a cube, each pointing out through the centre. So each one pushes on all three axes at once, with 0.577 of its thrust on each.

Fire the right two and the other two axes cancel. Take one row: for plus X I fire R3 and R4. Their Y parts are equal and opposite, so they cancel. Their Z parts cancel. What's left is pure X, at 1.155 times what one nozzle gives. 108.76 millinewtons.

They cancel because of where the nozzles are, not because a controller noticed and corrected. And look at the last column: the torque is exactly zero, because every thrust line passes through the centre of mass.

Six clean moves, two nozzles each, nothing left over. Moving genuinely is a lookup table.

Slide 14 Turning: one axis at a time 10:02 - 11:20

Turning is not that tidy, and I want to be straight about why.

The canted four are the only source of torque on this vehicle, because the radial four make exactly none. But they're mounted on measured hardware, not a perfect tetrahedron. My cant angles come out between 38.43 and 38.50 degrees, not all identical.

So look at the table. Only two of the six axes come from a clean pair. Plus and minus Y, that's T1 with T3 and T2 with T4, 7.45 millinewton-metres. The other four axes need three nozzles at partial throttle, and they only give 5.87.

Two things fall out of that. First, turning is not equally easy on every axis: 7.45 about Y, 5.87 about X and Z, a ratio of 0.79. Y is just the axis the mounts happen to favour.

Second, turning always pushes as well. A canted nozzle thrusts while it twists, so an attitude command leaks up to 156 millinewtons of side force, and the translation group has to absorb that.

And this is exactly why the vehicle has an allocator instead of a lookup table.

Slide 15 Why the extra efficiency does not pay 11:20 - 12:15

First, what the tilt costs. Two losses multiply, the tetrahedron and the cant, and eight nozzles end up delivering 36 percent of the twist twelve would give from the same gas. On its own that says go back to twelve.

Here is why I didn't.

Turning force per pulse is only half of it. What a mission spends is turning force times how long the tank can keep going. The four extra valves give 2.78 times the twist, and cost 100 grams straight out of propellant.

The curves cross at 14.7 grams per valve. Above 17.5 you can't build twelve at all. Every flight valve I can actually buy is 20 to 30 grams. All of them are in the grey band, and in there, eight wins.

Slide 16 Solving the redundancy 12:15 - 13:17

So why is there an allocator at all.

Four nozzles, three equations. That means infinitely many firing combinations give exactly the same torque. The answers form a line, not a point. So the allocator isn't there to find an answer, it's there to pick the cheapest one out of infinitely many, 200 times a second, without ever stopping to iterate.

And cheapest matters, because gas is the mission. Every gram wasted on a clumsy firing choice is flight time gone.

Three steps. Firing all four equally does nothing at all, it just burns gas. The total gas used is a straight line in the one free parameter. And a straight line over an interval is smallest at one end.

So it's a formula, not a search. Under a microsecond, every cycle. An optimiser has no place in a real-time loop, and here it doesn't need one.

Slide 17 The smallest pulse we can make 13:17 - 14:22

The valves can't be opened halfway. Shut or open, 1 millisecond to open cleanly. So there's a smallest push this vehicle can ever deliver: thrust times that time, 94 micronewton-seconds. Every command is built out of whole copies of it.

And three things follow from one pulse. It leaves the drone drifting at 0.085 degrees per second. At that rate it takes 35 and a half seconds to cross the deadband. And it costs 0.41 milligrams of gas.

Those three lines are the whole mission-life argument in miniature.

Because firing without pause empties the tank in 48 seconds. So instead, while the drone points within one and a half degrees of target, every thruster is off and it just drifts. Only at the edge does it spend one pulse to turn itself around.

If that sawtooth looks like a badly tuned controller, it isn't. That's the propellant budget being protected.

Slide 18 How long does Sphere-1 actually fly? 14:22 - 14:55

So how long does it fly.

Firing non-stop, the tank is empty in 48 seconds. With the deadband, holding attitude on all three axes lasts 3.3 days. Same ten grams, about 5,900 times longer.

And what ten grams buys: 4.5 newton-seconds. All on moving, that's 9 metres per second, or 46 hops of ten metres. Or three days of pointing.

How far that goes depends on the flight path, and that's where we won the most.

Slide 19 Circles cost, straight lines do not 14:55 - 16:02

Flying the same scan in straight lines instead of circling the target dropped one full pass from 73% of the tank to 24%.

The reason is simple. Going round a curve means pushing sideways the whole time. Nothing holds you in the turn except your own thrusters, so a curve charges you for every second you're turning, and five laps is 31 radians. A straight coast is free, however long it is. A leg costs twice the speed, and that's the whole bill.

[pause]

One tank is four passes now, instead of one.

And we didn't buy it with a worse scan. Coverage went from 99.1% to 98.7%. Turning got cheaper too, because flying parallel to a flat face with the camera already square to it needs no turning at all.

One thing I'll say myself. This ran on a heavier model, 4.55 kilos. But it compares two flight paths on the same vehicle, so the ratio holds.

Slide 20 The straight-line pattern, flown 16:02 - 16:53

[the video starts and loops on its own, talk over it]

Watch the plumes. Short burn, long silent glide, burn to stop, then nothing. Every valve is shut for 88% of the flight.

That's the real 25-minute flight at 15 times speed. A 6.6-metre satellite, 84 metres of path, eight straight legs.

Two things before you ask. We planned a one-metre standoff and got to 38 centimetres. Too close. The legs need replanning with a hard keep-out sphere.

And the usable speed budget is about 10 metres per second, not the 25 the rocket equation gives. The nozzles sit on cube corners, so when the direction I want isn't on an axis, some of them fire against each other. 41%. Same effect as the 36% I showed you earlier.

Slide 21 Making the ten grams last 16:53 - 17:50

So if we want more, where does it come from. Chain those three lines together and you get the burn rate, and the valve time in it is squared.

That's the whole answer. Halving the valve opening time cuts the burn by four, so the same ten grams last four times as long. One millisecond to half a millisecond takes 3.3 days to 13. And a faster valve weighs nothing extra.

Widening the deadband only helps linearly, so it doubles. Both together, 26 days on the same ten grams.

And the honest version. What limits us is our own valve, not space. A sphere is barely pulled by gravity, and drag costs about 27 pulses per orbit. If we want it to fly longer we don't need orbital mechanics, we need a faster valve.

Slide 22 Hardware in the loop 17:50 - 18:37

Last part, the hardware. An ESP32 and a cheap gyro on a breadboard, running the real control loop at 200 hertz.

[pick up the board, turn it slowly, say nothing for three seconds]

Passing the gyro's raw rate straight into the controller lets it react while the motion is happening, instead of waiting for the error to build up. Average lag goes from 11 degrees to 1.2.

Under fifty dollars, testing a loop that normally needs an air-bearing table.

And the simulator underneath it was checked the same way. RK4 convergence came out at 4.006, and with every thruster off, angular momentum held to one part in ten to the fifteenth over twelve thousand steps while the body rates genuinely wandered.

Slide 23 What comes next 18:37 - 19:26

Three things stand between this and a vehicle that flies, and they're in this order because each one feeds the next.

First, flow-bench the injector. Measure how much gas it actually passes instead of trusting the datasheet. That one number sets the thrust, the mass flow and the minimum pulse, so everything else moves with it.

Second, measure the real inertia. Hang it on two strings, time how long it takes to twist back and forth, and recover the inertia from the period. Cheap, and far more trustworthy than what the CAD assumes.

Third, retune the gains against that measurement and close the loop on the real vehicle instead of the simulated one.

Slide 24 Thank you 19:26 - 19:30

Thanks. Happy to take questions.